Study of High Enthalpy Flows in Arc heated Wind Tunnels

A Dissertation
Submitted in partial fulfillment of
the requirements for the degree of
Master of Technology
by

Deepak Yadav
(Roll No. 173010011)

Supervisor:
Prof. Kowsik Bodi



Department of Aerospace Engineering
Indian Institute of Technology Bombay
Mumbai 400076 (India)

30 June 2019

Acceptance Certificate

Department of Aerospace Engineering

Indian Institute of Technology, Bombay

The dissertation entitled "Study of High Enthalpy Flows in Arc heated Wind Tunnels" submitted by Deepak Yadav (Roll No. 173010011) may be accepted for being evaluated.

Date: 26 June 2019

Prof. Kowsik Bodi

Approval Sheet

This dissertation entitled "Study of High Enthalpy Flows in Arc heated Wind Tunnels" by Deepak Yadav is approved for the degree of Master of Technology.

Examiners

Supervisor (s)

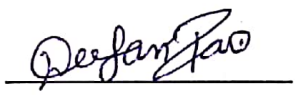
Chairman

Date: 26.6.19

Place: Mumbai

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I declare that I have properly and accurately acknowledged all sources used in the production of this report. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.



Deepak Yadav
(Roll No. 173010011)

Date: 26 June 2019

Acknowledgements

I take the opportunity to express my sincere gratitude and indebtedness to Prof. Kowsik Bodi, Department of Aerospace Engineering, Indian Institute of Technology Bombay, for his guidance and constant encouragement during the course of this project work without which it would not have been possible. I would also like to thanks all my friends (Ved Mukherjee, Aejit Tripathi, Ritu Rewar, Deepak Gujjar) for providing me valuable input and support for completion of this MTP.

Deepak Yadav
IIT Bombay
26 June 2019

Abstract

The increased speed requirements of missiles and aircraft from subsonic in 1930's to hypersonic in present era has demanded an evolution of wind tunnels to simulate the hypersonic flight conditions on ground. This has led to development of arc heated wind tunnels. In an arc-heater, direct observation of arc discharge between electrodes is practically impossible due to limited accessibility. So, experimental studies have been restricted to VI characteristics and thermal efficiencies. To analyse this complex mechanism of arc discharge, avoid run-time complexities and perform efficient test cycles computational results are necessary. The objective of present work is to simulate and understand the physics going on behind the Arc Heater and Nozzle by properly implementing the wall and electrode boundary conditions, accounting for convective cooling and radiative heating of wall. Open-source CFD software, OpenFOAM has been used as the simulation tool. Initially flow through a convergent divergent nozzle is studied with different wall boundary conditions, with and without shock in the divergent section. Then, we investigated the boundary conditions required to form a straight arc between two parallel Plates and a general methodology was developed. Also the effects of Lorentz force on the flow properties was studied. Lastly, we took the arc heater problem and analysed the arc behaviour in steady mode. A comparative study between the two and three dimensional modelling of arc discharge was done with and without Lorentz force. The effects of increasing voltage and mass flow rate were also investigated.

Table of Contents

Acknowledgements **ix**

Abstract **xi**

List of Figures **xv**

List of Tables **xix**

1 Introduction **1**

1.1 Motivation of the study 1

1.2 Overview of OpenFOAM 2

1.2.1 OpenFOAM general Structure 2

1.2.2 Typical file structure of OpenFOAM cases 3

1.3 Overview of Arc Heated Wind Tunnels 4

1.3.1 Electric Arc Heater / Plasmatron 4

1.4 Arc Physics 6

1.5 Magnetohydrodynamic model of ArcFoam 7

2 Numerical simulation of Flow through Nozzle **11**

2.1 Governing Equations 11

2.2 Conical Nozzle 12

2.2.1 Geometry and Mesh details for Case 1 12

2.2.2 Case Setup 13

2.2.3 Results 14

2.2.4 PostProcessing and Validation 15

2.2.5 Conclusion 17

2.3 Contoured CD Nozzle 17

2.3.1 Case 1 - Adiabatic walls with No Shock in Divergent section . . . 18

2.3.2 Case 2 - Adiabatic walls with Shocks in Divergent section 23

2.3.3 Case 3 - Isothermal walls with Shocks in Divergent section 30

3 Numerical simulation of Arc Heated Flow through Parallel Plates 37
3.1 Geometry and Mesh details 37
3.2 Case Setup 38
3.3 Results 42
3.3.1 Case 1 42
3.3.2 Case 2 44
3.4 Post Processing & Verification 47

4 2D simulation of Flow through Arc-Heater 51
4.1 Geometry and Mesh details 51
4.2 Case Setup 52
4.3 Results 55
4.3.1 Case 1 - with Argon 55
4.3.2 Case 2 - with Air 60
4.4 Post processing & Verification 65

5 3D simulation of Flow through Arc-Heater 69
5.1 Geometry and Mesh details 69
5.2 Case Setup 71
5.3 Results 74
5.4 Post Processing & Verification 80

6 Conclusion and Future Work 83
6.1 Conclusion 83
6.2 Future Work 84

List of Figures

1.1 OpenFOAM general structure [14] 3
1.2 Arc Heated Wind Tunnel [5] 4
1.3 Segmentation and assembly detail of NASA 1MW, multigas arc heater [6] 5
1.4 Zones near electron emitting cathode surface[3] 7

2.1 Conical Nozzle Geometry 13
2.2 Pressure contour for Inviscid Conical Nozzle 14
2.3 Temperature contour for Inviscid Conical Nozzle 15
2.4 Velocity contour for Inviscid Conical Nozzle 15
2.8 Contoured CD Nozzle geometry and Mesh for case 1 18
2.9 Pressure contour for CD Nozzle case 1 20
2.10 Temperature contour for CD Nozzle case 1 20
2.11 Velocity contour for CD Nozzle case 1 21
2.15 Mesh details for Case 2 24
2.16 Typical syntax for giving Wave tranmissive boundary condition 24
2.17 Pressure contour for CD Nozzle case 2 25
2.18 Temperature contour for CD Nozzle case 2 26
2.19 Velocity contour for CD Nozzle case 2 26
2.20 Pressure variation for CD Nozzle case 2 on top surface 27
2.21 Temperature variation for CD Nozzle case 2 on top surface 28
2.22 Grid Independence Results 29
2.23 Pressure contour for P_o = 3,08,885 Pa 32
2.24 Pressure contour for P_o = 10,38,350 Pa 33
2.25 Pressure contour for P_o = 17,49,200 Pa 33
2.26 Variation of convective Heat transfer coefficient along nozzle 35

3.1 Geometry for arcflow between parallel plates 37
3.2 Mesh Top View 38
3.3 Variation of Electrical conductivity of Argon gas with temperature 39

3.4 Variation of Net Emission coefficient of Argon gas with temperature . . . 39
3.5 Temperature distribution on Top and Bottom wall 40
3.6 Current density Magnitude (slice at z=0 mm) 42
3.8 Temperature Variation (slice at z=0mm) 43
3.9 Velocity Variation (slice at z=0mm) 43
3.10 Electric Potential(V) (slice at z=0 mm) 44
3.11 Current density Magnitude (slice at z=0 mm) 44
3.12 Velocity Variation (slice at z=0mm) 44
3.13 Temperature Variation (slice at z=0mm) 45
3.15 Lorentz Force Magnitude (N/m^3) (slice at z=0 mm) 45
3.16 Lorentz Force Magnitude (N/m^3) (slice at y=0 mm) 46
3.17 Magnetic Field Variation (slice at y=0mm) 46

4.1 2D view of modified Arc heater 51
4.2 Mesh near electrodes 52
4.3 Variation of Electrical conductivity of Air with temperature 53
4.4 Temperature Contour, 1.2 sec 55
4.5 Current Density near Electrodes, 1.5 sec 56
4.6 Electric Potential, 1.5 sec 56
4.7 Temperature Contour, 1.5 sec 56
4.8 Electrical Conductivity, 1.5 sec 57
4.9 Current Density, 2.5 sec 57
4.10 Temperature, 2.5 sec 57
4.11 Electric Potential, 2.5 sec 58
4.12 Electrical Conductivity, 2.5 sec 58
4.13 Velocity contour, 2.5 sec 58
4.14 Current Density contours for Argon gas at 2.5 sec; a) v=0.5m/sec, b) v=0.75m/sec, c) v=1m/sec, d) v=1.25m/sec, e) v=1.5m/sec, f) v=1.75m/sec 59
4.15 Temperature Contour, 1.2 sec 60
4.16 Current Density near Electrodes, 1.5 sec 60
4.17 Temperature Contour, 1.5 sec 61
4.18 Electric Potential, 1.5 sec 61
4.19 Electrical Conductivity, 1.5 sec 61
4.20 Current Density, 2.5 sec 62
4.21 Temperature, 2.5 sec 62
4.22 Electric Potential, 2.5 sec 62

4.23 Electrical Conductivity, 2.5 sec 63
4.24 Velocity Contour, 2.5 sec 63
4.25 Current Density contours for Air at 2.5 sec; a) v=0.5m/sec, b) v=0.75m/sec, c) v=1m/sec, d) v=1.25m/sec, e) v=1.5m/sec, f) v=1.75m/sec, g) v=2m/sec 64
4.26 Variation of Current and Enthalpy gain with mass flow rate for 100% Argon 66
4.27 Efficiency Vs Mass Flow rate for Argon case 67
4.28 Variation of Current and Enthalpy gain with mass flow rate for Air 68

5.1 2D view of the arc heater 69
5.2 Sliced view of mesh along XY plane 70
5.5 Temperature distribution on Main Anode 73
5.6 Temp. Contour,1.2 sec (sliced in xy plane) 75
5.7 Current Density Contour,1.5 sec (sliced in xy plane) 75
5.8 Temperature Contour,1.5 sec (sliced in xy plane) 76
5.9 Electrical Conductivity ,1.5 sec (clipped in xy plane) 76
5.10 Electric Potential,1.5 sec (clipped in xy plane) 76
5.11 Current Density, 2.3 sec (sliced in xy plane) 77
5.12 Temperature Contour, 2.3 sec (sliced in xy plane) 77
5.13 Electrical Conductivity Iso-Volume, 2.3 sec 78
5.15 Electrical Conductivity, clipped view, 2.3 sec 79
5.16 Electric Potential, clipped view, 2.3 sec 79
5.17 Velocity, 2.3 sec (sliced in xy plane) 80
5.18 VI Characteristics 82
5.19 Efficiency Vs Power Input 82

List of Tables

2.1 Inviscid Boundary conditions for Conical Nozzle 14
2.2 Boundary conditions for CD Nozzle case1 19
2.3 Boundary conditions for CD Nozzle case2 25
2.4 Boundary conditions for CD Nozzle case 3 31
2.5 Final yPlus Values for case3 34

3.1 Boundary Conditions for Pressure ,Temperature & Velocity 40
3.2 Boundary Conditions for J, Sie, Sigma and net Emission Coefficient . . . 41
3.3 Boundary Conditions for Magnetic Field and Magnetic Vector Potential . 42
3.4 Conservation of mass Flow rate (gm/s) 49
3.5 Conservation of electric Current(A) 50
3.6 A comparison between Case1 and Case2 50

4.1 Boundary Conditions for Pressure ,Temperature & Velocity 54
4.2 Boundary Conditions for J, Sie, Sigma and net Emission Coefficient . . . 54
4.3 Changes done in Boundary Conditions for shifting Arc 55
4.4 Mass Flow rate for Argon and Air with 0.5 m/sec inlet velocity 65
4.5 Current near Cathode and Anode for Argon & Air with 0.5 m/s inlet velocity 65
4.6 Effect of increasing inlet Velocity of Argon gas 66
4.7 Effect of increasing inlet Velocity of air 68

5.1 Boundary Conditions for Pressure ,Temperature & Velocity 72
5.2 Boundary Conditions for J, Sie, Sigma and net Emission Coefficient . . . 73
5.3 Changes done in Boundary Conditions for shifting Arc 74
5.4 Mass Flow rate at inlet and outlet 80
5.5 Current near cathode and anode 80
5.6 A comparison between axisymmetric and 3D results 81
5.7 Effect of changing Input Voltage with constant mass flow rate 81

Chapter 1

Introduction

1.1 Motivation of the study

During the 20th century the driving philosophy of main advancements in Aeronautics has been to fly more faster and higher. This thirst has led us to the onset of hyper-sonic era after the first manned supersonic flight in 1947 by Charles Yeager. Apart from a few test aircraft's and missiles, hypersonic flight is experienced during one particular application i.e during re-entry of any space vehicle to earth atmosphere or while entering into any other planet's atmosphere. To perform the flight tests of any supersonic vehicle on ground we have facilities like high speed wind tunnels or shock tube experimentation etc. but for hypersonic flight testing things are bit complex. Apart from speed, hypersonic flights are too complex to simulate because of aerodynamic heating, shock layer and other real gas effects. For example while re-entering earth's atmosphere for an altitude of 12,000 ft at 8 Mach stagnation pressure and temperature are of order 1000 psi and 5200 deg Rankine. However the vehicles can go well above Mach 20 at much more greater heights during its reentry. This aerodynamic heating requires special attention and so a Thermal Protection System (TPS) is needed. Typically a TPS consists of either ceramic tiles or ablative materials that can withstand high temperature and acts as a barrier to protect the vehicle during the searing heat of atmospheric re-entry.

For such experimentation on ground we need Hypersonic tunnels. A normal High speed tunnel can achieve High stagnation pressure of that order but prolonged operation at high temperature is not possible. This led to development of arc heated wind tunnels where an electric arc heater or Plasma-tron is used to heat the incoming air/gas which is then expanded in nozzle to obtain hypersonic flows.But an arc heated wind tunnels itself has issues like High Power Consumption (in Megawatts), Electrodes erosion, High maintenance etc. To overcome above complexities, avoid Experimental failures and perform

efficient test cycles, computational results are necessary. This became the major driving force behind arcFoam Development for simulating arc affected flows by Ved Mukherjee [4].The model was implemented using open-source CFD software OpenFoam.

The aim of this work is to efficiently and correctly simulate the flow within the Arc Heated wind Tunnels by implementing accurate boundary conditions for electrodes and to modify thermal boundary conditions of duct walls to account for radiative heating by arc & convective cooling by air and external coolant. Initially we have focused our attention on simulating the flow through two different nozzles. This is discussed in Chapter 2. Validation of Results was done analytically by comparing through quasi 1D theory as well as experimentally from results obtained by L.H Massier et. al[1]. In chapter 3, we investigated the flow between two parallel plates having circular electrodes at top and bottom. The effects of lorentz force were studied on the arc discharge. Lastly in Chapter 4 & 5, a comparative study between the two and three dimensional modelling of arc discharge inside the VSSC arc-heater was done.

1.2 Overview of OpenFOAM

OpenFOAM (Open Field Operation And Manipulation) is a free c++ based CFD simulation tool. It has large user base in areas of engineering and science in Commercial and academics world. Apart from basic CFD solvers and utilities it has widespread capabilities: like RANS, LES and DNS turbulent modelling, Buoyancy-driven flow solvers, Multiphase flow solvers, Particle-tracking solvers ,Solvers for combustion problems, Electromagnetics solvers etc.[14] Some major advantages of OpenFOAM are as below:

1. Complete freedom to customize and extend its existing functionality. Since the code is Open source and is visible to us, we can modify it to suit our requirements.

2. Most commercial software's cost heavily, and are compromise to national security when it comes to their application in DRDO, BARC, ISRO etc. So, we can not rely much on them

3. Over 80 standard solver in various field for simulation and nearly 180 utilities to help in pre and post processing tasks

1.2.1 OpenFOAM general Structure

OpenFOAM structure can be broadly divided into three parts as shown in Fig:1.1: Pre−processing, Solving & Post−processing where different utilities and solvers plays a key part. Main strength of openFOAM is that its Solvers and utilities can be modified by the user as per requirement.

Solvers are designed to solve a specific problem in computational continum mechanics. ex: sonicFoam, rhoCentralFoam for solving compressible flow and icoFoam, pimpleFoam for solving incompressible flow

Utilities are designed to perform preprocessing and post processing tasks with ease. ex: blockMesh utility for mesh generation, yPlus utility for y+ calculation

There are different predefined cases for each solvers in tutorials folder. The beauty is we can take any of these as per our requirement as our base case and do further modification as per our need

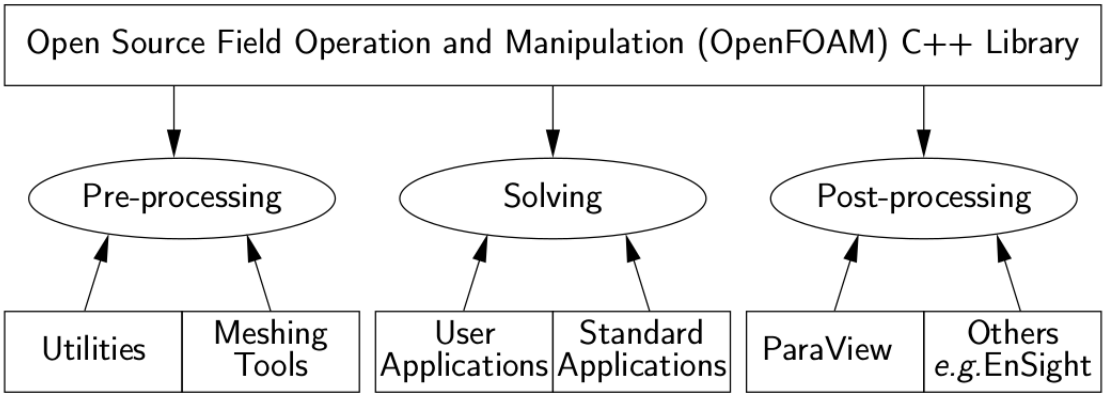


Figure 1.1: OpenFOAM general structure [14]

1.2.2 Typical file structure of OpenFOAM cases

A typical file structure of an OpenFOAM case consists of three different directories:

1.2.2.1 Time Directories (0 folder)

Initially time directories consist of **0** folder only but as the simulation is carried out different folders corresponding to different time intervals are created. A typical time directory consist of Pressure(P), Temperature(T), velocity(V) and other required folders which contain the boundary and initial conditions for these fields.

1.2.2.2 Constant Directory

It consist of a Polymesh folder and other constant files like transport properties, turbulent properties etc. All the data related to mesh generation like points, Boundary, Edges etc is stored here. Constant properties like viscosity, prandtl number, heat coefficient,tubulence models and other gas properties are defined in this folder.

1.2.2.3 System Directory

It contain three main files (other files may also be present). **Controldict** file for controlling the simulation parameters like start and end time, delta T, time step, write interval etc., **fvScheme** file for defining the choice of discretisation (like gradSchemes, interpolationSchemes, div Schemes, timeSchems, laplacianSchemes etc.) for solving a particular problem and **fvSolution** file to control equation solvers, tolerance , relaxation factors and other algorithm controls

1.3 Overview of Arc Heated Wind Tunnels

As briefly discussed earlier arc heated wind tunnels are required for hypersonic experimentation on ground. A typical tunnel consists of an electric Arc heater or Plasmatron which is used to heat the incoming gas/air and a nozzle which is then used to accelerate the flow to hypersonic speeds. Other main parts are identified in Fig:1.2. High Power supply is required and the current can be AC or DC. Any inert gas can be used along with air as working fluid, however argon is preferred above all. Arc Heaters can be of different types like segmented,linear coaxial,combined, multiarc and various other types. Once expanded in nozzle this high speed, high enthalpy flow is used to test various aspects of the sample model present in the test chamber. To decelerate the flow to low velocities a diffuser is used, from where flow is passed to heat exchanger which absorbs most of the heat and finally it to exhausted after passing through some pollutant removal system.

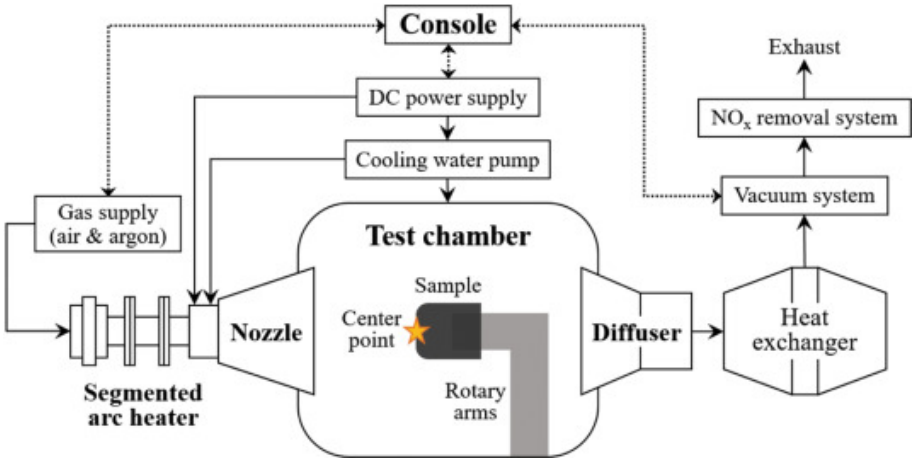


Figure 1.2: Arc Heated Wind Tunnel [5]

1.3.1 Electric Arc Heater / Plasmatron

This form the most important part of a hypersonic tunnel. Inside a Plasmatron heating of gas/air is accomplished by conductive, radiative and convective heat exchange be-

tween arc and gas/air. Arc heaters are relatively small compared to complete tunnel setup and process of gas heating is controlled by changing the input power supply which can vary from few KW to tens of MW. The main elements of a plasmatron are anode, cathode, a gas injection chamber and an Arc tunnel. A segmented type Arc heater is shown in Fig:1.3. which was developed by Nasa in late 1960's. The arc moves into the inner surface of chamber due to electrodynamic and aerodynamic forces. The main problem with such tunnels is high erosion rate of electrodes which depends on the current rate and is reduced by using argon as the working fluid.

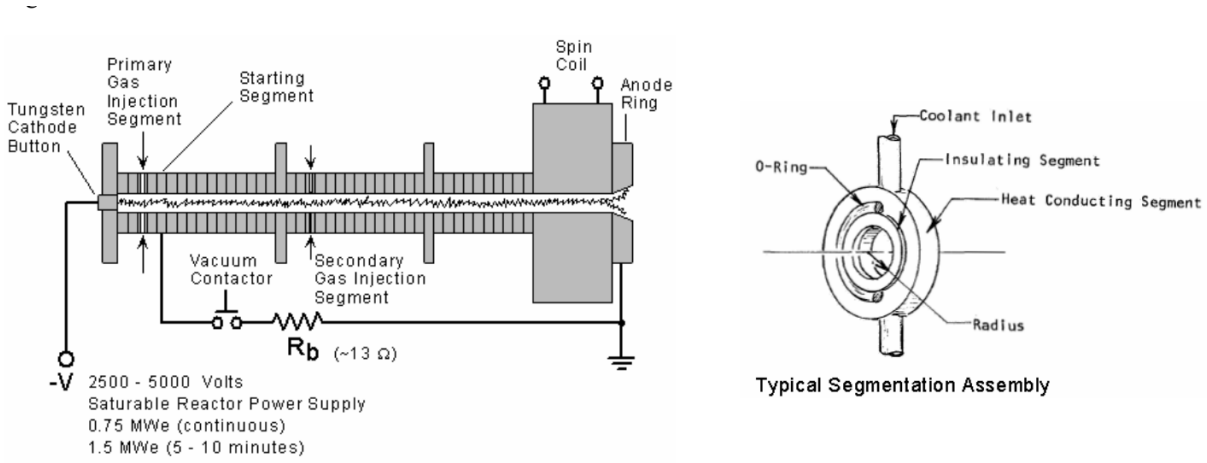


Figure 1.3: Segmentation and assembly detail of NASA 1MW, multigas arc heater [6]

1.3.1.1 VSSC Segmented Arc heater

Alike NASA ,Vikram Sarabhai Space Centre, Kerala has also developed its own segmented archeater based hypersonic tunnel. Here also the same traditional segmentation philosophy of having alternating conductor & insulator wafers stacking together to form the complete Arc tunnel is followed. Single Cathode of frustum shape ,made up of Tungsten is present at the central axis. The anode part is segmented and made of 10 mm copper rings separated by boron nitride insulators in stacked manner. Boron nitride acts as an electrical insulator between copper electrodes but does have a high thermal conductivity. The working gas (mixture of air and 20% argon) is injected tangentially (to introduce swirl in flow) through 5-6 different ports where the ports closer to electrodes have high percentage of argon to reduce their erosion rate. Cathode and Anode can go up to temperature of 2000 C & 1000 C respectively. So to maintain their temperature restrictions the walls and electrodes are cooled by water at atmospheric temperature.

First anode called as pilot anode is within 10 mm distant from the cathode. In order to start the device a modest flow of argon is introduced inside the heater and an arc is

formed between the Pilot anode as well as between the second anode and cathode. Then the power supply to first anode is switched off and to third anode is switch on. This will create an arc between cathode and 2,3 anode. This process is repeated to make the arc travel from the pilot anode to the main anode (last anode). The potentials of the segmented anode are in increasing order so that we have two arc at the same instant. In order to minimize local heat concentration over electrodes, the arc attachment points are rapidly moved either by applying external magnetic field or by using gas vortex. However because of high erosion rates electrodes need to be replaced after a few experiments. overall efficiency of the tunnel is 30-40% since most of the heat is lost either by radiation or through convective cooling of walls.

1.4 Arc Physics

An electric Arc is an electric discharge formed by applying a voltage between two electrodes. Following are the general characteristics of an arc:

1. Low Voltage(10-100V), High Current Density ($10^3 - 10^7$ A/cm^2) and power level greater than 1kW/cm between electrodes

2. High Temperature (5- 50k Kelvin) and moderate Ionization (of deg. 2-100%)

Above characteristics results in a very high temperature, which can melt or vaporize most materials. Basically an electric arc consists of three different regions: **Cathode layer**, **Thermal Plasma column**, and **Anode layer**. Cathode and Anode layers are further divided into sheath (space charge layer) and presheath (ionization Zone) as depicted in 1.4. Sheath zone is very thin, of order of one electron mean free path and is characterized by absence of thermal equilibrium . It is mostly composed of positive ions and so also named as space charge layer. Presheath zone is comparatively much thicker than sheath zone. It is composed of ions resulting from collisions between electrons and heavy particles, so also called as ionization zone. This zone is in partial thermal equilibrium and quasi neutral. It is followed by Arc Plasma column that is the main body of arc. This region is free of space charge, is in local thermal equilibrium(LTE), and both ionization and recombination occur here. LTE means that the temperature of electrons and heavy particales are of same order.

Similarly the Anode layer is subdivided into Sheath and Presheath zones [3].

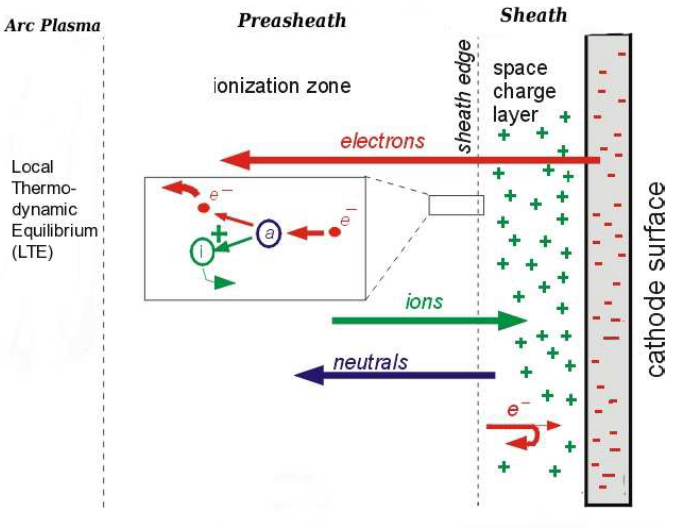


Figure 1.4: Zones near electron emitting cathode surface[3]

Due to flow of electric current between cathode and anode, a magnetic fields is induced. This magnetic flux density interact with electric field, giving rise to Lorentz force which acts as major source term in momentum equations governing Plasma flow.

1.5 Magnetohydrodynamic model of ArcFoam

To simulate the physics involved in archeater Ved Mukherjee [4] has created a solver arcFoam based on magnetohydrodynamic model proposed by Margarita Sass-Tisovskaya[3]. The base solver was taken as sonicFoam which is OpenFOAM inbuilt Transient solver for trans-sonic/supersonic, turbulent flow of a compressible gas. In arcFoam solver the navier stokes equation of hydrodynamics are coupled with the maxwell's equation of electromagnetics. Joule heating and radiation terms were added in energy equations along with Steady state Maxwell's equation. Other necessary modifications and alterations has been done to give necessary formulation. We reconstructed the solver and changed its work flow to calculate current based on electric potential. Also, the linear function of electrical conductivity and net radiation coefficient were changed to polynomial curve fits. The solver is also modified to take air as the working fluid.

In the present section governing equations as proposed by [3] are discussed. This mathematical model for arc column assumes that the plasma column is electrically neutral and in local thermodynamic equilibrium (LTE). LTE implies that temperature of electrons and heavy particles is nearly same. However it is not valid in the region very close to electrodes.

- Conservation of mass

$$\frac{\partial \rho}{\partial t} + \nabla(\rho \mathbf{V}) = 0 \tag{1.1}$$

where $\mathbf{V}$ is Velocity vector and ρ is density

- Conservation of momentum

$$\frac{\partial \rho \mathbf{V}}{\partial t} = -\nabla.[\rho \mathbf{V}\mathbf{V} + p] + \nabla.[\mu(\nabla \mathbf{V} + (\nabla \mathbf{V})^T) - \frac{2}{3}\mu(\nabla.\mathbf{V})I] + \mathbf{j} \times \mathbf{B} \tag{1.2}$$

where $\mathbf{j}$ is electric current density, $\mathbf{B}$ is magnetic field, p is static pressure μ is dynamic fluid viscosity and I is unit tensor

$\mathbf{j} \times \mathbf{B}$ is Lorentz force which arises due to electromagnetic interaction

- Enthalpy Conservation equation

$$\frac{\partial \rho h}{\partial t} + \nabla.(\rho \mathbf{V} h) - h\nabla.(\rho \mathbf{V}) - \nabla.(\alpha \nabla h) = \nabla.(\mathbf{V} p) - p\nabla.\mathbf{V} + \mathbf{j}.\mathbf{E} - 4\pi\epsilon_N \tag{1.3}$$

where h is enthalpy, E is electric field,ϵ_N is net emission coefficient.

$\mathbf{j}.\mathbf{E}$ is joule heating and $-4\pi\epsilon_N$ is the radiation loss term.

- Simplified equations for electromagnetic phenomenon within Plasma

Assuming Electro−neutrality, Quasi−steady electromagnetic phenomenon and magneto−hydrodynamic approach is valid[3], electromagnetic equations can be simplified as below:

$$\nabla.\mathbf{j} = 0, \qquad Conservation\ of\ electric\ charge \tag{1.4}$$

$$\nabla.\mathbf{B} = 0, \qquad Gauss's\ law\ (magnetic) \tag{1.5}$$

$$\nabla.\mathbf{E} = 0, \qquad Gauss's\ law\ (electric) \tag{1.6}$$

$$\nabla \times \mathbf{B} = \mu_o \mathbf{j}, \qquad Ampere's\ law \tag{1.7}$$

$$\nabla \times \mathbf{E} = 0, \qquad Faraday's\ Law \tag{1.8}$$

$$\mathbf{j} = \sigma E, \qquad Ohm's\ Law \tag{1.9}$$

where σ and μ_o are electrical conductivity and permeability of freee space.

Electric potential ϕ is defined as $E = -\nabla\phi$, substituting it in Ohm's law

$$\mathbf{j} = -\sigma \nabla \phi$$

Combining it with Charge Conservation equation

$$\nabla.(\sigma \nabla \phi) = 0 \tag{1.10}$$

Gauss's law (magnetic) can be satisfied if we define a vector potential **A** such that **B**=$\nabla \times \mathbf{A}$, putting this and **j** in Ampere's law yields:

$$\nabla \times \nabla \times \mathbf{A} = -\mu_o \sigma \nabla \phi$$
$$\nabla(\nabla.\mathbf{A}) - \nabla^2 \mathbf{A} = -\mu_o \sigma \nabla \phi$$
$$\nabla^2 \mathbf{A} = \sigma \mu_o \nabla \phi \tag{1.11}$$

Eq.1.10 & Eq.1.11 are desired Maxwell's equations in potential formulation which are solved along with the navier stoke equations.

Chapter 2

Numerical simulation of Flow through Nozzle

A hypersonic expansion nozzle is the second most important part of an arc heated wind tunnel responsible for accelerating the high enthalpy flow to hypersonic speeds at its outlet. Generally convergent divergent axisymmetric nozzles are preferred to obtain required mach number in the test section. We chose this problem because of two particular reasons; firstly we need to integrate nozzle with the arc Heater, secondly we want to study and get familiar with different wall boundary conditions within OpenFOAM. The study of different boundary conditions also holds an important place because we have to account for the radiative heating of arc heater walls in our Second Phase.

2.1 Governing Equations

- For an Isentropic flow: $\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}$
- Total conditions in a flow are related to static conditions via:

$$\left(\frac{T_0}{T}\right) = 1 + \frac{\gamma-1}{2}M^2 \tag{2.1}$$

$$\left(\frac{P_0}{p}\right) = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\gamma/(\gamma-1)} \tag{2.2}$$

$$\left(\frac{\rho_0}{\rho}\right) = \left(1 + \frac{\gamma-1}{2}M^2\right)^{1/(\gamma-1)} \tag{2.3}$$

- For steady adiabatic flow, energy equation can be written as: $h_o = h + \frac{V^2}{2} = constant$

- For a calorically perfect gas total temperature is constant across a normal shock wave while there is a loss in total pressure. Basic Normal Shock equations are:

$$M_2^2 = \frac{1 + [(\gamma - 1)/2]M_1^2}{\gamma M_1^2 - (\gamma - 1)/2} \tag{2.4}$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1) \tag{2.5}$$

- Quasi One Dimensional Flow

  If area variation is moderate, the components other than axial direction are often neglected. This assumption give rise to Quasi 1D flow where all flow field variables vary primarily along one direction. we have taken this assumption to obtain the analytical results. The isentropic flow of a calorically perfect gas through a CD nozzle is governed by following equation:

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma + 1} + \frac{\gamma - 1}{\gamma + 1}M^2\right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \tag{2.6}$$

  This relation is called as area Mach number relation. It tells us that Mach no. inside a duct is governed by the ratio of local duct area(A) to the sonic throat area(A^*). The closed form expression for mass flow through a choked nozzle is given by:

$$\dot{m} = \frac{p_0 A^*}{\sqrt{T_0}}\sqrt{\frac{\gamma}{R}\left(\frac{2}{\gamma + 1}\right)^{(\gamma+1)/(\gamma-1)}} \tag{2.7}$$

  Clearly mass flow rate depends on the Total Pressure, Temperature, Sonic Throat area and Gas properties.

2.2 Conical Nozzle

Conical shaped nozzle were used in old time because of their simple configuration and ease to fabrication. However they are longer, heavier & suffer from flow separation issues. To get things started we initially chose a conical nozzle because of its ease in construction and then moved to contoured convergent divergent nozzle in the later part.

2.2.1 Geometry and Mesh details for Case 1

A 3° wedge of a Conical CD nozzle is created using the inbuilt 'BlockMesh' utility of the OpenFoam. Geometry details are as below:
Length of Nozzle: 0.65m
Inlet and Outlet Radius: 0.2m & 0.175m
Throat Radius and Location: 0.1m & 0.25m

Convergent and Divergent Half angle: 21.8^o & 10.6^o

Expansion and Contraction area Ratio: 3.0625 & 4

Number of Cells:

20 along radial direction × (80+120) along symmetry axis, 1 layer is taken along Y axis.

3D elements: 3800 Hexahedra and 200 Prisms

Mesh Quality

Max aspect Ratio: 3.08

Max skewness :0.945

Max nonorthogonality: 21.31 (Avg. 9.33)

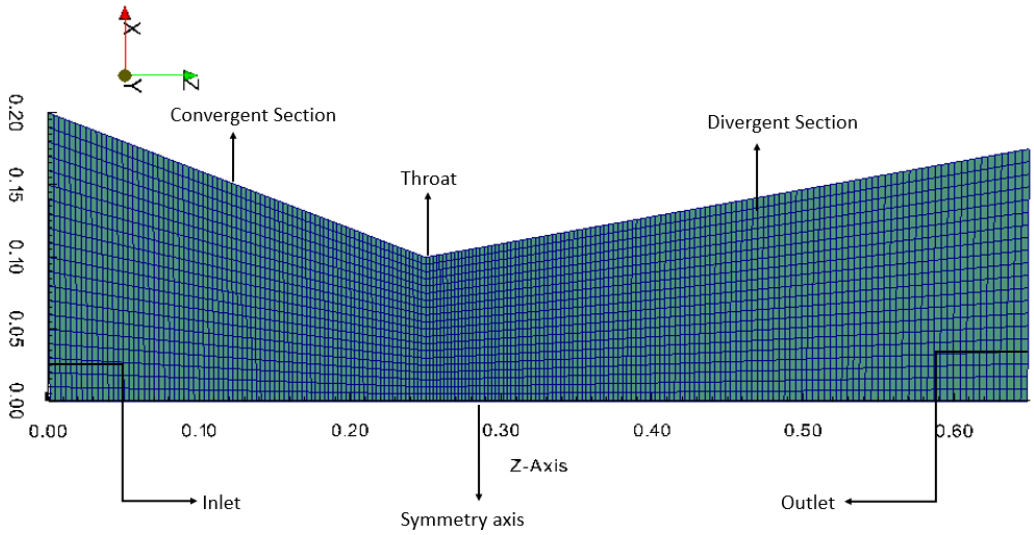


Figure 2.1: Conical Nozzle Geometry

2.2.2 Case Setup

Two different laminar test cases were setup with same boundary conditions; one having viscous effects and other Inviscid. Perfect gas is taken as the fluid with molecular weight 28.9 . Firstly we considered the inviscid flow using the inbuilt compressible solvers rhoCentralFoam and sonicFoam. Then we move on to Viscous simulations with dynamic viscosity coeffiecient $\mu = 18.37 \times 10^{-6}$ and Prandtl no. = 0.7; Boundary conditions as specified are summarised in table below:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	1,20,000	298	Zero gradient
Outlet	Zero gradient	Zero gradient	Zero gradient
Top_wall	Zero gradient	Zero gradient	slip
Front_face	wedge	wedge	wedge
Back_face	wedge	wedge	wedge
Internal fields	101325	298	(0,0,0)

Table 2.1: Inviscid Boundary conditions for Conical Nozzle

For adding viscosity, velocity boundary condition on Top wall is changed from slip to noSlip.

2.2.3 Results

In case of inviscid simulations Steady results were obtained within 0.06 sec while it took nearly 0.04 sec to get steady results for Viscous simulations. However simulations were run for 0.3 sec to avoid any later complications. Following Contour plots of Pressure, Velocity and Temperature for sonicFoam were generated using ParaView after obtaining steady results.

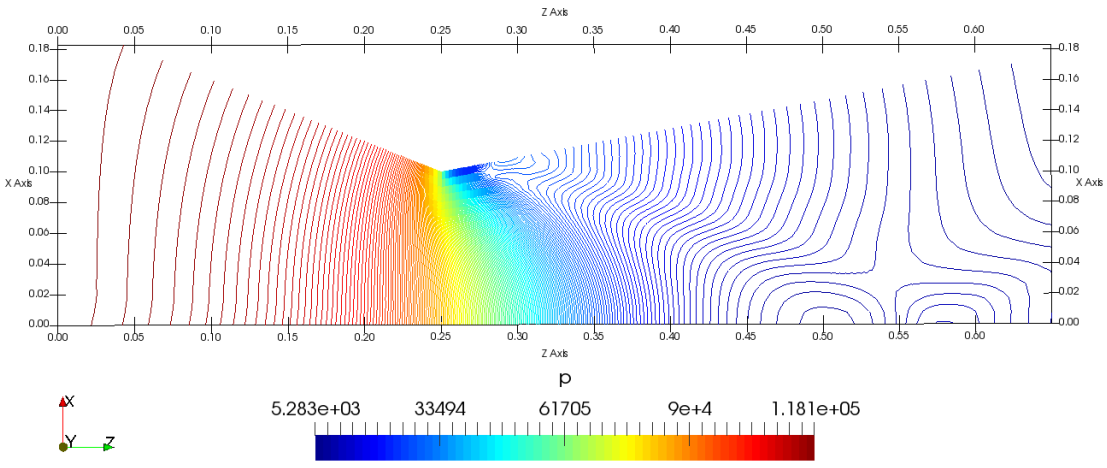


Figure 2.2: Pressure contour for Inviscid Conical Nozzle

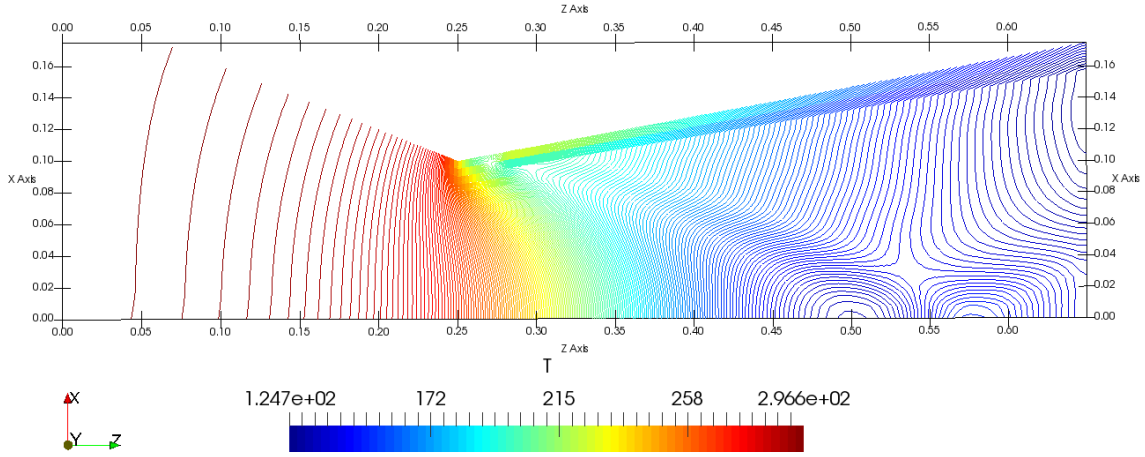


Figure 2.3: Temperature contour for Inviscid Conical Nozzle

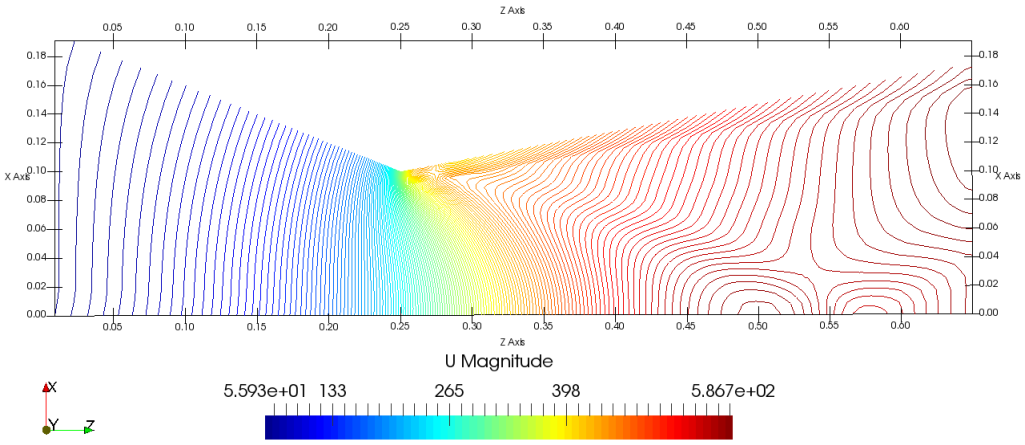
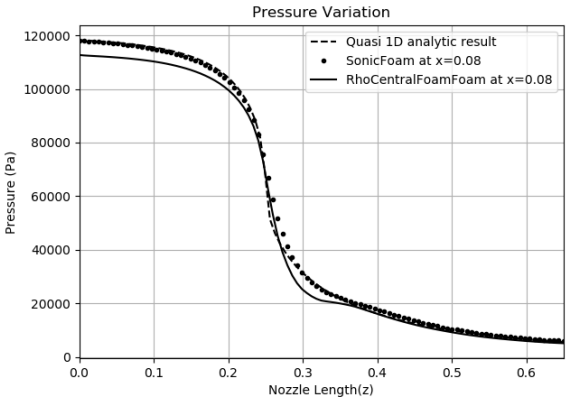


Figure 2.4: Velocity contour for Inviscid Conical Nozzle

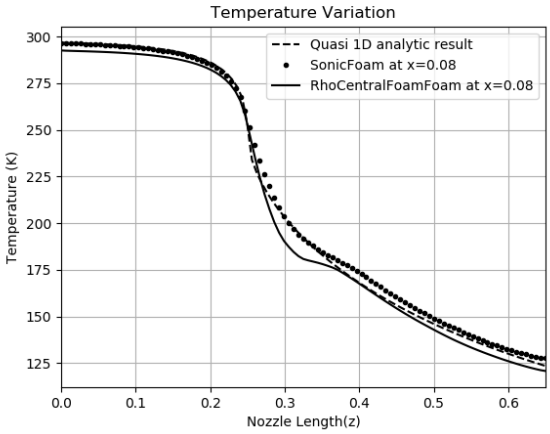
Similar results were observed for rhoCentralFoam too. We can observe some unphysical contours at z=0.55. This has its origination from the sharp divergence at Throat. This became the main reason for change of our geometry at later stage.

2.2.4 PostProcessing and Validation

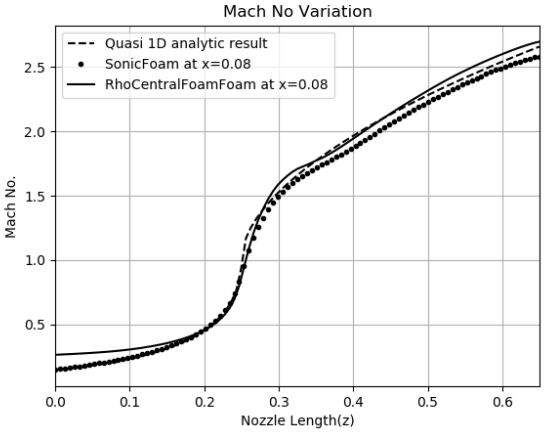
For post-processing purpose data along zaxis at x=0.08 was extracted as .csv file using "Export scene" feature of Paraview. It turns out the x=0.08 (which is slightly away from throat wall) behaves as the averaged position to compare with the analytical 1D results. so, a python script was written for Quasi One Dimensional Flow using The governing equations as discussed in section 3.1. Then analytical results were compared with rhoCentralFoam and SonicFoam in Python for case1 i.e inviscid flow.



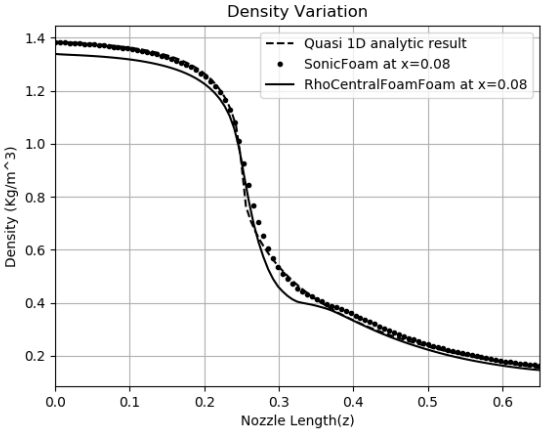
(a) Pressure Variation for Inviscid case



(b) Temperature Variation for Inviscid case



(a) Mach No. Variation for inviscid case



(b) Density Variation Inviscid case

2.2.4.1 Mass flow Conservation

Using sample file defined in the system directory and using postprocessing command **"postProcess -func sample"**, we extracted Pressure & Temperature cell values at inlet and outlet. These values are utilized to calculate mass flow rate at both these locations using following expression:

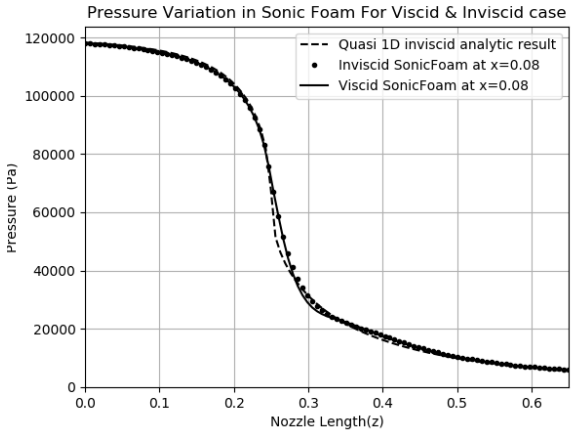
$$\dot{m} = \int_0^R 2\pi \times \rho(r, x) \times u(r, x).dr \tag{2.8}$$

The calculated mass flow rate at inlet and outlet comes out to be 8.794 kg/sec and 8.880 kg/sec respectively. Using Analytical expression for mass flow rate calculation through a choked nozzle as given by Eq.2.7, it comes out to be 8.827 kg/sec.

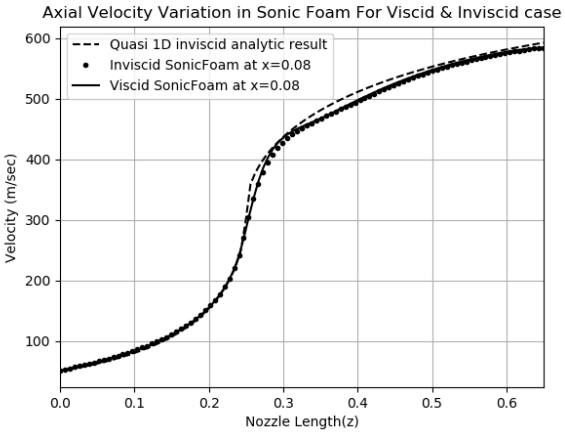
Percentage error in inlet mass flow rate calculation: 0.37%

Percentage error in outlet mass flow rate calculation: 0.6%

Similarly, for the case setup 2 i.e for viscous flow, the data was extracted at x=0.08 using "Export scene" feature of paraView. A python script was written to compare the results of sonicFoam for Viscous and Inviscid cases with the Quasi 1D flow. Following plots were obtained:



(a) Pressure Variation for Viscous & Inviscid case



(b) Velocity Variation for Viscous & Inviscid case

2.2.5 Conclusion

Both the schemes are in strong agreement with the the analytical results. At Subsonic speeds sonicFoam is giving same results as quasi 1D. However we cannot generalize which one is better since quasi Dimensional flow is only an assumption and does not consider the 3D effects however small they might be. Viscous effects are very limited to the surface of the Top wall only. Very minute changes can be seen at x=0.08 in pressure and Velocity plots. However their is a re circulation behind the throat. Also the mesh was not fine enough to capture the near wall effects quite perfectly. This all prompted us to change the geometry and work on a contoured convergent divergent Nozzle as discussed in next section.

2.3 Contoured CD Nozzle

As previously discussed conical nozzle suffers from serious re circulations issues and was not properly designed. So we changed our geometry for further analysis. We picked the nozzle dimensions from [1] where he discussed about the convective hat transfer through a CD nozzle. We have subdivided our work in three different cases:

Case 1: Adiabatic walls without any shock inside the divergent section.

Case 2: Adiabatic walls with Shock inside the divergent section.

Case 3: Isothermal walls with Shock inside the divergent section.

2.3.1 Case 1 - Adiabatic walls with No Shock in Divergent section

2.3.1.1 Geometry and Mesh details

A 3° wedge of a Contoured CD nozzle is created using **Gmsh** which is a open source 3D finite element mesh generator with builtin CAD engine. Geometry details are as below:
Length of Nozzle: 0.150495m
Inlet and Outlet Radius: 0.064m & 0.037296m
Throat Radius and Location: 0.0229m & 0.09075m
Convergent and Divergent Half angle: 30° & 15°
Expansion and Contraction area Ratio: 2.68 & 7.75
Radius of curvature along inlet: 0.036068m
Radius of curvature along throat: 0.04572m
Number of Cells:
50 along radial direction × (12+35+25+32) along symmetry axis × 1 layer along Z axis.
3D elements: 4800 Hexahedra and 100 Prisms

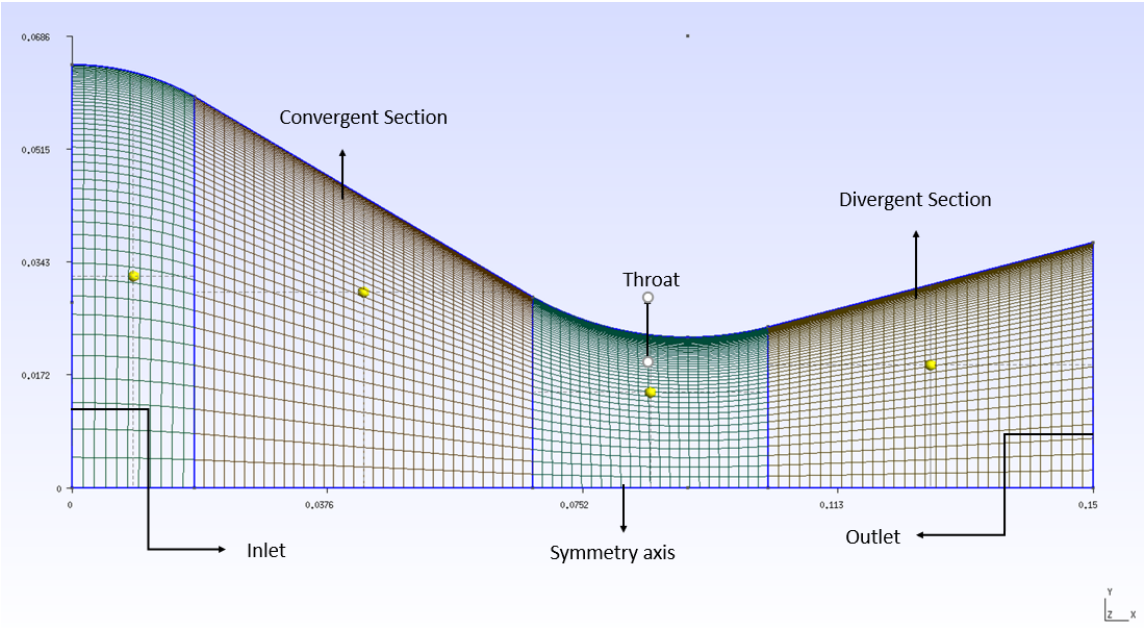


Figure 2.8: Contoured CD Nozzle geometry and Mesh for case 1

Mesh Quality:
Max aspect Ratio: 19.92
Max skewness: 0.668
Max nonorthogonality: 31.42 (Avg. 17.92)

2.3.1.2 Case Setup

The case is setup using sonicFoam; OpenFoam's inbuilt solver for transient, steady, compressible or incompressible flows. Perfect gas is taken as the fluid with molecular weight 28.9. Dynamic viscosity coeffiecient and Prandtl No. are kept to be of air at atmospheric conditions i.e $\mu = 18.37 \times 10^{-6}$ and Prandtl no. = 0.7. Boundary conditions for our case are summarised in table below:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	P_o =3,08,885	T_0 = 835	Zero gradient
Outlet	Zero gradient	Zero gradient	Zero gradient
Top_wall	Zero gradient	Zero gradient	noslip
Front_face	wedge	wedge	wedge
Back_face	wedge	wedge	wedge
Internal fields	101325	298	(0,0,0)

Table 2.2: Boundary conditions for CD Nozzle case1

The zero gradient boundary conditions in pressure will ensure that there is no shock inside the nozzle since the pressure at the second last cell is copied as the outlet pressure, so uniformity at the outlet pressure is maintained.

Spalart Allmaras model is used for capturing the turbulence effects. It is a low Reynolds number, one-equation model that solves a modelled transport equation for kinematic eddy turbulent viscosity. It was specifically designed for aerospace applications involving wall-bounded flows and has been shown to give good results for boundary layers subjected to adverse pressure gradients. In our case we considered 5% turbulent intensity with proper wall functions for different coefficients at Wall.

2.3.1.3 Results

Steady results were obtained within 0.01 sec with a time step of 10^{-6}. However simulations were run for 0.25 sec to avoid any later complications. To get a clear view of variation in Pressure, Temperature and Velocity following contour Plots are drawn in paraView:

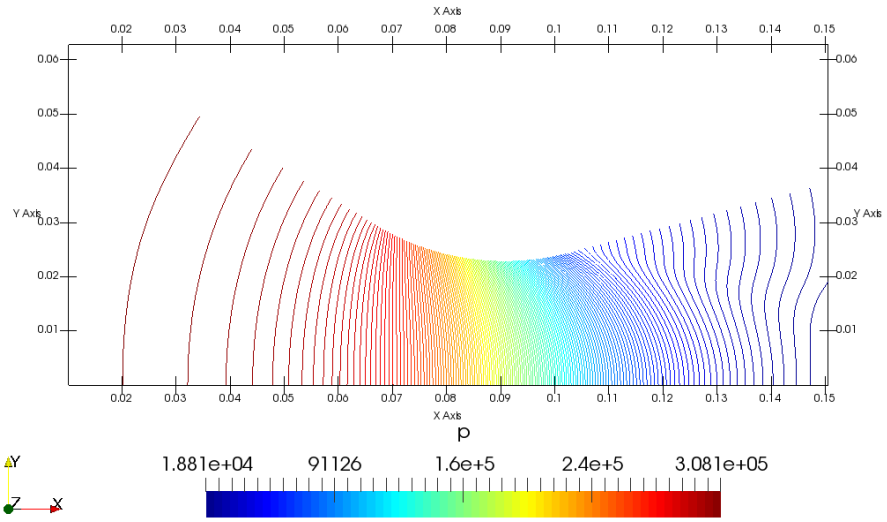


Figure 2.9: Pressure contour for CD Nozzle case 1

One can see from Pressure and Velocity contour that as expected there are no shocks formed inside the nozzle. There is uniform expansion and no sudden jumps are observed.

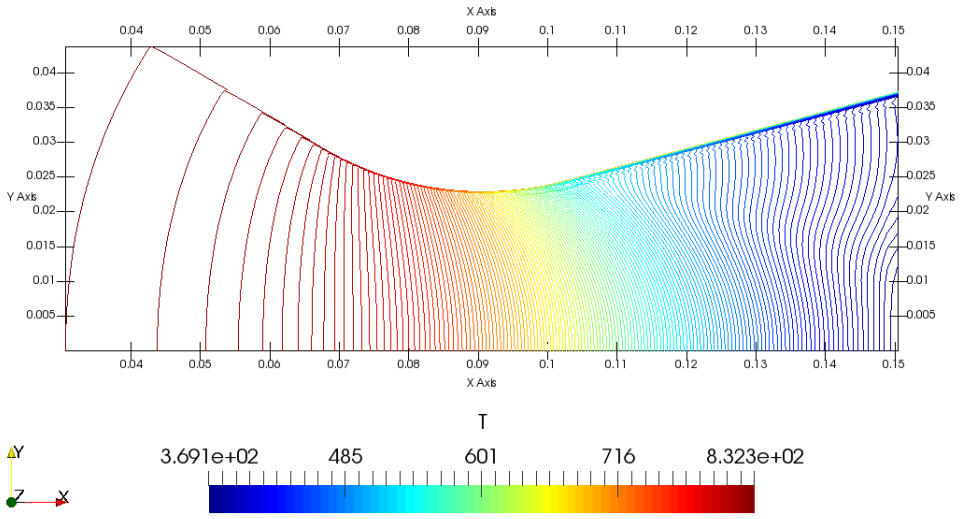


Figure 2.10: Temperature contour for CD Nozzle case 1

Temperature varies from 832K near inlet to 369 K at outlet without any sudden jump verifying nonexistence of any shock.

Velocity is accelerated from nearly 17m/sec at inlet to supersonic velocity of closer to 950m/sec at its outlet. We saw something wierd here, inspite of putting noSlip boundary condition at the wall we are getting some velocity at top wall. However when we plotted velocity over Top wall to see validity of noSlip Boundary condition. We observed it

is coming zero. So hereby we suspect it might be some paraview issues since it is not accounting for zero velocity in its contour plots.

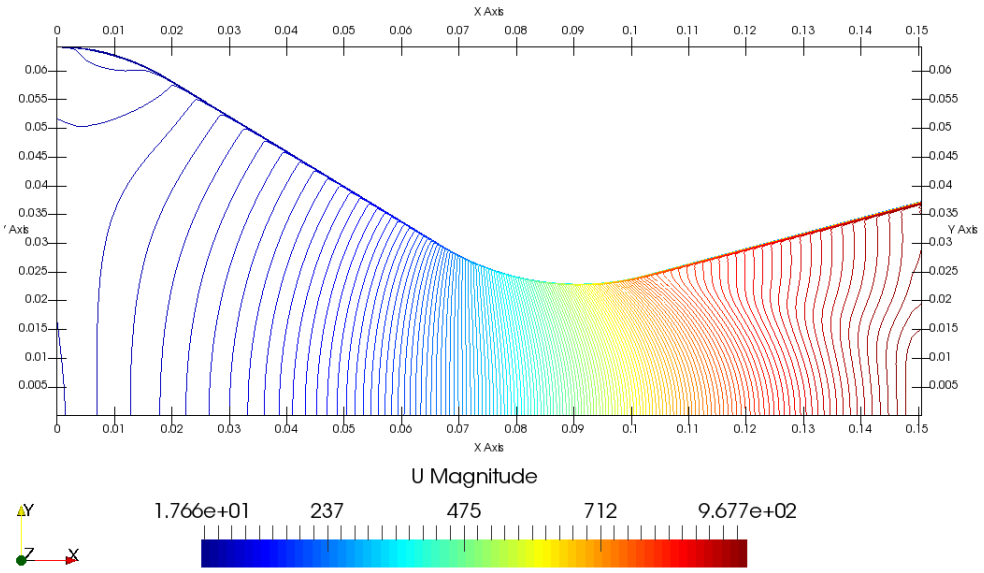
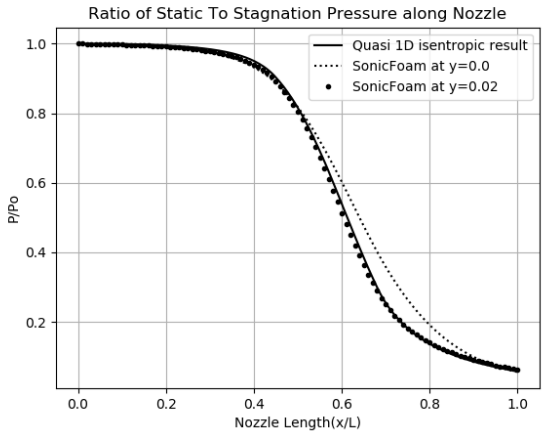


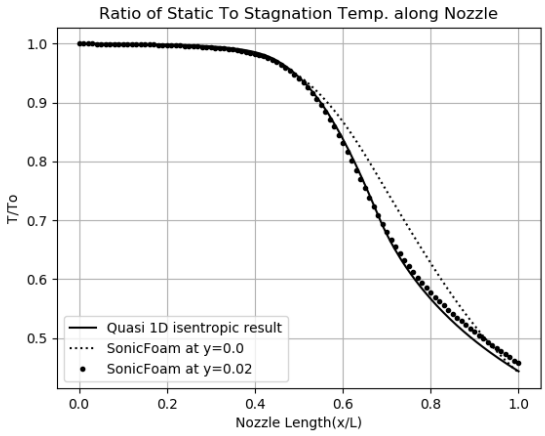
Figure 2.11: Velocity contour for CD Nozzle case 1

2.3.1.4 PostProcessing and Validation

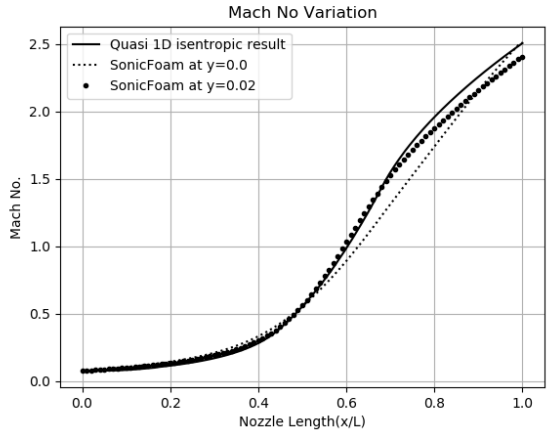
For post-processing purpose data along Xaxis at y=0 (symmetry line) and y=0.02 was extracted as .csv file using "Export scene" feature of Paraview. It turns out that y=0.02 (which is near to throat wall) behaves as the averaged position to compare with the analytical 1D results. So, a python script was written for Quasi One Dimensional Flow using the governing equations as discussed in section 3.1. Then analytical results were compared with the simulation results at y=0.0 and y=0.02 locations in Python.



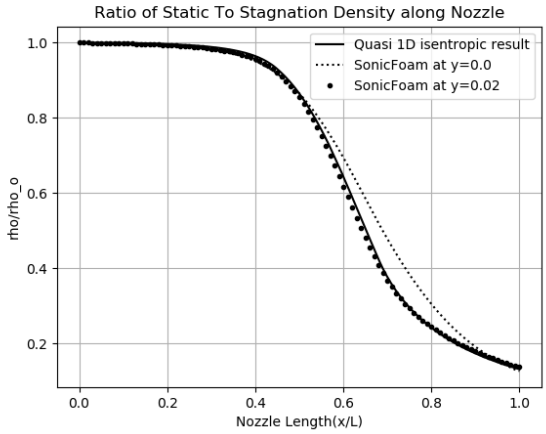
(a) Pressure Variation for Case 1



(b) Temperature Variation for Case 1



(a) Mach No. Variation for Case 1



(b) Density Variation for Case 1

Mass flow Conservation

As discussed in section 3.2.4, Using sample file defined in the system directory and using postprocessing command **"postProcess -func sample"**, we extracted Pressure & Temperature cell values at inlet and outlet. These values are utilized to calculate mass flow rate at both these locations using Eq.2.8.The calculated mass flow rate at inlet and outlet comes out to be 0.705 kg/sec and 0.7109 kg/sec respectively.

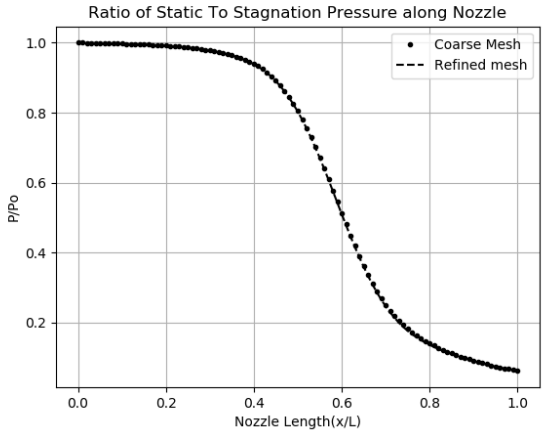
Analytical mass flow rate as given by Eq.2.7, for a choked nozzle comes out to be 0.711 kg/sec.

Percentage error in inlet mass flow rate calculation: 0.91%

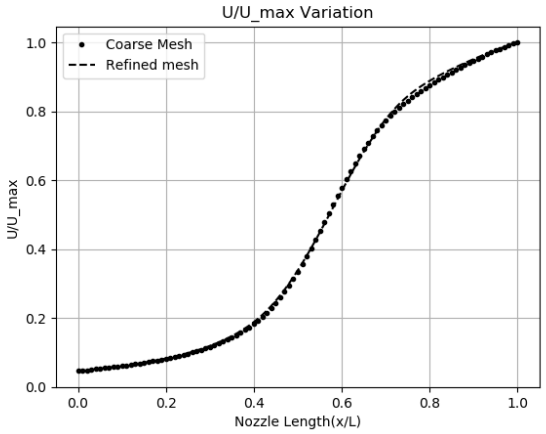
Percentage error in outlet mass flow rate calculation: 0.018%

Grid Independence Results

A more refined mesh was created to study the grid independence results. Number of cells are doubled and are increased to 99 along radial direction $\times$ (23+69+49+63) along symmetry axis $\times$ 1 layer along z axis. The graphs are plotted at y=0.02 since as we have seen in earlier graphs at y=0.02, the results are in close proximity to the quasi 1D isentropic results. Graphs below show that the results are independent of the grid size and there is no or very slight variation with doubling the number of cells



(a) Pressure Variation for Coarse and Fine mesh



(b) Velocity Variation for Coarse and Fine mesh

2.3.1.5 Conclusion

In this Section we have successfully simulated flow inside a Contoured CD nozzle and verified our results with Quasi 1D theory. Flow is accelerated to supersonic speed with Mach number 2.4 at its outlet. The outlet flow velocity, Pressure, Temperature and density comes out to be 942 m/sec, 19337 Pa, 382K and 0.176 kg/m^3 respectively. The inflow and outflow mass flow rate are within 1% error range from analytically calculated mass flow through choked nozzle. Finally a Grid independent study was done by doubling the number of cells in radial and axial direction. As can be seen from plots above graphs for Pressure and Velocity are coincident on each other for Coarser and Finer Mesh.

2.3.2 Case 2 - Adiabatic walls with Shocks in Divergent section

2.3.2.1 Mesh details

Shocks are highly sensitive in the diverging section, so in same geometry mesh is further refined so that it can capture the shock perfectly.

Number of cells:

90 along radial direction × (15+38+35+85) along symmetry axis × 1 layer along z axis. The bias factor or progression in gmsh has been changed to 0.98 along radial direction to have more number of cells near wall.

3D elements: 14872 Hexahedra and 169 Prisms

Mesh Quality:

Max aspect Ratio: 9.74

Max skewness: 0.668

Max nonorthogonality: 31.34 (Avg. 14.13)

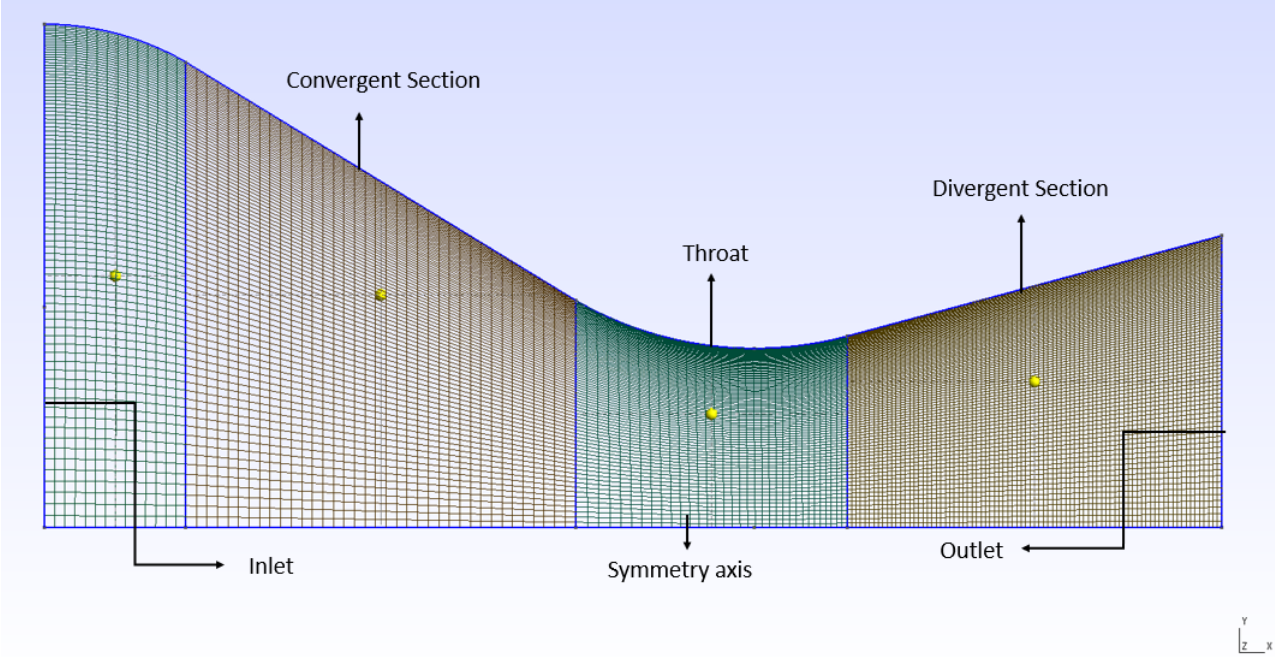


Figure 2.15: Mesh details for Case 2

2.3.2.2 Case Setup

Everything is kept same as was in Case 1; i.e perfect gas with molecular weight 28.9, Dynamic viscosity coeffiecient $\mu = 18.37 \times 10^{-6}$ and Prandtl No. 0.7. Only pressure boundary condition at outlet is changed. Wave transmissive boundary conditions are used for setting p to set atmospheric boundary condition of 101325 pascal at outlet. The results are highly affected by the LInf value that is a measure of how far away the far-field condition is applied. At Linf value of 0.0001 steady results were obtained. However if some lesser value is given than reflections inside the nozzle were observed and the solutions could not become steady. This value is obtained after certain hit and trial for our case and might vary from case to case.

```
outlet
{
    type            waveTransmissive;
    field           p;                  //the name of the field that we are working on
    value           uniform 101325;     //important for correct I/O
    gamma           1.4;                //the ratio of specific heats
    phi             phi;                //volumetric flux field
    rho             rho;                //the name of the density field
    psi             thermo:psi;         //field that is the deriv. of density with respect to pressure
    lInf            0.0001;             //a measure of how far away the far-field condition should be
    fieldInf        101325;             //the far-field value to be applied to p
}
```

Figure 2.16: Typical syntax for giving Wave tranmissive boundary condition

Same as earlier, Spalart Allmaras model is used for capturing the turbulence effects which is a low Reynolds number, one-equation model that solves a modelled transport equation for kinematic eddy turbulent viscosity. In our case we considered 5% turbulent intensity with proper wall functions for different coefficients at Wall. Boundary conditions as given are summarised in table below:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	P_o =3,08,885	T_0 = 835	Zero gradient
Outlet	WaveTransmissive uniform 101325	Zero gradient	Zero gradient
Top_wall	Zero gradient	Zero gradient	noslip
Front_face	wedge	wedge	wedge
Back_face	wedge	wedge	wedge
Internal fields	101325	298	(0,0,0)

Table 2.3: Boundary conditions for CD Nozzle case2

2.3.2.3 *Results*

Steady results were obtained within 0.04 sec with a time step of 5×10^{-7}. However simulations were run for 0.2 sec to avoid any later complications. To get a clear view of variation in Pressure, Temperature and Velocity following contour Plots are drawn in paraView:

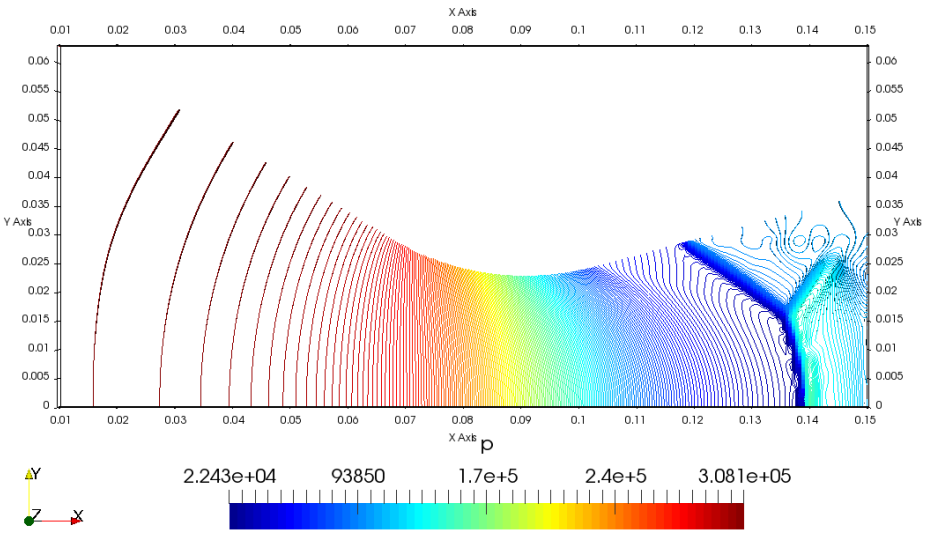


Figure 2.17: Pressure contour for CD Nozzle case 2

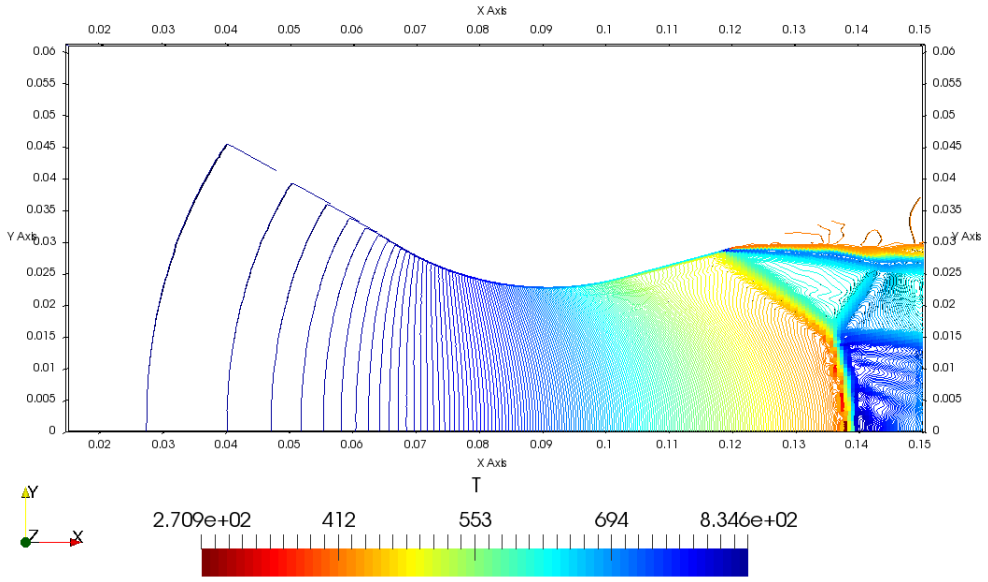


Figure 2.18: Temperature contour for CD Nozzle case 2

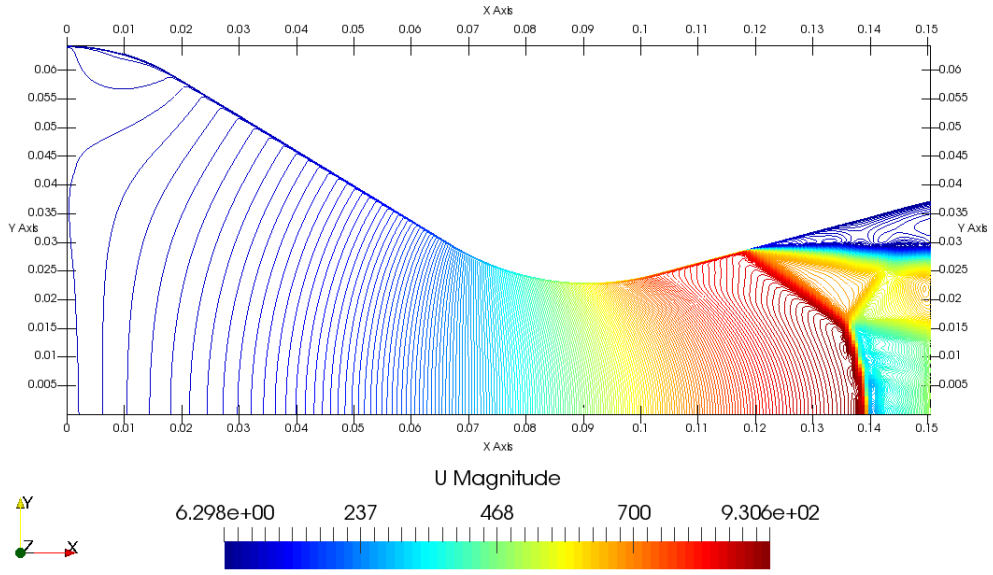


Figure 2.19: Velocity contour for CD Nozzle case 2

One can infer from the contour plots that as expected there is shock formed inside the divergent section of the nozzle at nearly x=0.14. A complex phenomenon of interaction of shock wave with boundary layer can be seen leading to Mach reflection. Shock impinges on the top wall at x= 0.12. Because of the large pressure rise across the shock which acts as severe adverse pressure gradient imposed on boundary layer, flow is separated and can be clearly observed in the velocity contour plot. One slip line originates as a result of this interaction. Slip lines (or slip planes) can be thought of lines (or planes) across which pressures are constant but magnitude of velocity is different (direction is same i.e parallel

to slip line). All other properties like density, temperature, entropy are different across slip lines. To keep the velocity of the flow parallel to the nozzle surface a Mach reflection is formed leading to formation of second slip plane. However because of the divergence in area again leading to adverse pressure gradient, the flow never reattach with the surface and is passed out of the nozzle as it is. Clearly in this case flow through the nozzle is not properly accelerated and the back pressure is needed to be lowered to get a shock free flow.

2.3.2.4 PostProcessing and Validation

For post-processing purpose data along the Top_wall was extracted as .csv file using "Export scene" feature of ParaView. For getting data along a curved line on the top surface, filters ⇒ Alphabetical ⇒ Plot data, function is used. Extracted data is than plotted and compared with experimental results using a python script. To extract data from [1] WebPlotDigitizer, an online tool is used.

In experiment done by L.H Back et al.[1] the static ports are present at the outer wall of the nozzle, so in this way they are not accounting for the variation in static pressure along the radial direction. In figure below we can see that shock location is captured exactly with only a little variation with experimental results

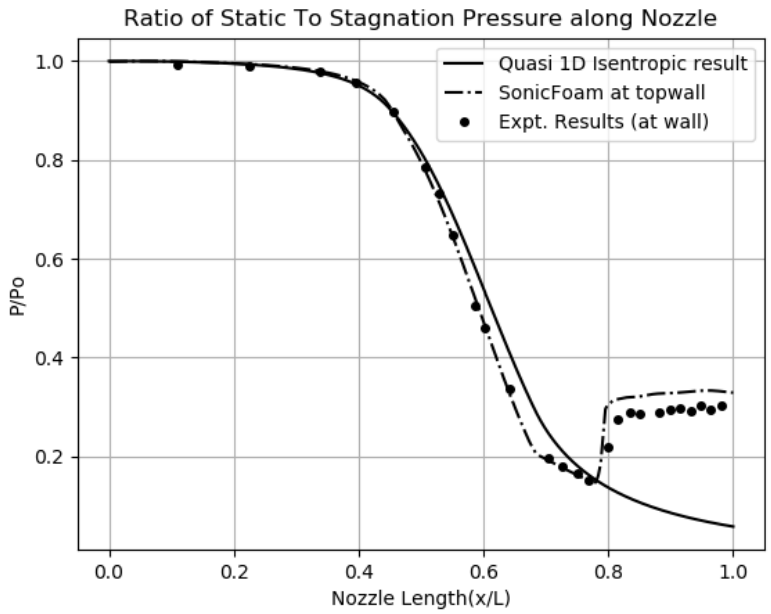


Figure 2.20: Pressure variation for CD Nozzle case 2 on top surface

Temperature on the top surface rises suddenly after shock but due to flow separation it drops down discontinuously. Fig:2.21 clearly shows this trend.

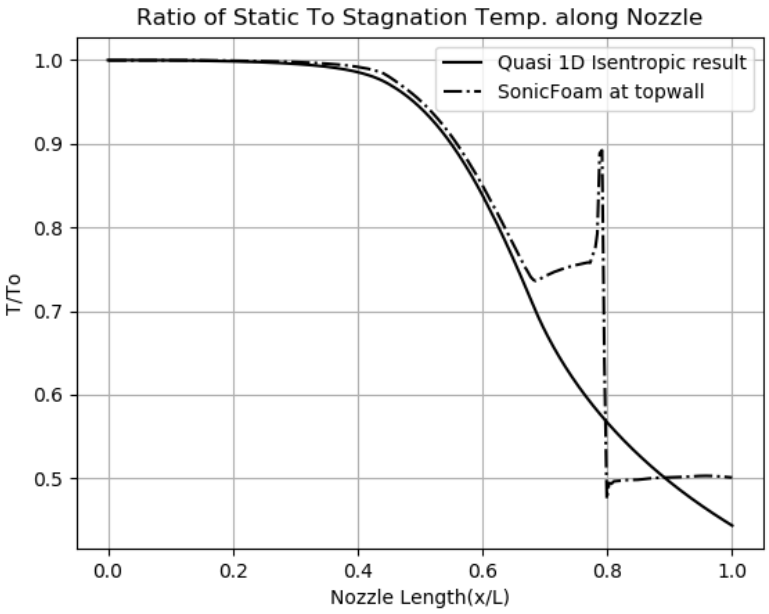


Figure 2.21: Temperature variation for CD Nozzle case 2 on top surface

Algorithm for capturing shock location inside the divergent part

Given total pressure at inlet, static pressure at outlet and the area variation of nozzle one can locate the shock position using the algorithm below:

- Guess Shock location, with known A_1/A_1^*, M_1 i.e mach no. before shock is known.
- Calculate after shock mach number using Eq.3.4. Total pressure before shock is known (same as total inlet pressure), calculate static pressure before shock i.e p_1 using Eq. 3.2 then Using expression 3.5 static pressure after shock i.e p_2 is calculated.
- Since $A_1 = A_2$, calculate sonic throat area (A_2^*) using expression 3.6. Once (A_2^*) is known calculate mach no at exit. Using after shock mach no M_2 and static pressure calculate P_{o2}.
- From P_{o2} and M_e calculate exit pressure of nozzle. Compare it with back pressure provided. If not equal go to first step.

Grid Independence Results

Shocks are highly dependent on the grid size. To study the grid independence, Four more different meshes were created i.e Two Coarser and two finer ones. In all the meshes 1 layer is taken along z axis. Initially considered Mesh is termed as Mesh 2. Number of cells in five meshes are as below:

Mesh 0: 70 along radial direction × (12+35+30+60) along symmetry axis

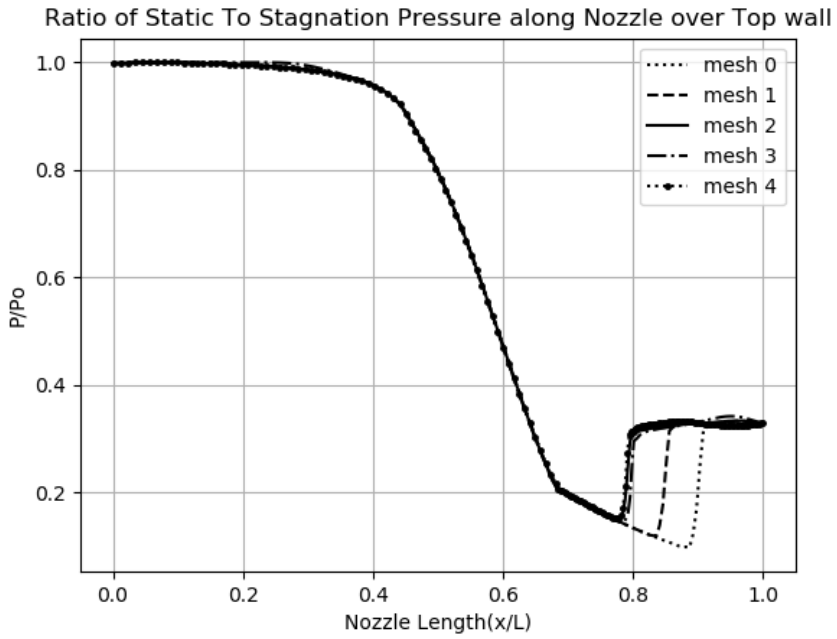


Figure 2.22: Grid Independence Results

Mesh 1: 80 along radial direction × (15+38+30+65) along symmetry axis
Mesh 2: 90 along radial direction ×(15+38+35+85) along symmetry axis
Mesh 3: 90 along radial direction ×(20+52+47+113) along symmetry axis
Mesh 4: 90 along radial direction ×(12+35+30+100) along symmetry axis

Above Plot 2.22 shows that the results are independent of the grid size and there is very slight variation with any further change in mesh. Here we observed that increasing the number of cells greater than 90 in radial direction cause the mesh to capture transient fluctuation's resulting in non steady oscillations in residuals.

2.3.2.5 Conclusion

In this section we have successfully simulated flow inside a CD nozzle having shock in the divergent part and validated our results experimentally[1]. Because of Shock and boundary layer interaction there is reverse flow and boundary layer separation at the rear part of nozzle. Shock location is very sensitive to Mesh size, however we obtained the grid independence by fixing the no. of cells to be 90 along radial direction.

2.3.3 Case 3 - Isothermal walls with Shocks in Divergent section

In this section we will focus on the Heat transfer within isothermal walls and calculate convective heat transfer coefficient coefficient(h). Previously considered adiabatic wall is made isothermal with approximating wall Temperature as 0.5To.

2.3.3.1 Mesh details

Same Mesh as earlier is used i.e. 90 along radial direction × (15+38+35+85) along symmetry axis × 1 layer along z axis. The bias factor is increased from 0.98 to 0.96 so as to maintain a y-Plus of less than 5.
3D elements: 14872 Hexahedra and 169 Prisms
Mesh Quality:
Max aspect Ratio: 29.97
Max skewness: 0.669
Max nonorthogonality: 31.996 (Avg. 15.955)

2.3.3.2 Case Setup

Keeping everything same as was in Case 2, Except Total Pressure, Temperature and a variable value of Dynamic viscosity coefficient μ. The change in viscosity of air for our Temperature ranges is small but to keep things in order this variation is captured using Sutherland's Formulation. Sutherland's viscosity law can be specified using two or three coefficients. However OpenFOAM considers two coefficient model only.

$$\mu = \frac{A_s \sqrt{T}}{1 + T_s/T}$$

where $A_s = 1.458 \times 10^{-6}$ and $T_s = 110.4k$
Three different test case with different Total pressure and temperature are considered. Wall temperature is assumed to be 0.5 × Total temperature given at the inlet. It is just an approximation to study the effect of change in Total Pressure on wall heat transfer coefficient.
Boundary conditions as given are summarized in table below:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	1) P_o =3,08,885 2) P_o =10,38,350 3) P_o =17,49,200	1) T_0 = 835 2) T_0 = 824 3) T_0 = 572	Zero gradient
Outlet	WaveTransmissive Uniform 101325	Zero gradient	Zero gradient
Top_wall	Zero gradient	0.5 T_0	noslip
Front_face	wedge	wedge	wedge
Back_face	wedge	wedge	wedge
Internal fields	101325	298	(0,0,0)

Table 2.4: Boundary conditions for CD Nozzle case 3

Two different turbulent models were tested i.e. Spalart Allmaras and k-Epsilon. The problem with Spalart Allmaras is that the yplus utility to calculate y+ works fine for the incompressible flow but when compressible flow is considered it appears that it give 0 y+ values. Also the convective heat transfer values seem to come a bit low with spalart-allmarus. So after initial works we need to switch to k Epsilon model which is more robust and seems to work quite satisfactory. K−Epsilon is a two equation model which gives a general description of turbulence by means of two transport equations (PDEs).

To introduce K-Epsilon we have assumed a reference value of U_ref i.e initial flow velocity to be 40m/sec. This value is assumed taking in mind the results of earlier simulation. As a thumb rule following formulas were used to calculate turbulent intensity, turbulent kinetic energy(k) and turbulent dissipation(ϵ):

$$\begin{aligned} Turbulent\ Intensity\ (T_{int}) &= 0.16 R_e^{\frac{-1}{8}} \\ Turbulent\ Kinetic\ Energy\ (k) &= \frac{3}{2}(U_ref \times T_{int})^2 \\ Turbulent\ Dissipation(\epsilon) &= 0.09^{3/4} \times \frac{k^{3/2}}{0.07L} \end{aligned}$$

where L is characteristic inlet scale i.e 0.064m. Example: For case3 i.e $P_o = 17,49,200$ and $T_o = 572$, taking 40m/sec as U_ref, Approximate Reynolds number is 8,79,700. This gives a turbulent intensity of 2.8 percent. So we initialize Turbulent kinetic energy K to be 2 m^2/sec^{-2} and turbulent dissipation ϵ equals to 103 m^2/sec^{-3} in our case setup.

We have ensured that our yPlus should remain within 5 initially but due to change in flow field when solution progress it can exceed beyond 5 so we have used proper wall function for different coefficients. yPlus is a dimensionless wall distance given by:

$$y^+ = \frac{yU_T}{\nu}$$

where y is absolute distance from wall, ν is kinematic viscosity. U_T is friction velocity calculated as $U_T = \sqrt{\frac{\tau_w}{\rho}}$; τ_w is shear stress at wall given by:

$$\tau_w = \frac{1}{2} C_f \rho u^2$$

where C_f is skin friction approximated as $0.058 R_e^{-0.2}$ for external flows and $0.079 R_e^{-0.25}$ for internal flows.

2.3.3.3 Results

For three different boundary conditions of Total pressure and Total Temperature, following contour plots are drwan using ParaView:

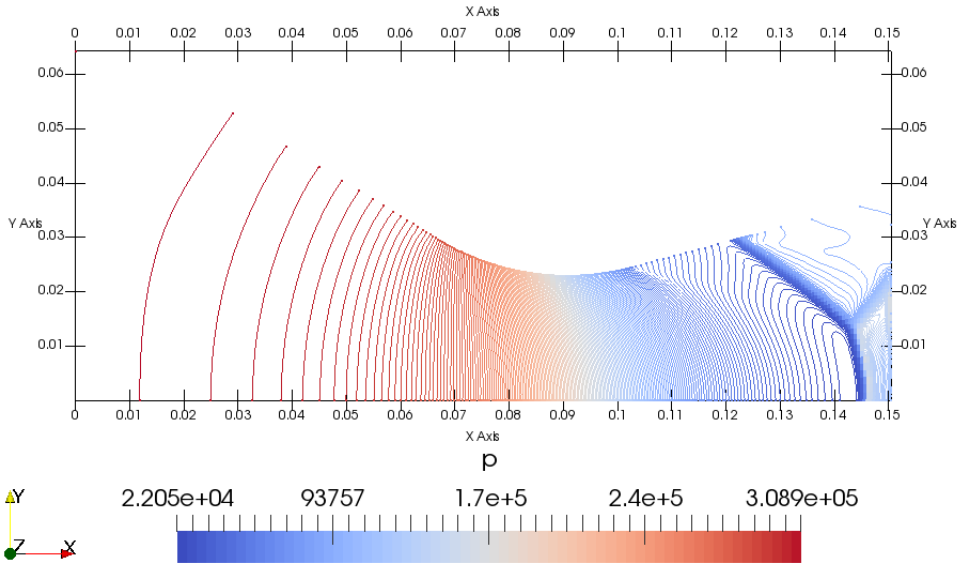


Figure 2.23: Pressure contour for P_o = 3,08,885 Pa

As seen in adiabatic wall case also total pressure condition of P_o = 3,08,885 Pa is creating shock in the divergent section.

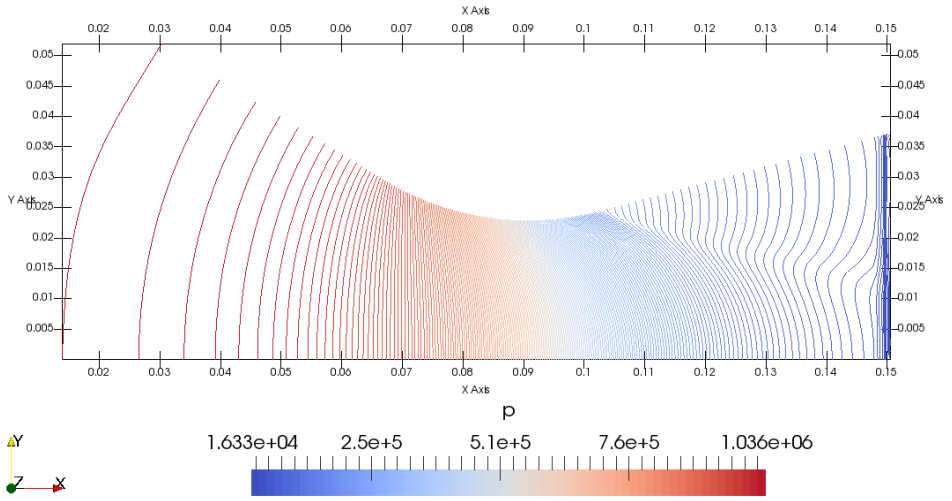


Figure 2.24: Pressure contour for P_o = 10,38,350 Pa

For P_o = 10,38,350 Pa and 17,49,200 Pa the flow is uniformly accelerated in the divergent section with the exit mach number of 2.6 and 2.74 respectively.

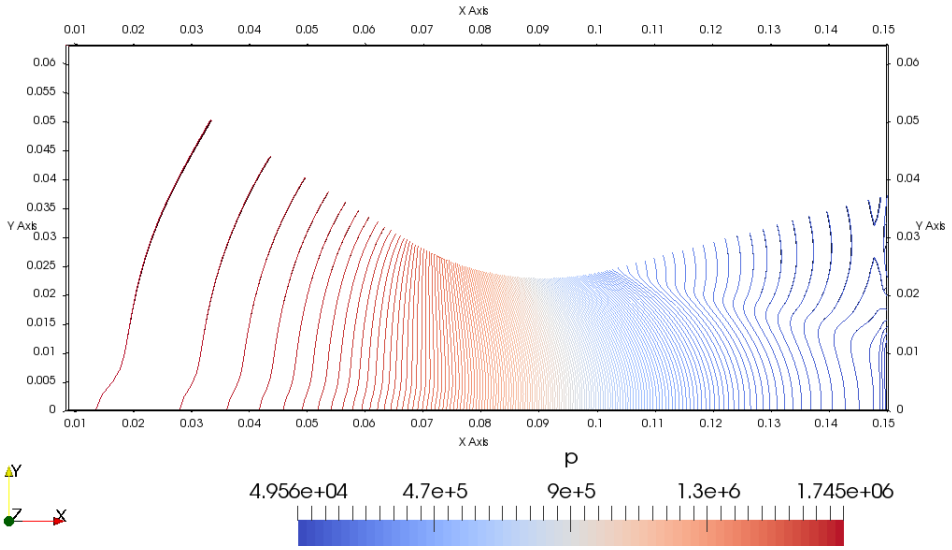


Figure 2.25: Pressure contour for P_o = 17,49,200 Pa

2.3.3.4 PostProcessing and Validation

To calculate the yplus value, inbuilt utility of OpenFoam is used. For using it following command need to be given in terminal:

sonicFoam -postProcess -func yPlus

Similarly, to calculate the heat flux along the wall **wallHeatFlux** command needs to be given in the terminal after solutions are achieved.

Total Pressure(P_o)	min	max	average
3,08,885 Pa	2.40	22.31	9.36
10,38,350 Pa	1.8	29.4	12.88
17,49,200 Pa	2.93	38.7	18.7

Table 2.5: Final yPlus Values for case3

Initially we kept all the yPlus values below 2 but with change in flow properties the calculated yPLus values had risen significantly. We tried reducing it further by either increasing number of divisions along y axis or by decreasing the bias factor. But in both the circumstances the delta T required is less than 10^{-9} to satisfy the CFL number requirement which is not possible to do with the available computation resources.

2.3.3.5 Convection Heat Transfer coefficient calculation procedure

- After the residuals come down to the set limits and a steady stable solution is obtained, openFoam post process utility **wallHeatFlux** is used to calculate the heat flux at wall.

- T_adiabatic is calculated using temperature (T_{ref}) and M_{ref} at the symmetry line (i.e y=0.0) using below expression:

$$T_{adiabatic} = T_{ref}(1 + r \times \frac{\gamma - 1}{2} \times M_{ref}^2)$$

- Wall temperature is assumed to be half of the total temperature provided at inlet and recovery factor(r) of 0.89 is used to calculate the connective coefficient. Gamma value is taken to be 1.345.

- Finally convective coefficient is calculated using below expression:

$$h(x) = \frac{-\frac{KdT}{dy}}{T_{adiabatic}(x) - T_{wall}(x)}$$

  Numerator value is provided by the openfoam wall heat flux utility and the denominator part was calculated using excel.

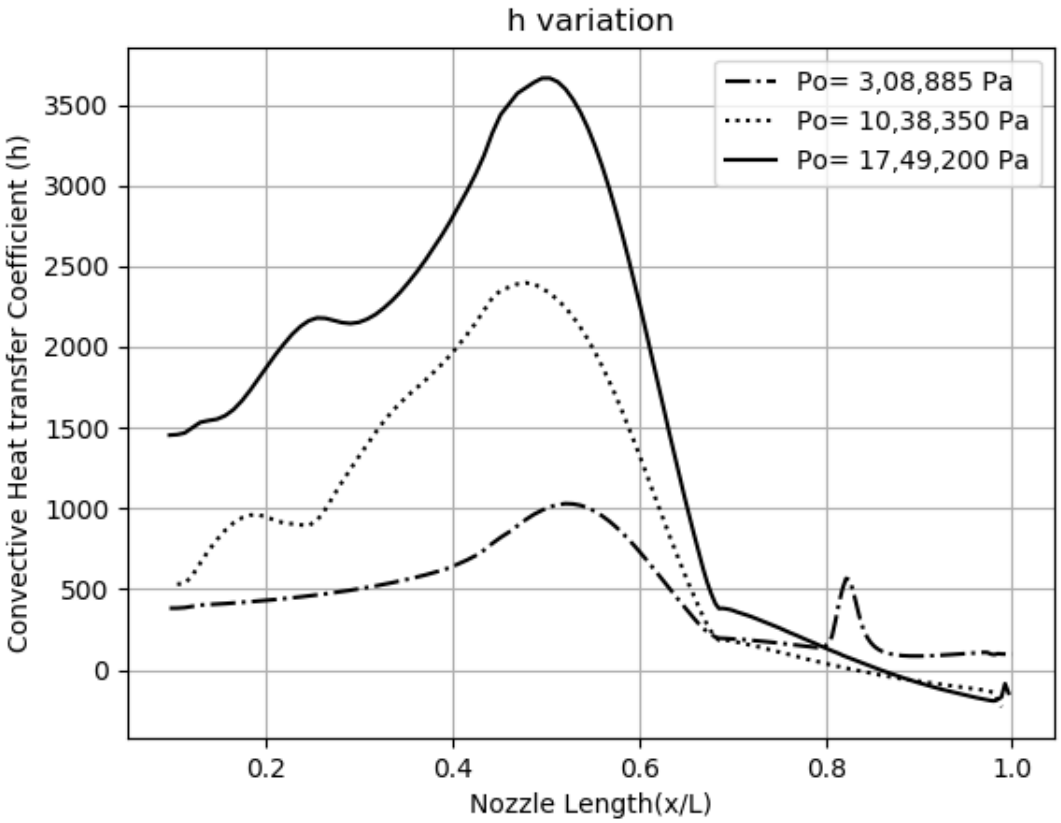


Figure 2.26: Variation of convective Heat transfer coefficient along nozzle

Heat transfer coefficient variation for the three cases discussed above is plotted in Fig: 2.26. We can see that the maximum value of h is achieved just near the nozzle throat and this value increases as we increase the total pressure. The variation of changing the total temperature is very little and doesnt have much pronounced effects. The sudden bump at 0.83 x/L in case1 is arising because of shock formation in divergent section.

We are unable to validate our results with the experiment done by L.H Back Et al.[2] because of unavailability of sufficient information regarding wall temperature and turbulence effects. However as it should be heat transfer coefficient(h) is maximum just upstream of the throat where the Mass flux is maximum. Increase in Mass flux can be attributed to increase in Stagnation pressure.

In all the above cases sonicFoam is perfectly capturing the flow Physics, as it should be. Even the shock location in case two is captured properly. SonicFoam is giving fairly accurate results and we have validated that with either quasi 1D theory or with experimental data.

Chapter 3

Numerical simulation of Arc Heated Flow through Parallel Plates

Before jumping directly to the arc heater problem we took the arc Heated flow through parallel plates as our intermediate stage. Here we tried to study the Boundary conditions required to form a straight arc, the effects of high input voltage and lorentz force.

3.1 Geometry and Mesh details

Using Gmsh, a Cuboid (20mm × 4mm × 20mm) was created through extrusion of a 2D square (having circular patches at centre) in y direction. Diameter of circular Patches is 1 mm, and will be treated as Electrodes in later work.

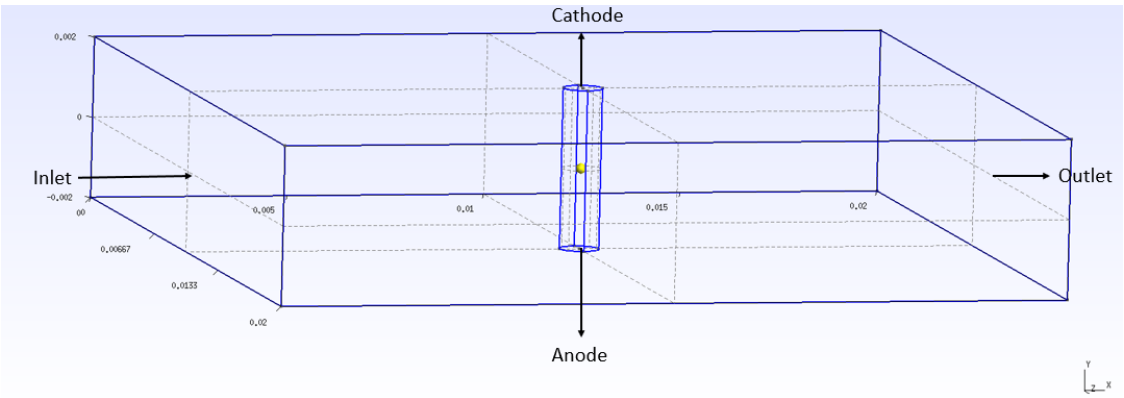


Figure 3.1: Geometry for arcflow between parallel plates

Number of Cells:

A structured mesh was created within gmsh using the Transfinite option. There are 45 cells at each edge, 60 in the radial direction and 25 uniform layers in y direction. A total of 3,08,000 hexahedras were created.

Mesh Quality

Mesh is checked using the checkMesh command and below parameters are crucial to determine whether mesh is good or bad.

Max Aspect Ratio: 150.109

Max Skewness :0.9

Max NonOrthogonality: 73.12 (Avg. 19.99)

Aspect Ratio is the length of longest and shortest length. Aspect ratio of 1 is considered to be best. Skewness measures the distance between the centre of the common face and intersection of the line connecting two cell centres with their common face. A smaller value of skewness is good. Non orthogonality is the measure of the angle between the line connecting two cell centres and the normal of their common face. A value of 0 is considered best. In our case, the mesh is highly refined near the electrodes and so its showing a high value of skewness and non orthogonality in its vicinity.

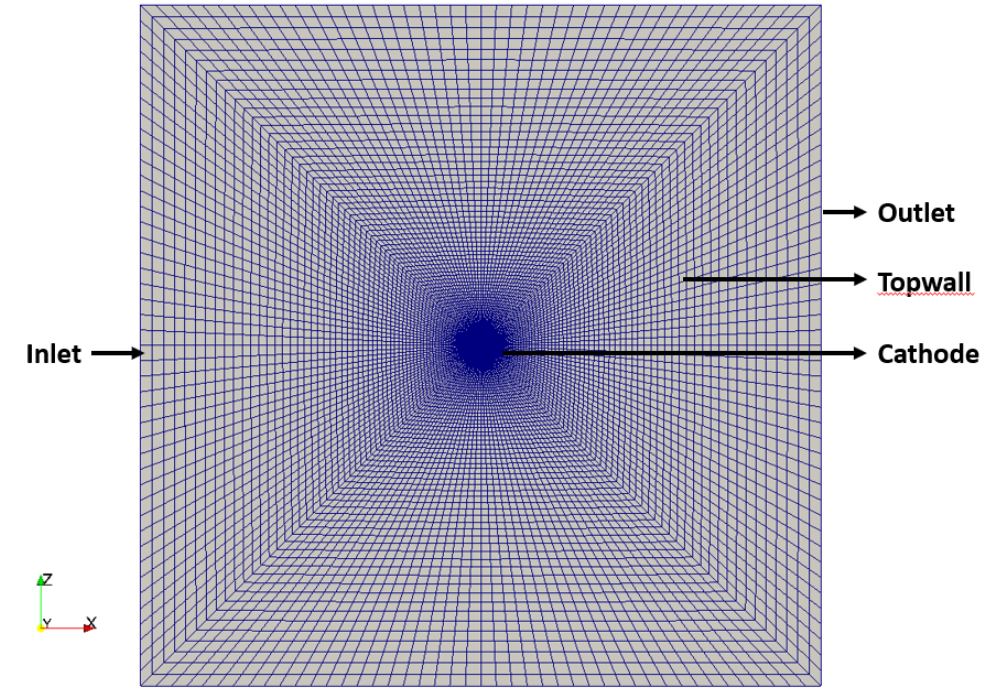


Figure 3.2: Mesh Top View

3.2 Case Setup

A modified version of ArcFoam solver is used to simulate Laminar flow through an arc with Argon gas as working fluid. Perfect gas model is taken for density calculation and Tabular Formulations are used for Viscosity, Enthalpy, Specific Heat and Thermal conductivity. The linear curve fits of electrical conductivity and net radiation coefficients

were changed to polynomial curve fits. A 3rd order polynomial for electrical conductivity and a 6th order polynomial for net emission coefficient of Argon are used.

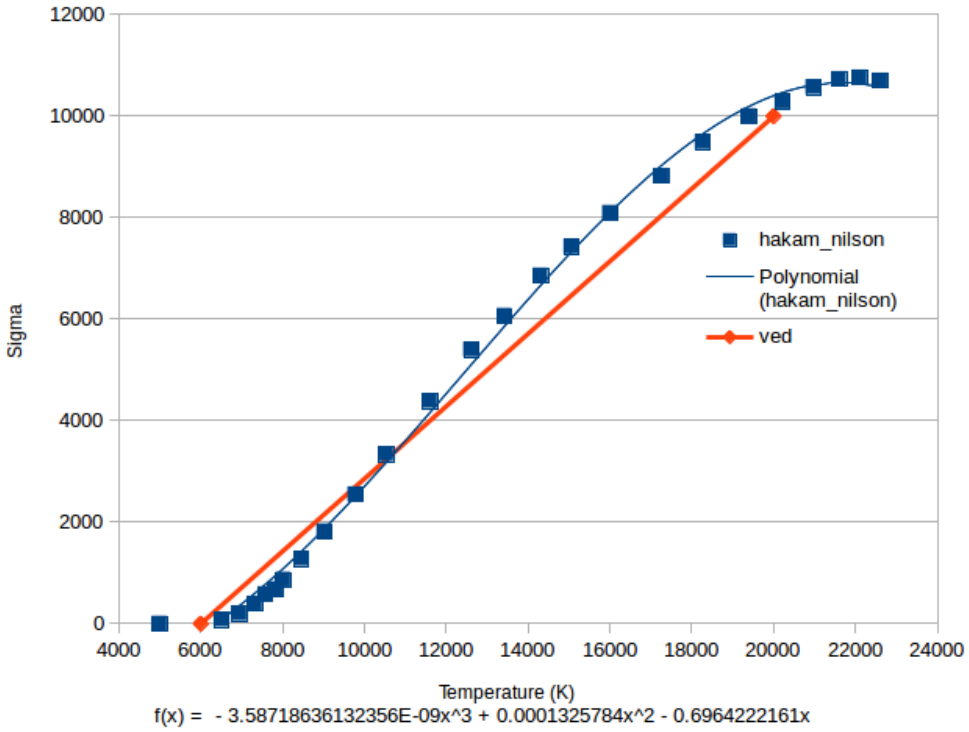


Figure 3.3: Variation of Electrical conductivity of Argon gas with temperature

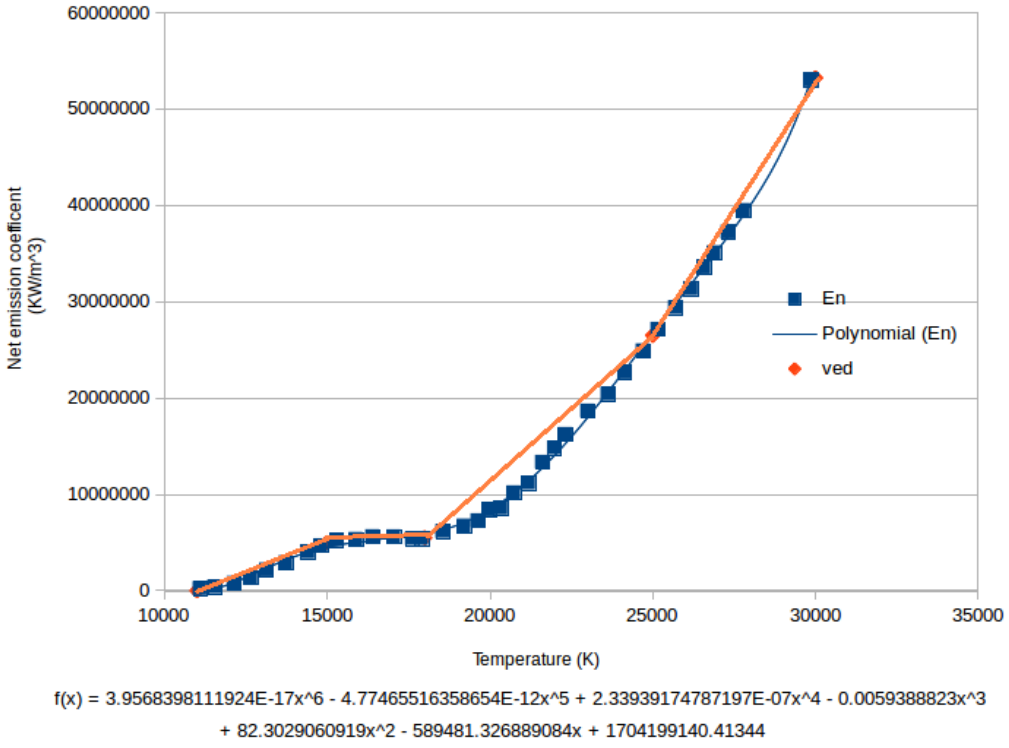


Figure 3.4: Variation of Net Emission coefficient of Argon gas with temperature

Work Flow

1. Initially, the flow has been heated using HT_sonicFoam which is an extension of sonicFoam solver working on tabular formulation to extend temperature ranges upto 24,000K. Electrodes are kept at high temperature of 10,000K while the internal temperature is set initially to 500K. Over such a wide range of temperature thermodynamic properties like heat capactity, enthalpy, entropy etc. and transport properties like viscosity vary significantly. GroovyBC Boundary conditions are used to give expressions of temperature at Topwall and Bottomwall 3.5. The temperature is varying from 500-1000K in X direction and is kept 500K at walls. Calculations are done at atmospheric pressure condition with an inlet velocity of 1m/s. Below are the boundary conditions used for Pressure, Temperature and Velocity:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	Z.G	500	(1,0,0)
Outlet	10^5	Z.G	Z.G
Cathode	Z.G	10,000	noSlip
Top & Bottom wall	Z.G	groovyBC	noSlip
Anode	Z.G	10,000	noSlip
Sidewalls	Z.G	Z.G	noSlip
Internal field	101325	500	(0,0,0)

Table 3.1: Boundary Conditions for Pressure ,Temperature & Velocity

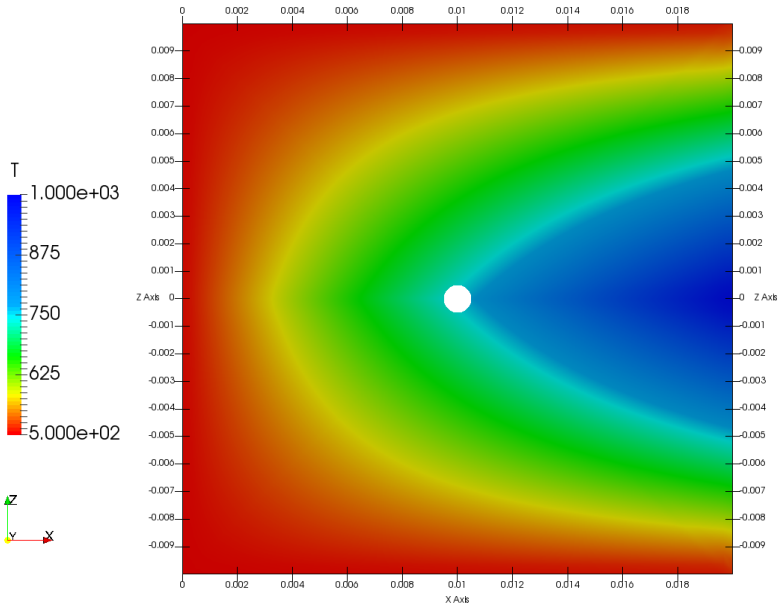


Figure 3.5: Temperature distribution on Top and Bottom wall

In Table: 3.1, Z.G implies zero gradient boundary condition. The time step Δt is set to 0.1 μs and the calculations were done upto 0.35 sec.

2. The steady state solution of HT_sonicFoam is passed on to argon_arcFoam which uses 100% argon as working fluid. This solver is accounting for joule heating and radiation cooling in the energy equation with no addition of lorentz force in momentum equation. An arc is formed first between Cathode and Pilot Anode with a potential drop of 100V. Before jumping to 100 V, an attempt to form the arc with 50, 60 and 70 V potential difference bear no fruit. Below are the boundary conditions of Current Density(J), Potential(Sie), Sigma(σ) and Net Emission Coefficient (ϵ):

	Current Density (Amp/m^2)	Potential (Volt)	Sigma (S/m)	Net Emission Coefficient (W/m^3)
Inlet	(0,0,0)	Z.G	Z.G	Z.G
Outlet	Z.G	Z.G	Z.G	Z.G
Cathode	Calculated	-50	Z.G	Z.G
Top & Bottomwall	(0,0,0)	Z.G	0	Z.G
Anode	Calculated	50	Z.G	Z.G
Sidewalls	(0,0,0)	Z.G	0	Z.G
Internal field	(0,0,0)	0	1000	0

Table 3.2: Boundary Conditions for J, Sie, Sigma and net Emission Coefficient

3. Before the solution can converge, at 0.7 sec we applied a linear potential ramp to decrease the potential difference from 100 volt to 20 volt in 0.08 sec. The iterations were continued for 0.17 sec. The solutions got steady by 0.95 sec. We then doubled the number of cells in the domain to 6,14,250 hexahedras, took the earlier solution as intial solution and mapped that into newer mesh using mapFields command. Solution was then allowed to converge till 1.15 sec.

4. Lastly, to check the influence of Lorentz force we mapped that solution into loarg_arcFoam in which neceesary changes were done to calculate magnetic field based on the magnetic potential. Iterations were run upto 1.35 sec. Below are the Boundary conditions of magnetic field(B) and magnetic vector potential(A).

	Magnetic Field (B)	Magnetic vector potential (A)
Inlet	Z.G	Z.G
Outlet	Z.G	(0,0,0)
Cathode	Calculated	Z.G
Top & Bottomwall	Z.G	Z.G
Anode	Calculated	Z.G
Sidewalls	Z.G	Z.G
Internal field	(0,0,0)	Z.G

Table 3.3: Boundary Conditions for Magnetic Field and Magnetic Vector Potential

3.3 Results

For ease in Post processing, we subdivide our work into following two cases:

Case 1: without Lorentz Force

Case 2: Considering Lorentz Force

3.3.1 Case 1

For the first case i.e neglecting Lorentz Force, following plots were obtained:

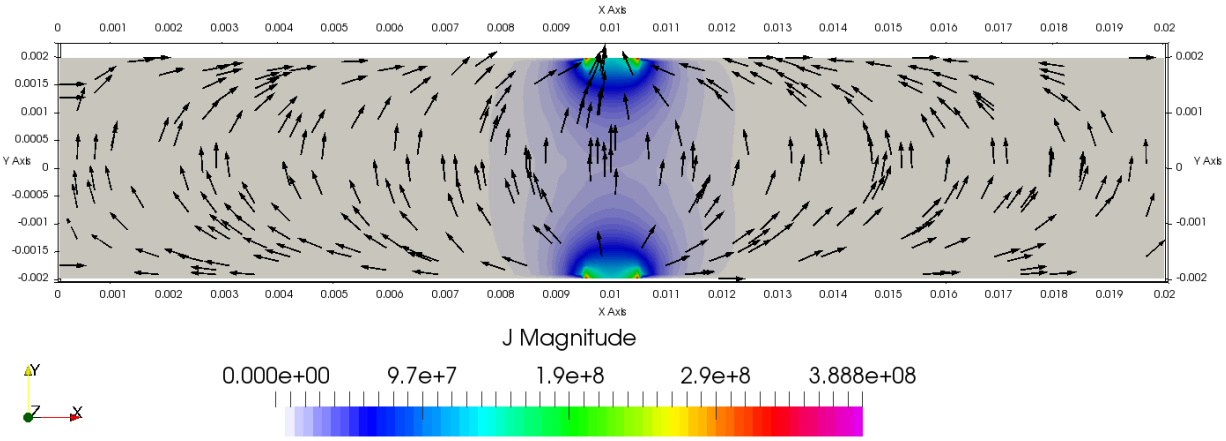
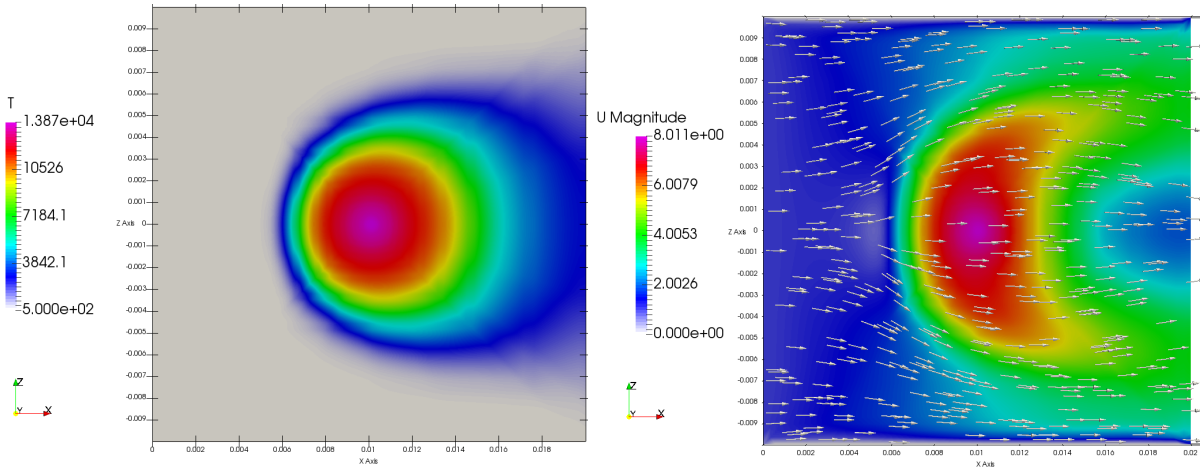


Figure 3.6: Current density Magnitude (slice at z=0 mm)

As can be seen maximum current density is obtained near electrodes having a magnitude of 3.88×10^{8}. Maximum Velocity is nearly 8 m/s. The upper limit of Temperature is set to 20,000K using fvOptions.



(a) Temperature variation (slice at y=0mm) (b) Velocity Variation (slice at y=0mm)

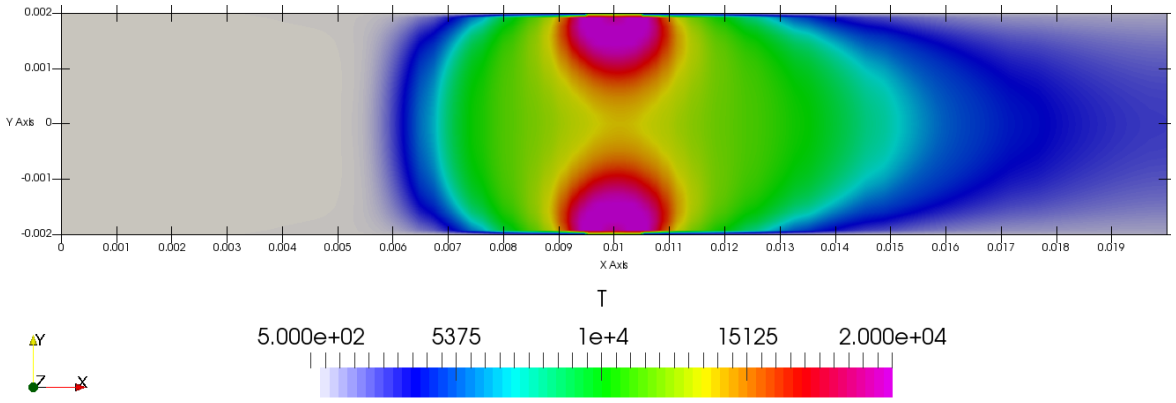


Figure 3.8: Temperature Variation (slice at z=0mm)

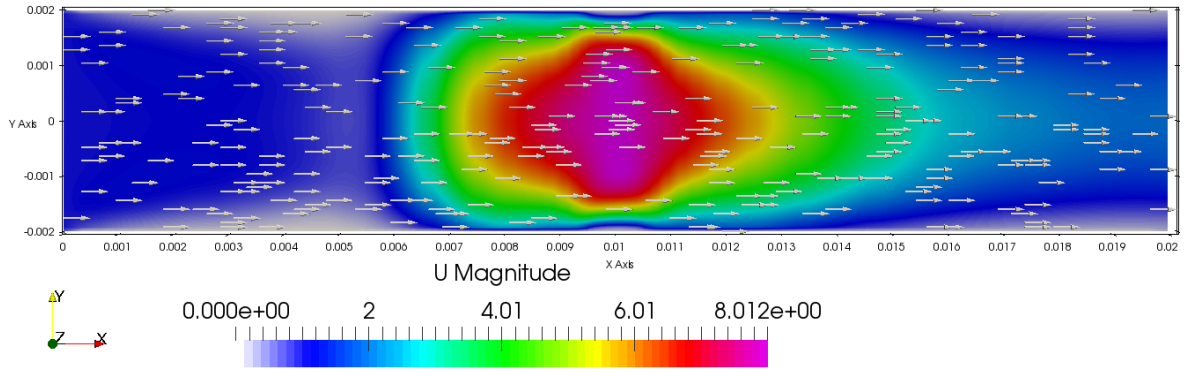


Figure 3.9: Velocity Variation (slice at z=0mm)

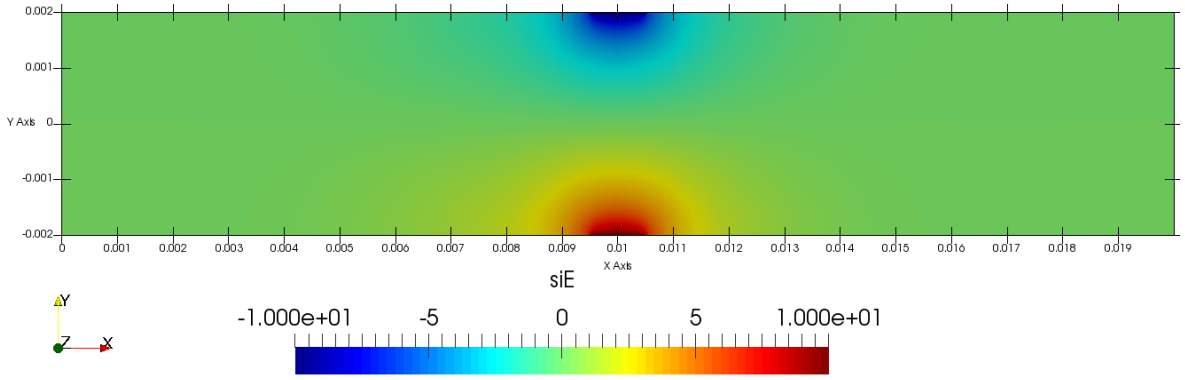


Figure 3.10: Electric Potential(V) (slice at z=0 mm)

3.3.2 Case 2

For the second case i.e taking Lorentz force into account following contour plots were obtained:

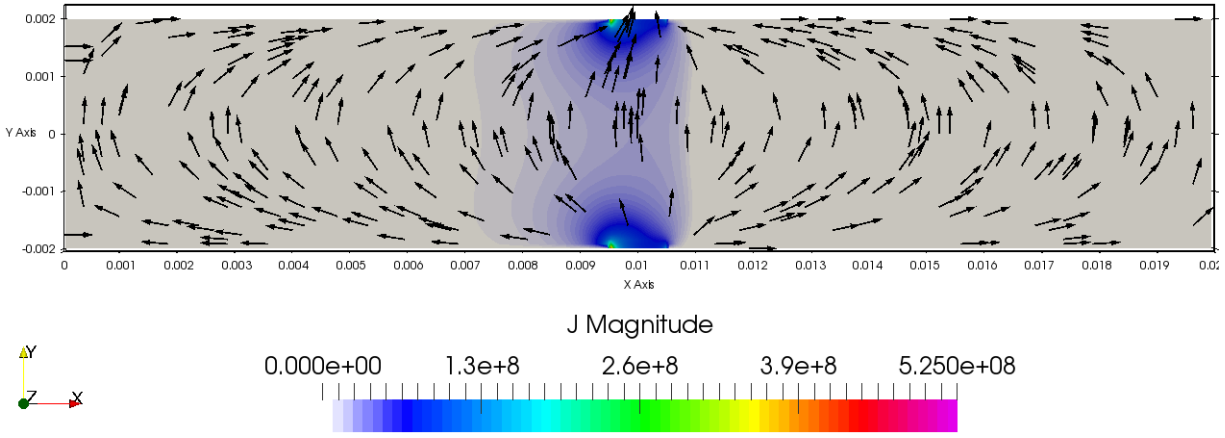


Figure 3.11: Current density Magnitude (slice at z=0 mm)

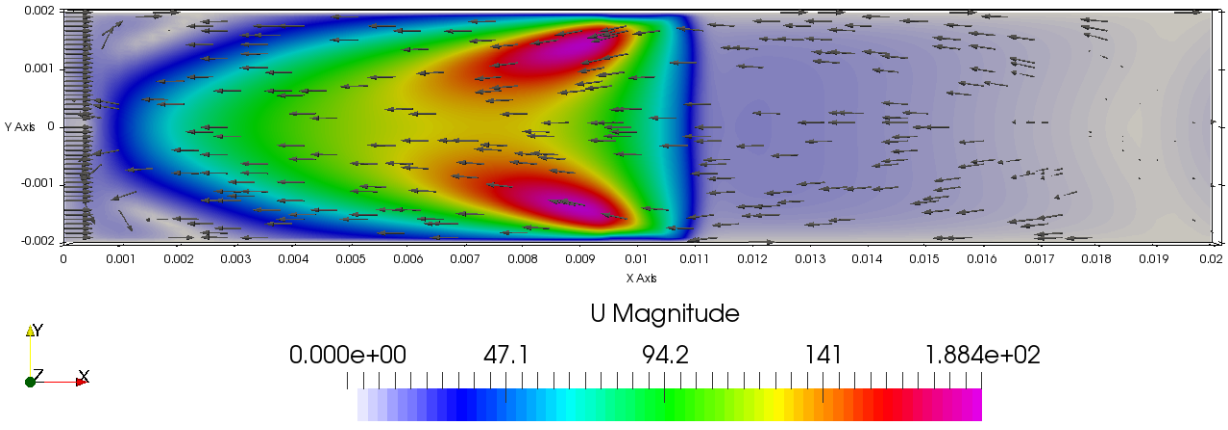


Figure 3.12: Velocity Variation (slice at z=0mm)

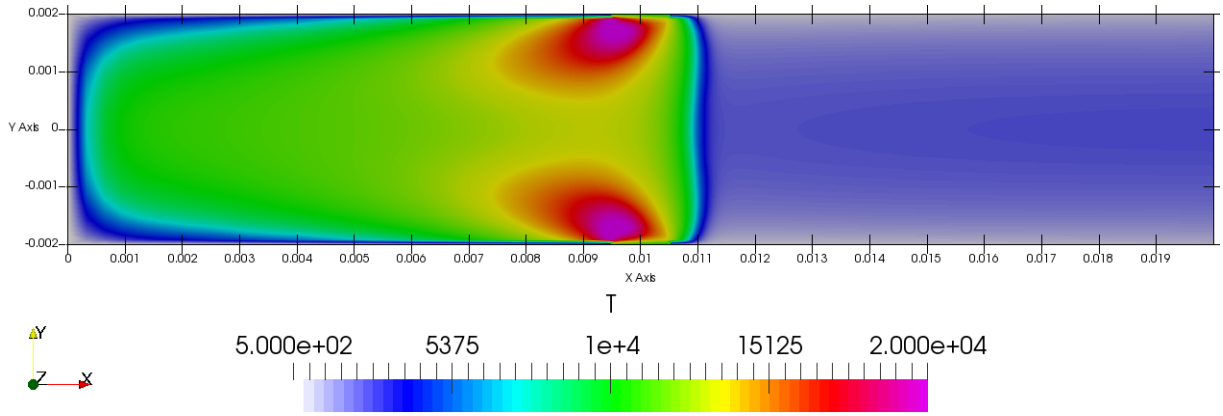
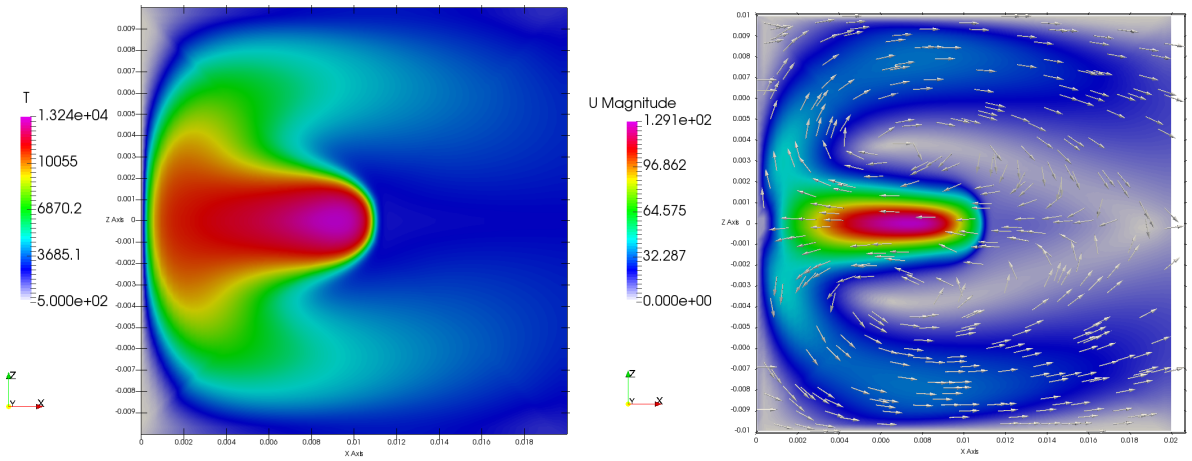


Figure 3.13: Temperature Variation (slice at z=0mm)

Maximum current density is obtained near electrodes having a magnitude of 5.25×10^8. Maximum Velocity is nearly 188 m/s, more than 20 times that of case 1.



(a) Temperature variation (slice at y=0mm)

(b) Velocity Variation (slice at y=0mm)

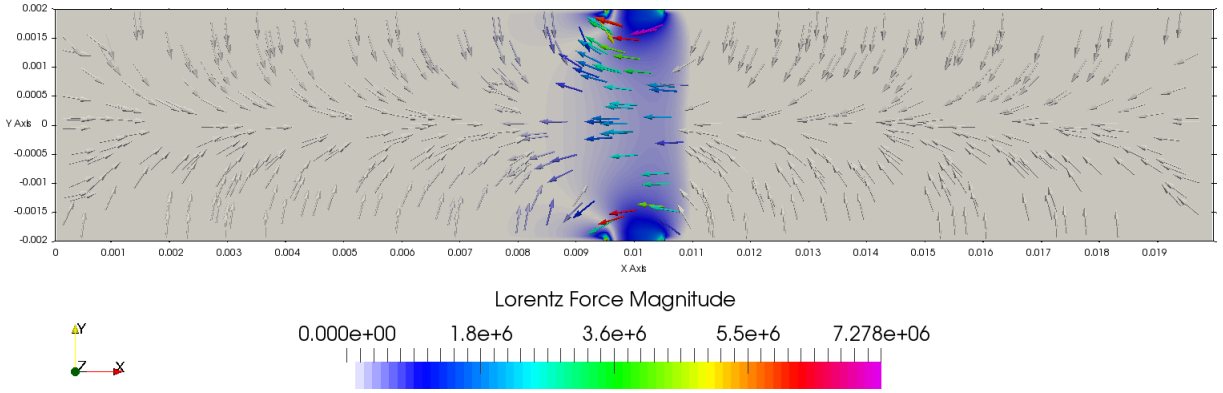


Figure 3.15: Lorentz Force Magnitude (N/m^3) (slice at z=0 mm)

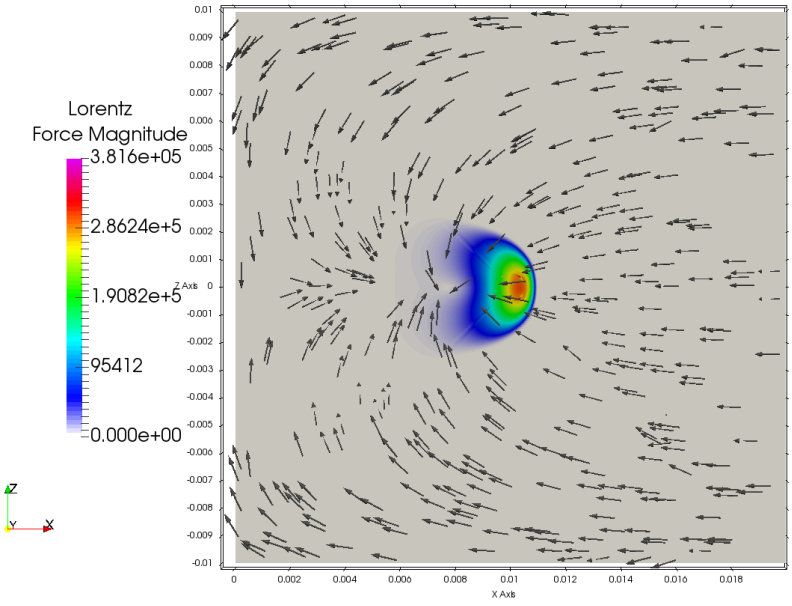


Figure 3.16: Lorentz Force Magnitude (N/m^3) (slice at y=0 mm)

It can be clearly seen from Figures above that two jets are arising from electrodes and making the flow to reverse ahead of the arc. This leads to heating of flow upstream. It appears as a shocking results but flow reversal is happening because of large value of Lorentz force which is turning the flow towards inlet. We probed it further and figured out that it is non uniform magnetic field(B) magnitude that is causing this Lorentz force to direct towards inlet in case 2. As of till now, the reasons are uncertain for this behaviour of magnetic field density(B).

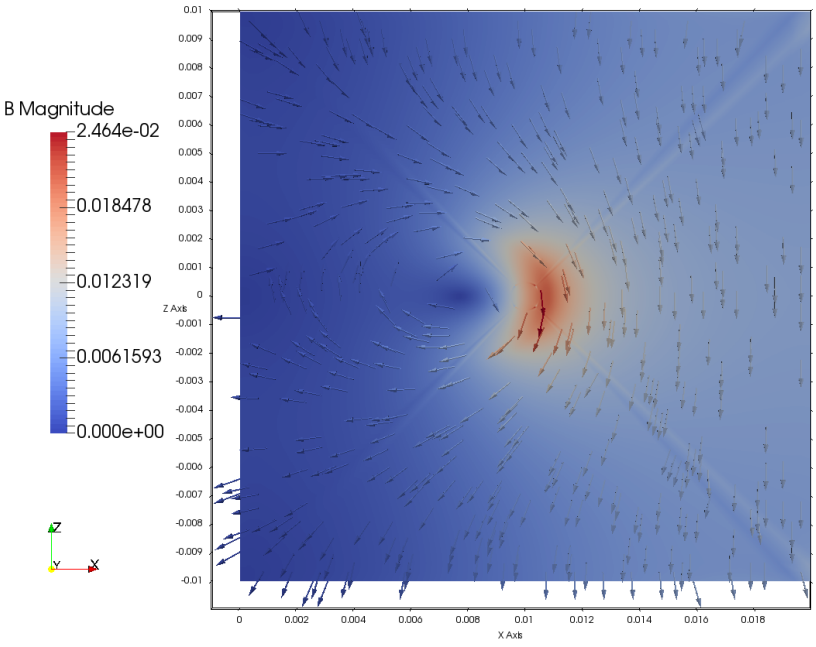
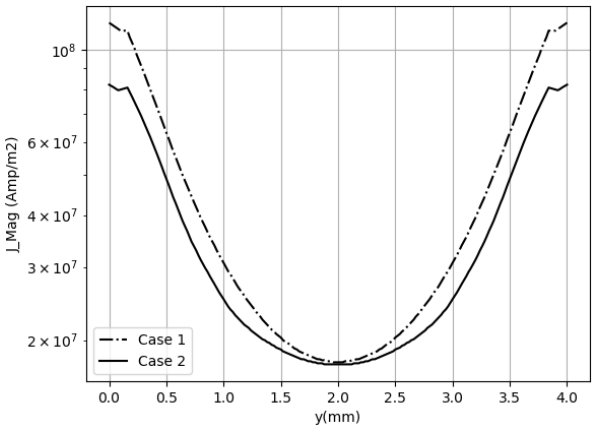


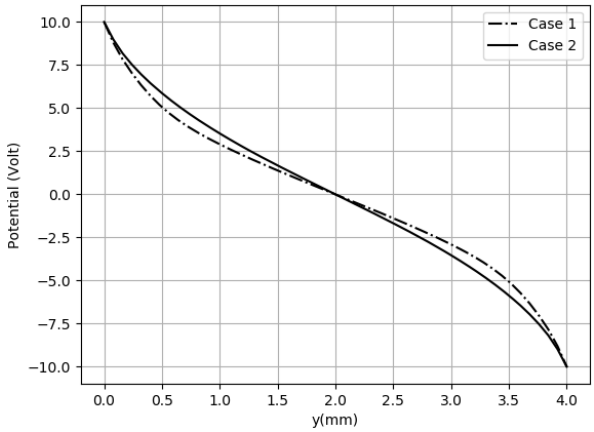
Figure 3.17: Magnetic Field Variation (slice at y=0mm)

3.4 Post Processing & Verification

For Post processing purpose '.csv' files along a line between arc (between two electrodes) and through the arc (at y=0, along x axis) were extracted using 'Export scene' feature of ParaView. A python script was written to plot pressure,Temperature, electric Potential, Current Density along the arc for both cases. Obtained plots are as below:

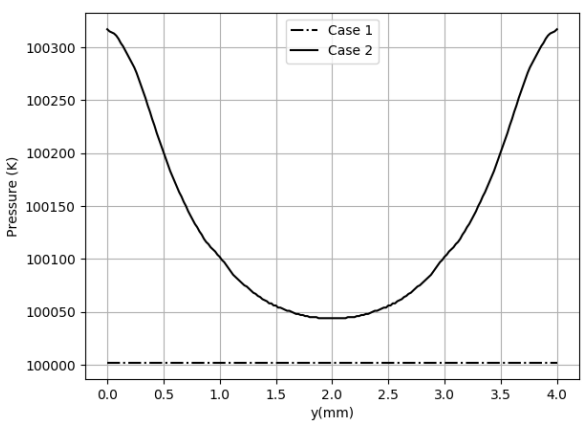


(a) Current density variation along Arc

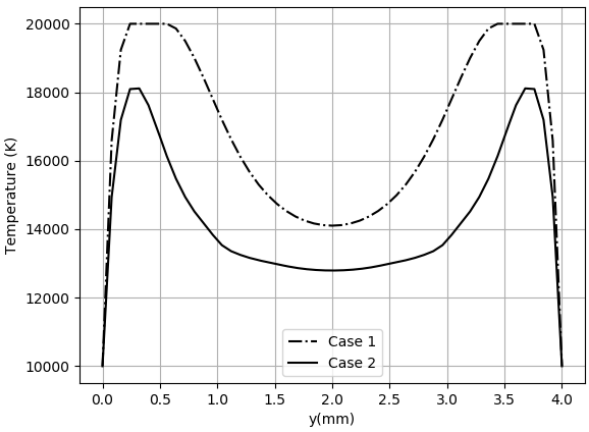


(b) Electric Potential variation along Arc

As expected, Electric potential(ϕ) varies almost linearly from a maximum of +10 volt (at anode) to a minimum of -10 volt (at cathode) in both the cases. Ideally ϕ should vary steeply near anode and then have a constant slope and then again steep slope near cathode. This trend is slightly captured in case1. Trends of Current density plots for both the cases are identical, with only difference in magnitude. In fig. above J_y is plotted on semilog scale in y direction. For both cases it have its maximum near electrodes and then decreases smoothly. Maximum obtained value of J for case 1 and 2 are 11,60,00,000 Amp/m^2 and 8,23,00,000 Amp/m^2 respectively.

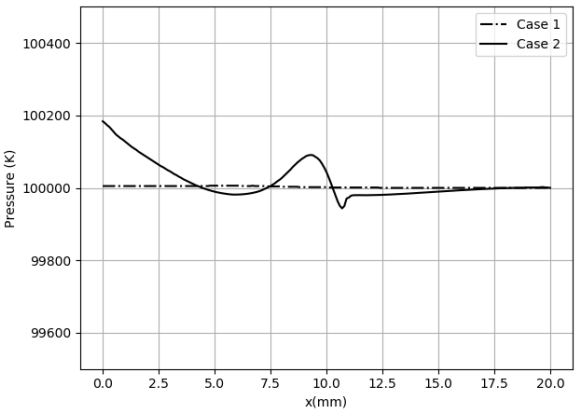


(a) Pressure variation along Arc

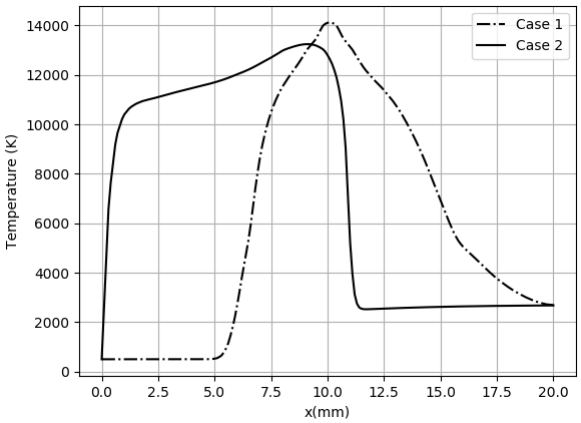


(b) Temperature variation along Arc

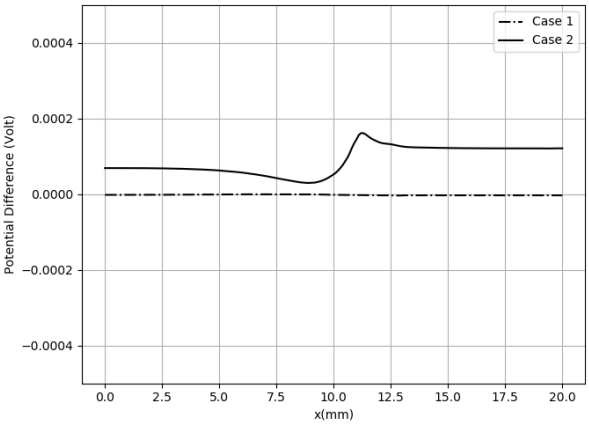
For both cases, temperature along the arc follow the same trends only difference in their magnitudes. In case 2, it is coming slightly lower because the peak position of temperature is shifted towards inlet and arc is not axisymmetric. Pressure for case 1 is constant having a value of 100,000 Pa along the arc, while for case 2 it achieves its maximum of 100300 Pa slightly more than case 1. Similar to as done above, A python script was written to plot pressure, Temperature, current Density, electric Potential, Lorentz Force and Magnetic Flux density, across the arc (at y=0.0, along x axis) for both cases. Obtained plots are as below:



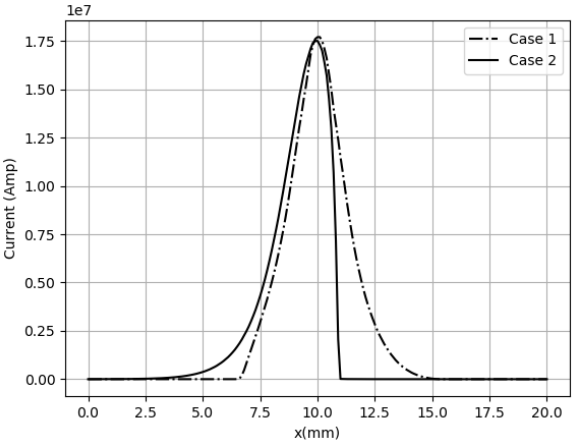
(a) Pressure variation across Arc



(b) Temperature variation across Arc



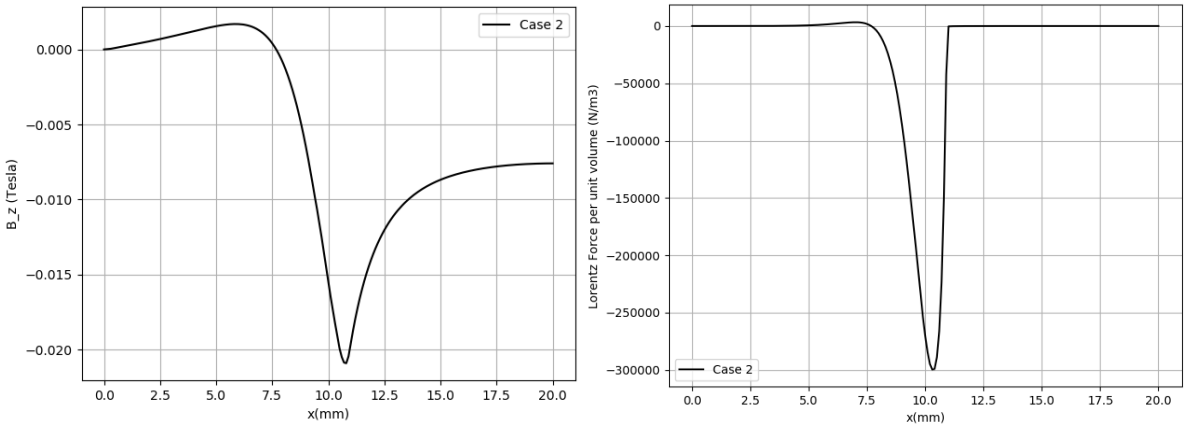
(a) Electric Potential variation across Arc



(b) Current Density variation across Arc

For case 2, a small potential of order 0.0001 is there at the XZ plane at y=0. Current Density plot is symmetrical for case1 and asymmetrical for case 2. Because of flow reversal flow in case 2, the flow leaving the inlet is immediately heated up . A similarity between temperature plot and Electrical conductivity plot is seen. This is intuitional too, Electrical Conductivity being maximum where the temperature is maximum and

minimum where temperature is minimum.



(a) Magnetic Field (B_Z) variation across Arc (b) Lorentz Force(F_x) variation across Arc

As can be seen that the lorentz force per unit volume for case 2 is too high. The asymmetric Magnetic field can be held responsible for this. Before arc Lorentz force is having a small magnitude directed towards positive X direction while after arc it is having a high magnitude towards negative X direction resulting in Flow reversal. Also It was found that near electrodes the magnetic field is symmetric and it turned asymmetric away from electrodes.

For Verification Purposes, mass and electric current conservation were checked. Mass flow rate was evaluated at inlet & outlet using Cell data and Integrate variable filter of paraView. Inbuilt calculator was used & following formula was created within paraView:

$$\iint_S \rho(y,z) \times V_x(y,z)\, dy\, dz \qquad Mass flowrate\ calculation \tag{3.1}$$

	Case 1	Case 2
Inlet	0.07659	0.07664
Outlet	0.07655	0.07660
Percentage difference	0.05%	0.052%

Table 3.4: Conservation of mass Flow rate (gm/s)

In a similar manner current at cathode and anode is evaluated.

$$\iint_S J(y,z)\, dy\, dz, \qquad Current\ Calculation \tag{3.2}$$

	Case 1	Case 2
Cathode	175.267	114.937
Anode	175.267	114.936

Table 3.5: Conservation of electric Current(A)

A comparison between the results of both cases was done. Power input i.e Voltage × Current, is always greater than power added to flow because of radiation and conduction losses. Because of the greater average velocity at outlet, more heat is carried away in case2, leading to higher Avg temperature and more enthalpy gain.

	Case 1 (without Lorentz Force)	Case 1 (with Lorentz Force)
Power Input(Watts)	3,505	2,300
Avg. T at outlet(K)	1404	2182
Avg. U at outlet(m/s)	2.15	5.24
Avg. ρ at outlet(kg/m^3)	0.45	0.25
Enthalpy gain (W)	44	75.13
Radiation Losses (W)	1372	987.65

Table 3.6: A comparison between Case1 and Case2

Chapter 4

2D simulation of Flow through Arc-Heater

In this Chapter, We performed the axisymmetric simulations through a modified geometry of Arc Heater. Biggest advantage of a 2D model is in terms of computing time but it lacks in properly reproducing the arc root attachment point. Local thermodynamic equilibrium (LTE) was assumed and calculations are done with 100% argon and Air. The flow was considered to be entering axially rather than entering tangentially through the inlet as it happens in actual. Electrodes dimensions were also altered to facilitate arc formation. Effects of mass flow rate with a constant Voltage was studied in this section.

4.1 Geometry and Mesh details

A 3 Deg. wedge of the Arc heater was created using Gmsh, an opensource meshing tool compatible with OpenFOAM. Cathode tip is made more pointed than in actual with a 1.2 mm radius and 45 Deg angle. The anode length is also reduced from 9 mm in actual to 5 mm. These modifications were made to reduce the electrodes area such that arc can strike at a point. These alterations were removed in the 3D geometry which will be discussed in the next chapter.

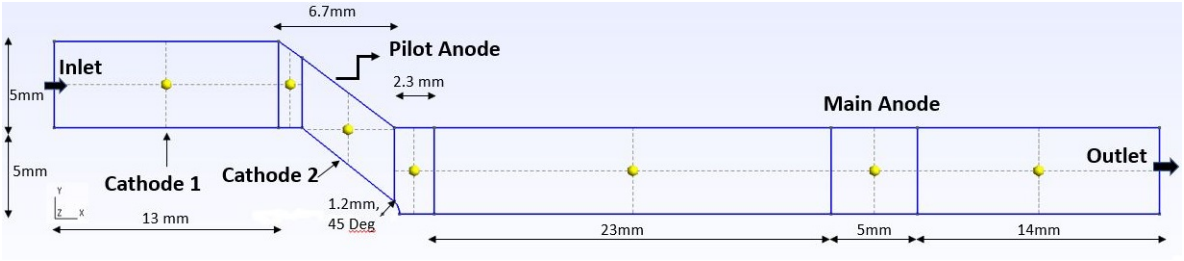


Figure 4.1: 2D view of modified Arc heater

Number of cells:

A structured mesh was created within Gmsh using the transfinite option, with 47,220 hexahedra and 391 prisms. Number of divisions are 550 in axial direction and 94 in radial direction. It is properly refined near the electrodes and walls to account for the boundary layer.

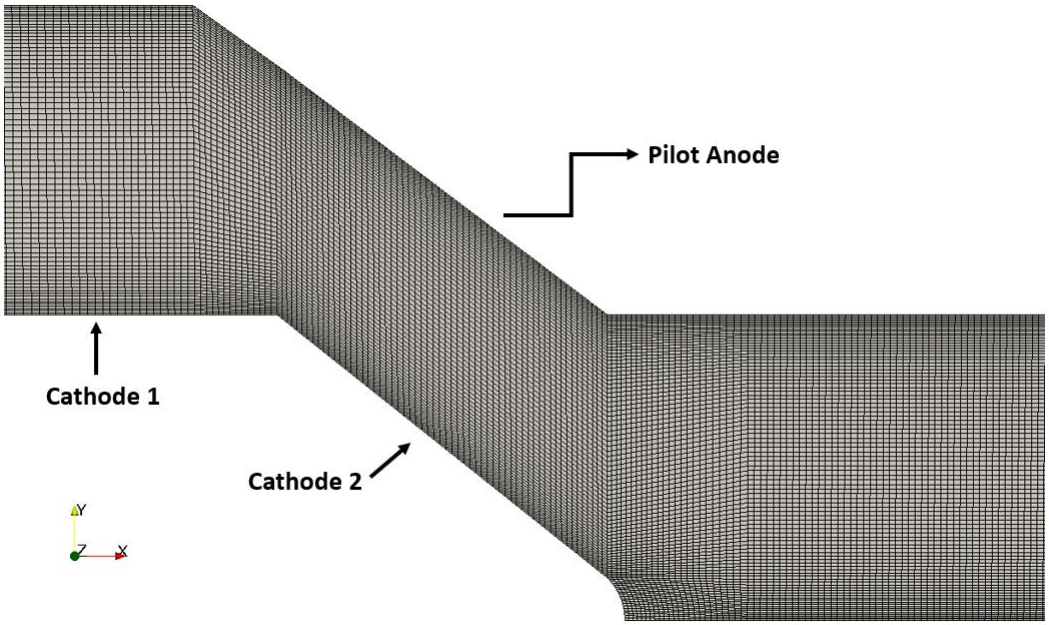


Figure 4.2: Mesh near electrodes

Mesh Quality

Max aspect Ratio: 16.33

Max skewness : 1.26

Max nonorthogonality: 43.18 (Avg. 11.9)

4.2 Case Setup

A modified version of ArcFoam solver is used to simulate flow through the arc-heater. Two different cases are setup, Case1 with 100 %Argon gas and Case2 with Air. Perfect gas model is taken for density calculation and Tabular Formulations are used for Viscosity, Enthalpy, Specific Heat and Thermal conductivity. As previously discussed like Argon, the linear curve fits of electrical conductivity and net radiation coefficients were changed to polynomial curve fits for Air also. a $4th$ order polynomial for electrical conductivity is used. Lorentz force is removed from the momentum equation due to instabilities it was causing in computations. It was later shown in chapter 5 that effect of lorentz force can be neglected in this case. K-Epsilon is taken as the turbulent model, however since inlet Reynolds number is too low, it can be neglected.

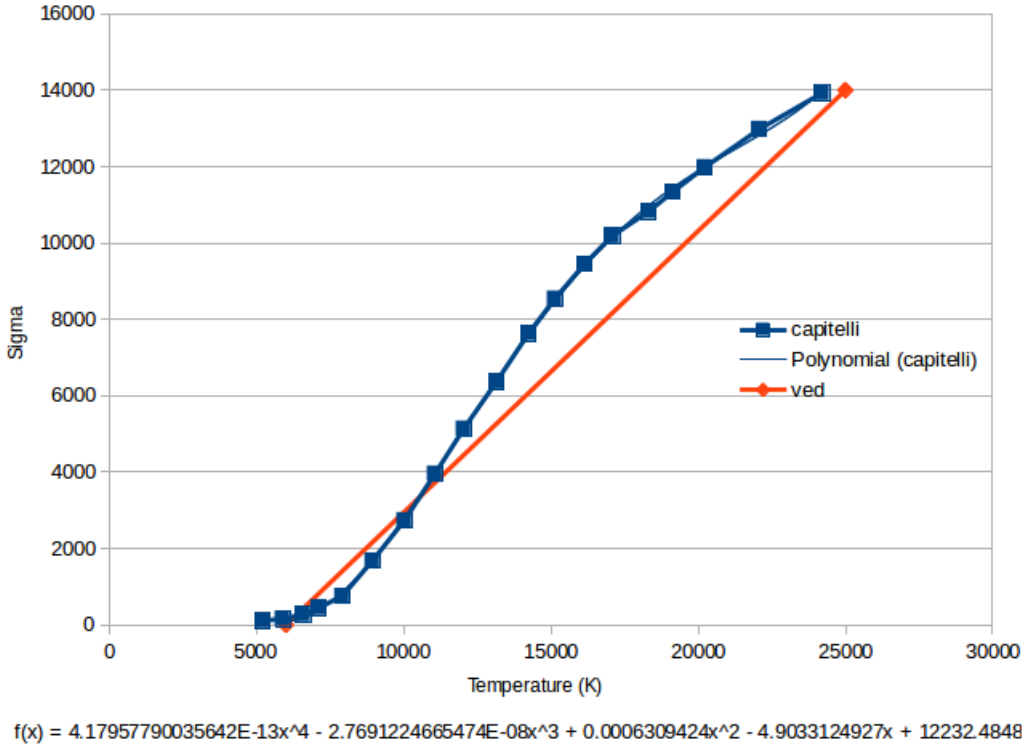


Figure 4.3: Variation of Electrical conductivity of Air with temperature

Work Flow

1. Initially, the flow has been heated in tunnel using HT_sonicFoam which is an extension of sonicFoam solver working on tabular formulation to extend temperature ranges upto 24,000K. At first a lesser velocity of 0.5m/sec is used to facilitate more heating inside tunnel.

   GroovyBC Boundary conditions are used to give expressions of temperature at Top_wall, Cathode1, Cathode2, Pilot Anode and main anode. Calculations are done at atmospheric pressure condition. Gaussian distribution is used for temperature at both the Cathode2 and Anode. Since specific heat (C_P) of air is more, Maximum anode temperature is taken more in air i.e 10,000K as compared to argon i.e 8,000K. The time step Δt is set to 0.1 μs and the calculations were done upto 1.2 sec. Below are the boundary conditions used for Pressure, Temperature and Velocity:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	Z.G	300	(0.5,0,0)
Outlet	10^5	Z.G	Z.G
Cathode1	Z.G	Linear Variation from [500-1000]	noSlip
Cathode2	Z.G	Gaussian Distribution Min:1000, Max:10,000, Centre: 0.02	noSlip
Topwall	Z.G	Linear Variation from [300-1000]	noSlip
pilot_anode	Z.G	Gaussian Distribution Min:800, Max:10,000, Centre: 0.0197	noSlip
main_anode	Z.G	Gaussian Distribution, Min:2500 Max: 8,000 for Ar & 10,000 for Air Centre: (0.0454, 0.005)	noSlip
Internal field	101325	300	(0,0,0)

Table 4.1: Boundary Conditions for Pressure ,Temperature & Velocity

2. The steady state solution of HT_sonicFoam is passed on to argon_arcFoam and air_arcFoam which are using 100% argon and air respectively as working fluid. These solvers are accounting for joule heating and radiation cooling in the energy equation with no addition of lorentz force in momentum equation. A short arc is formed first between Cathode and Pilot Anode. A potential drop of 50V is kept for Argon and 60V for Air. Below are the boundary conditions of Current Density(J), Potential(Sie), Sigma(σ) and Net Emission Coefficient (ϵ):

	Current Density (Amp/m^2)	Potential (Volt)	Sigma (S/m)	Net Emission Coefficient (W/m^3)
Inlet	(0,0,0)	Z.G	Z.G	Z.G
Outlet	Z.G	Z.G	Z.G	Z.G
Cathode1	Calculated	-25	Z.G	Z.G
Cathode2	Calculated	-25	Z.G	Z.G
Topwall	(0,0,0)	Z.G	0	Z.G
Pilot_anode	Calculated	25 for Ar 35 for Air	Z.G	Z.G
Main_anode	Calculated	5	Z.G	Z.G
Internal field	(0,0,0)	0	1000	0

Table 4.2: Boundary Conditions for J, Sie, Sigma and net Emission Coefficient

3. At 1.5 sec the arc was shifted from pilot anode to main anode. Steady Solution was obtained at 2.15 sec, with a time step of 5 μs. For Air, Arc was not forming with a potential difference of 50 volt so we increased it to 100 volts but still it wasn't able to sustain 1 m/s velocity. so we further increased it to 150 volts. Below are the modifications done in the boundary conditions:

	Temperature	Potential	Current Density
Cathode 1 & 2	No change	No change for Ar -75 for Air	No change
Pilot_anode	Consistent with Top wall	Z.G	(0,0,0)
Main_anode	No Change	25 for Ar & 75 for Air	No change

Table 4.3: Changes done in Boundary Conditions for shifting Arc

4. Lastly for a constant voltage drop of 50 for Argon and 150 for Air, we increased the mass flow rate by increasing the velocity in the range of 0.5-2 m/sec and studied the effect of mass flow rate on current and enthalpy gain. For this we took the 1.6 timestep directory, increased the velocity and run the simulations till 2.5 sec.

4.3 Results

4.3.1 Case 1 - with Argon

As already discussed initially we heated the flow inside the tunnel, below is the contour plot of temperature at 1.2 sec:

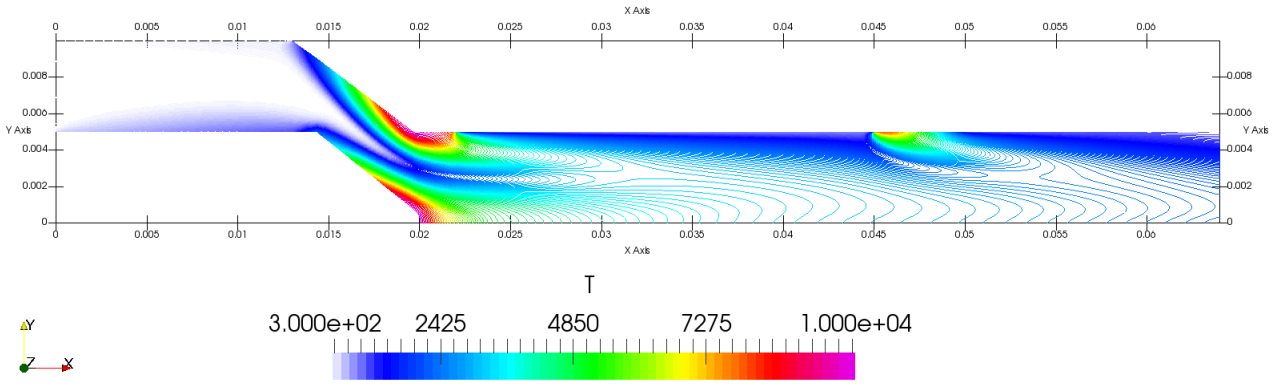


Figure 4.4: Temperature Contour, 1.2 sec

The flow is entering at 300K and is leaving at an average temperature and velocity of 1658K and 9.62 m/sec. Boundary conditions of current density, potential, Electrical Conductivity and Net Emission coefficient are then applied and a short arc is made between

cathode and pilot anode. This will heat up the tunnel, such that we can transfer the arc successfully to main anode. Below are the contours of Current density, Electric Potential, Temperature and Electrical conductivity at 1.5 sec:

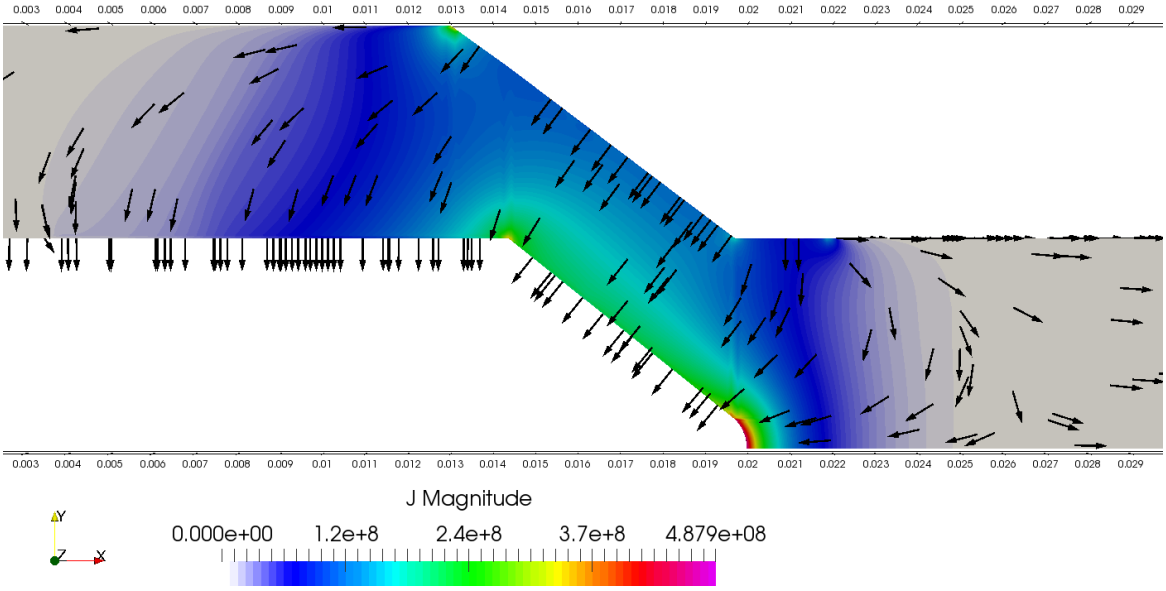


Figure 4.5: Current Density near Electrodes, 1.5 sec

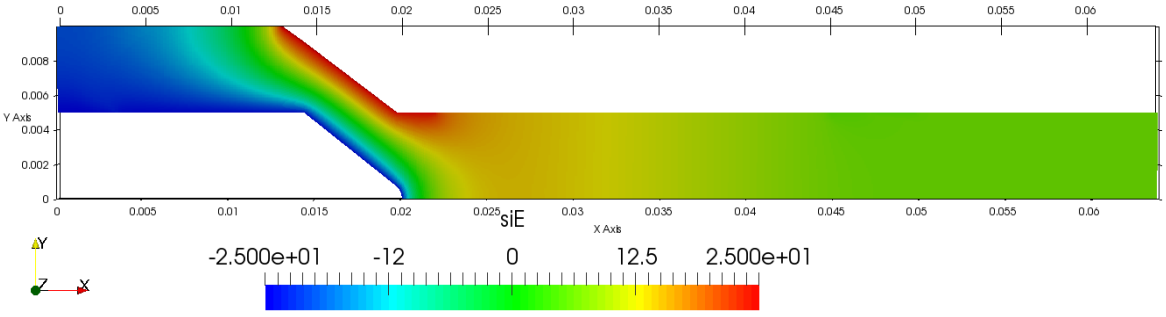


Figure 4.6: Electric Potential, 1.5 sec

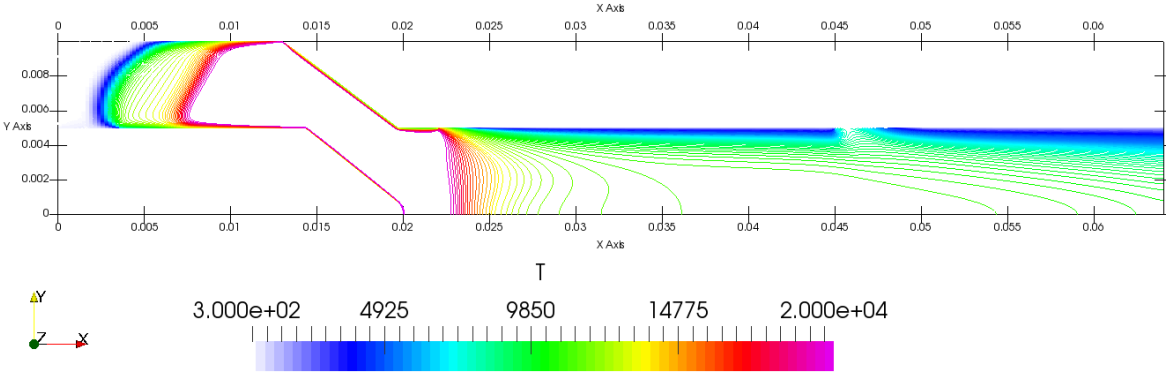


Figure 4.7: Temperature Contour, 1.5 sec

A Temperature limit is applied from 300-20,000K using fvOptions. This was necessary since the temperature near electrodes was going more than 24,000K which is the maximum limit for the tables. Thus blowing the solution in the runtime.

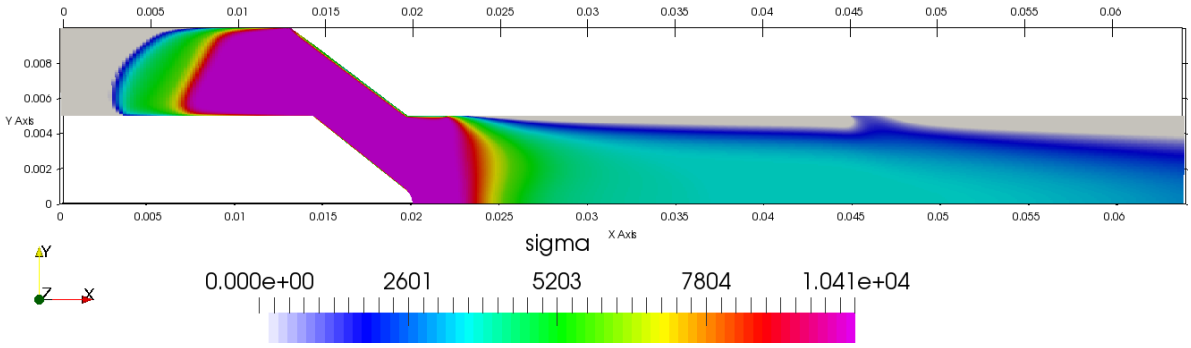


Figure 4.8: Electrical Conductivity, 1.5 sec

After this, the Arc has been shifted to main anode by modifying the boundary conditions. Below are the contours of Current Density, Temperature, Electric Potential, Electrical conductivity and velocity for inlet velocity of 0.5m/sec:

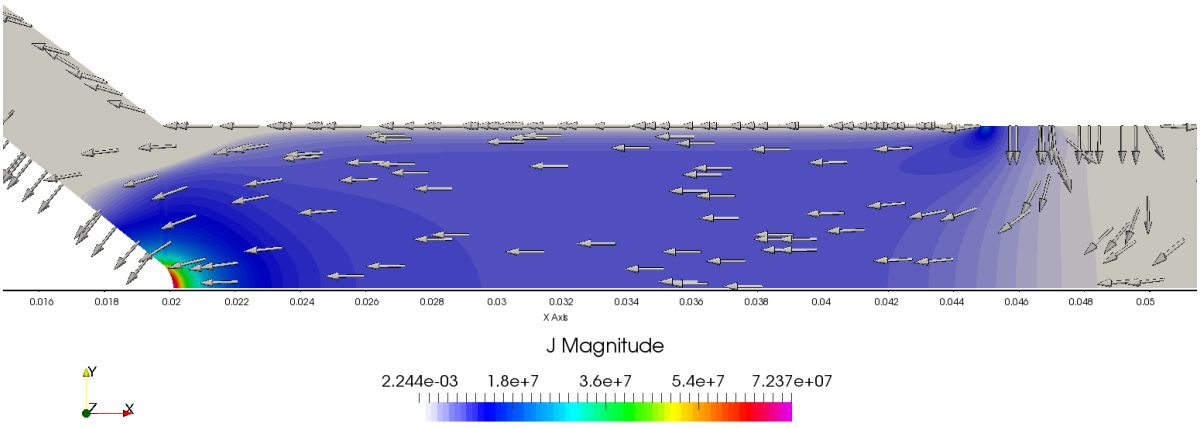


Figure 4.9: Current Density, 2.5 sec

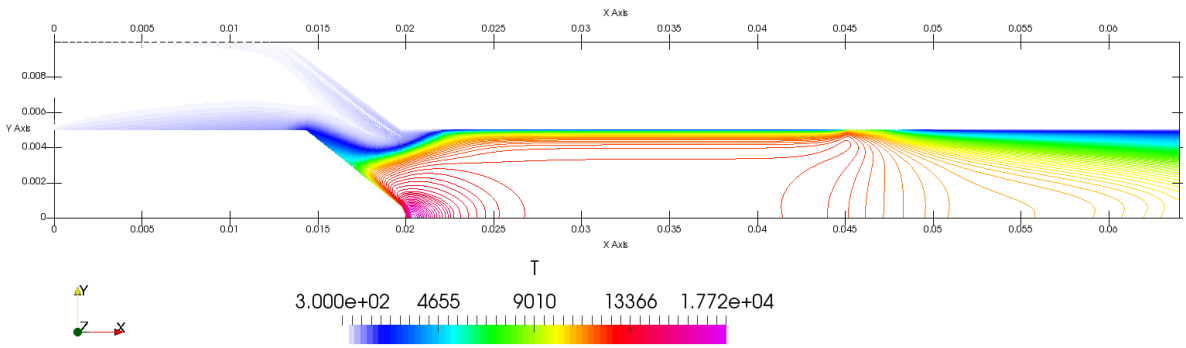


Figure 4.10: Temperature, 2.5 sec

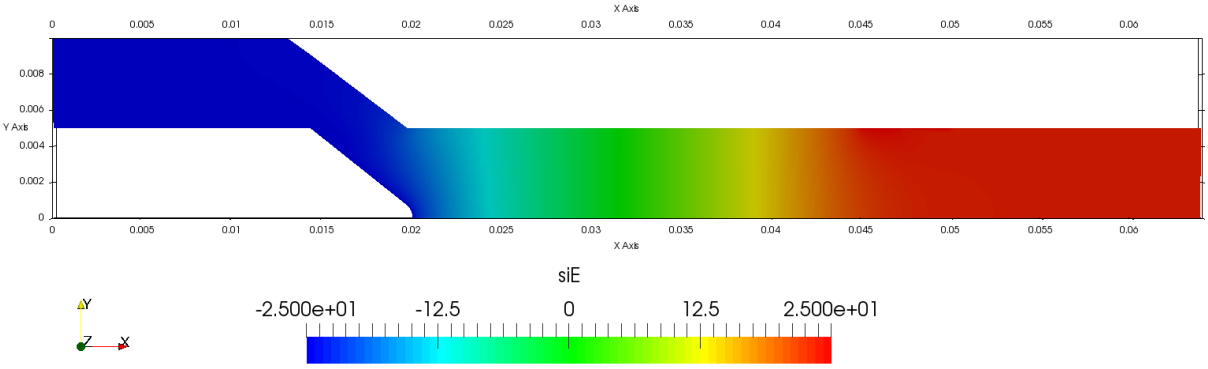


Figure 4.11: Electric Potential, 2.5 sec

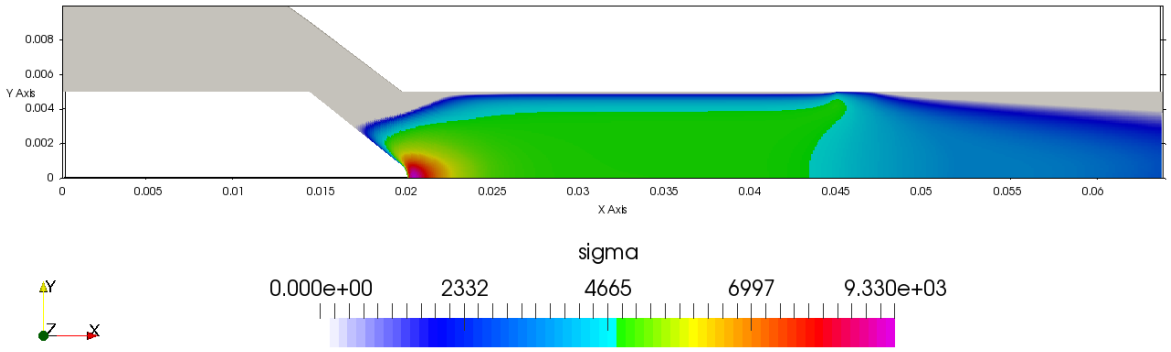


Figure 4.12: Electrical Conductivity, 2.5 sec

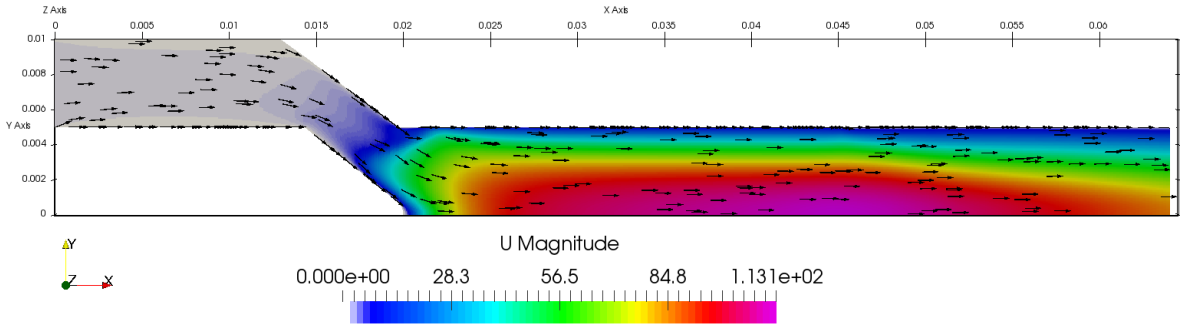


Figure 4.13: Velocity contour, 2.5 sec

It can be seen that Maximum temperature occurs close to cathode and is as high as 17,720K. High sigma indicates arc zone which will shrink as the inlet velocity is increased. Maximum velocity in this case is coming out to be 113 m/sec with a exit pressure of 1,00,000 Pa. Next, A comparison of current density for a range of inlet velocity from 0.5 - 1.5m/sec is made. It is seen that current value decreases with an increment of mass-flow rate while the Enthalpy gain increases. At 1.75m/sec Arc could not sustain itself and blows away. Below are the contours of Current density with different values of inlet velocity:

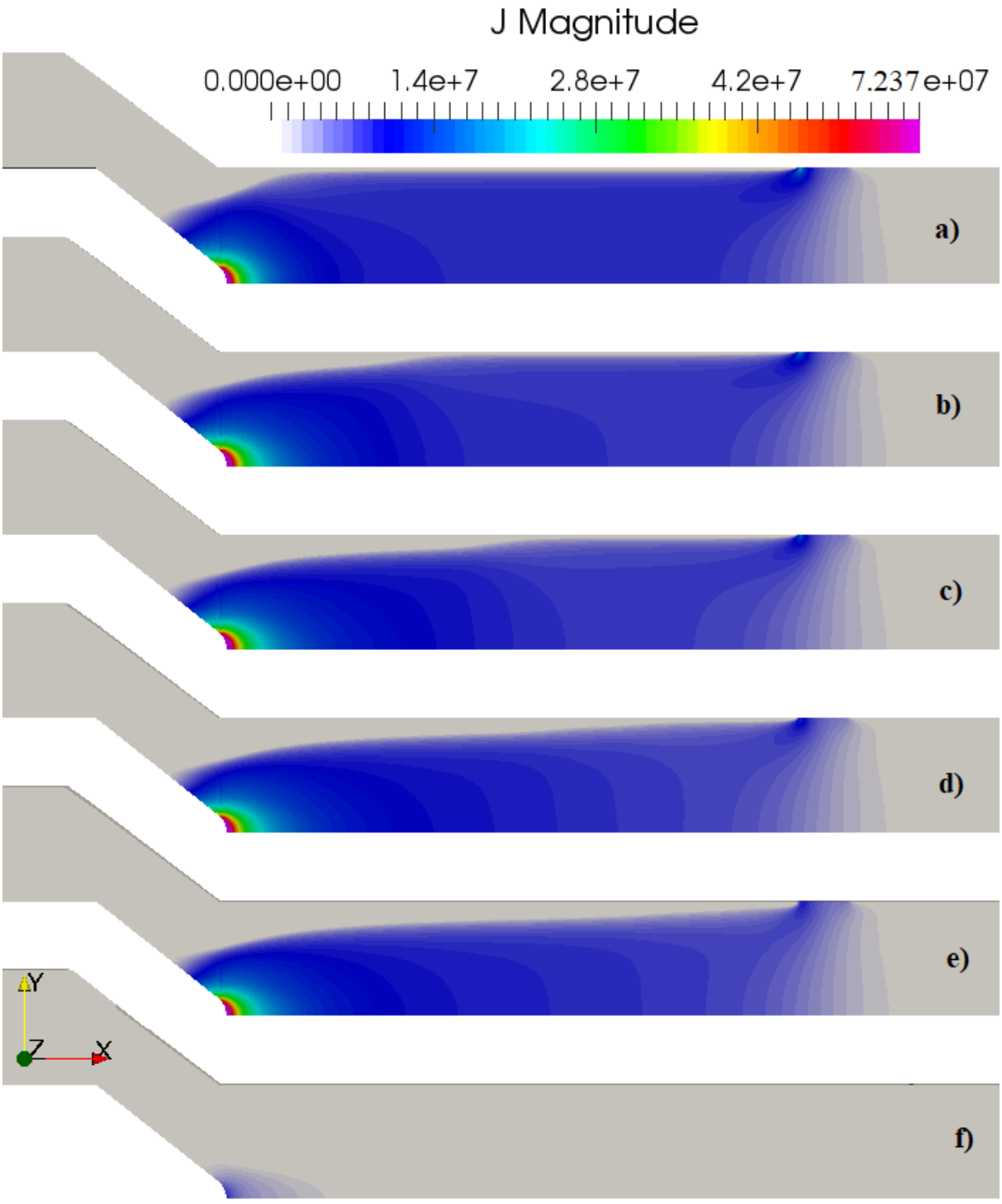


Figure 4.14: Current Density contours for Argon gas at 2.5 sec; a) v=0.5m/sec, b) v=0.75m/sec, c) v=1m/sec, d) v=1.25m/sec, e) v=1.5m/sec, f) v=1.75m/sec

4.3.2 Case 2 - with Air

A similar process, as done with Argon gas was followed for Air. Only difference in the HT_sonicFoam Boundary conditions is that of main anode temperature which is kept at 10,000K due to high specific heat of Air. Below is the temperature contour plot at 1.2 sec.

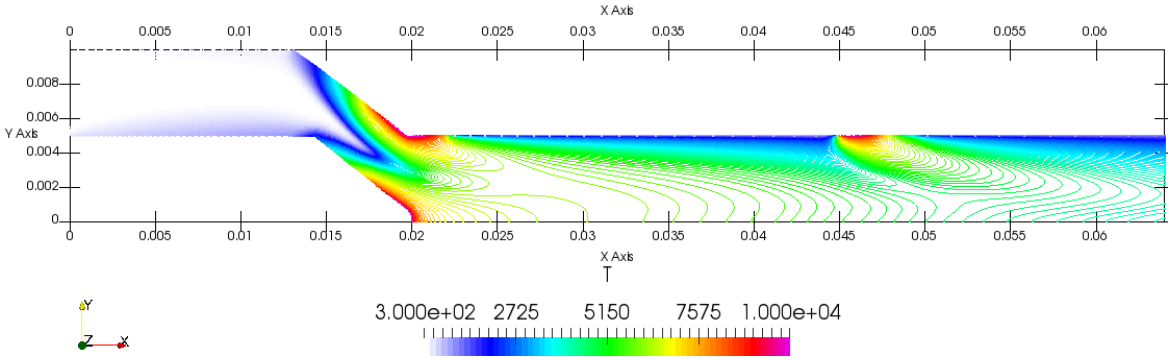


Figure 4.15: Temperature Contour, 1.2 sec

The flow is leaving at an average temperature and velocity of 321K and 18.17 m/sec. The flow fields were then inputted into air_arcFoam and a short arc is formed with a voltage difference of 60 volts. Below are the contours of Current density, Temperature, Electric Potential and Electrical conductivity at 1.5 sec:

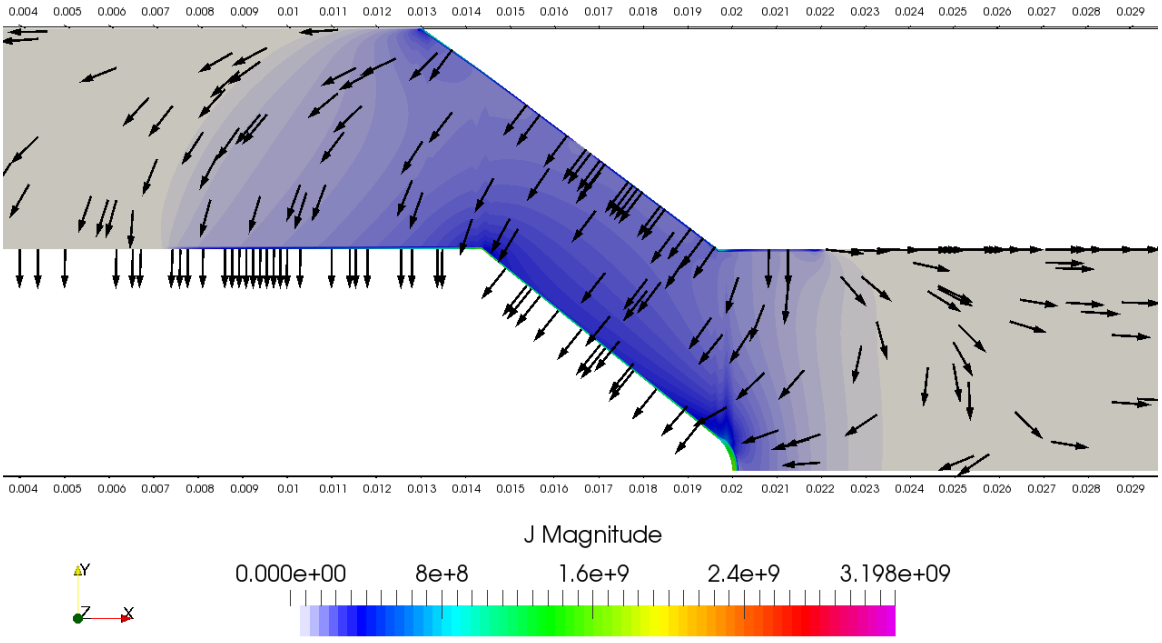


Figure 4.16: Current Density near Electrodes, 1.5 sec

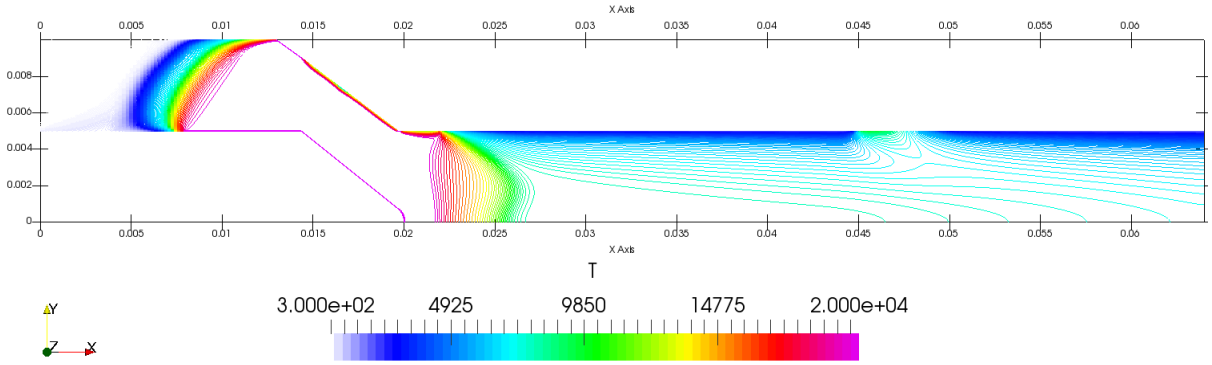


Figure 4.17: Temperature Contour, 1.5 sec

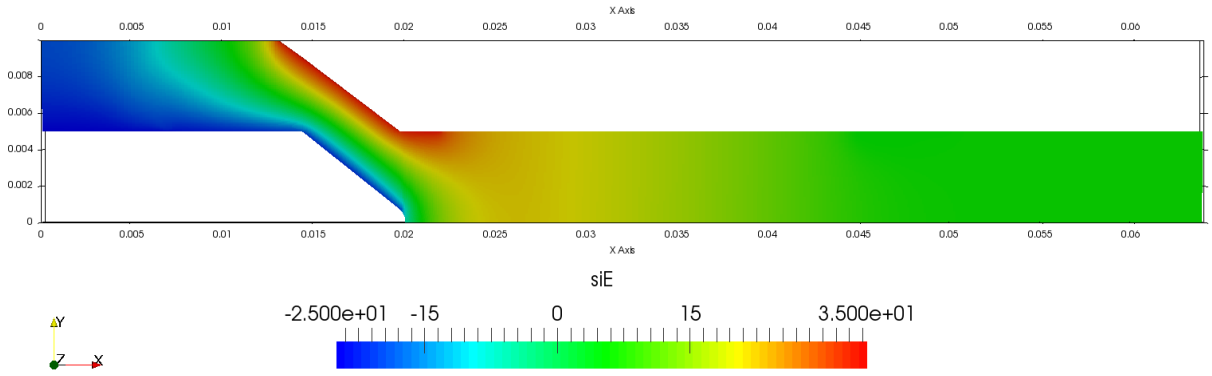


Figure 4.18: Electric Potential, 1.5 sec

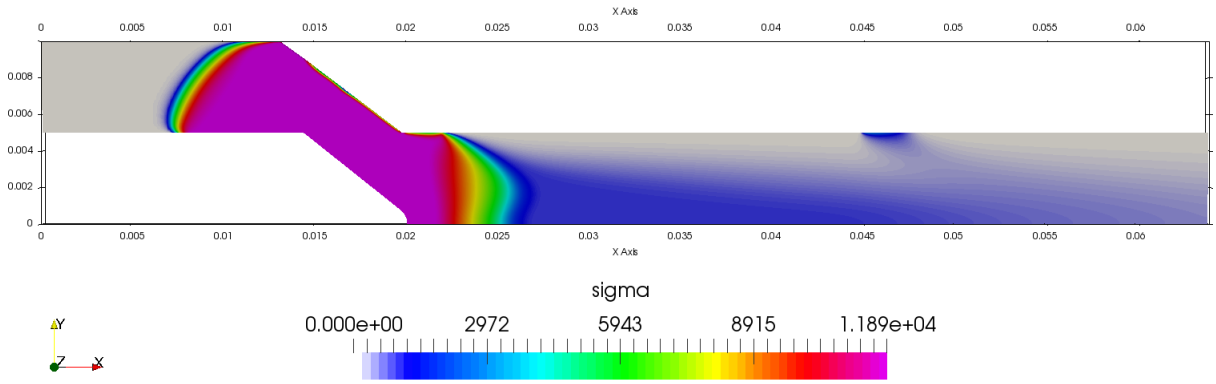


Figure 4.19: Electrical Conductivity, 1.5 sec

Alike the previous case, here too we are using an upper cap of 20,000 K on temperature, since the temperature was exceeding 24,000 limit of tables and blowing up the solution. Comparing the contour of Electrical conductivity for Argon and Air at 1.5 sec, the need for a high Temperature at main anode can be clearly seen. Still with 10,000K at main anode, sigma is not as continuous as for argon. So we increased the potential difference to 100V. Arc was formed for 0.5m/sec but it could not withstand 1m/sec inlet

velocity. So we increased the potential difference Further to 150V. Below are the Contours of Current Density, Temperature, Potential, Conductivity and Velocity obtained at 2.5 sec for an inlet velocity of 0.5m/sec with a potential difference of 150V:

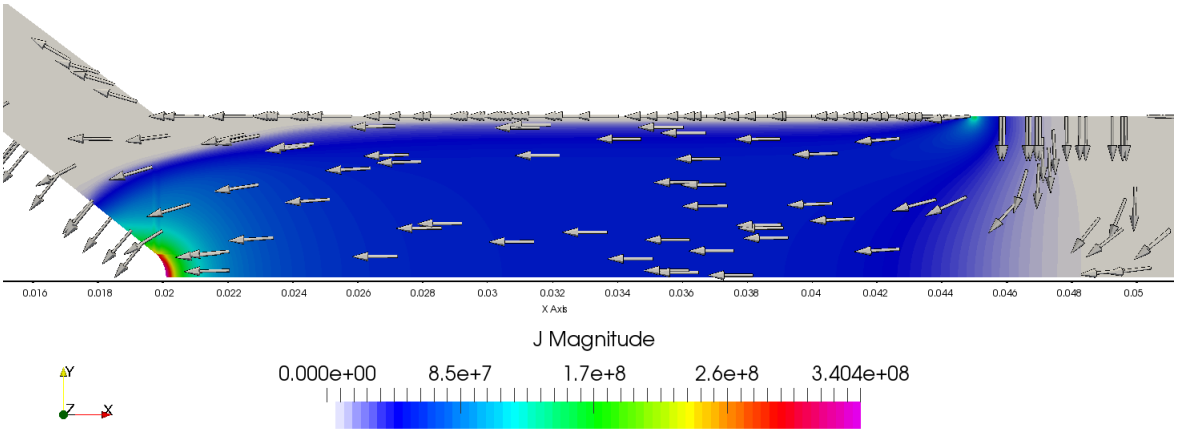


Figure 4.20: Current Density, 2.5 sec

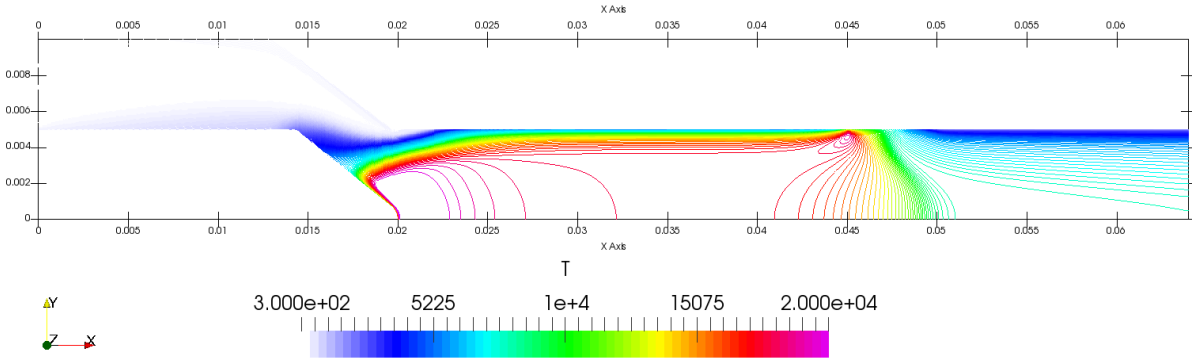


Figure 4.21: Temperature, 2.5 sec

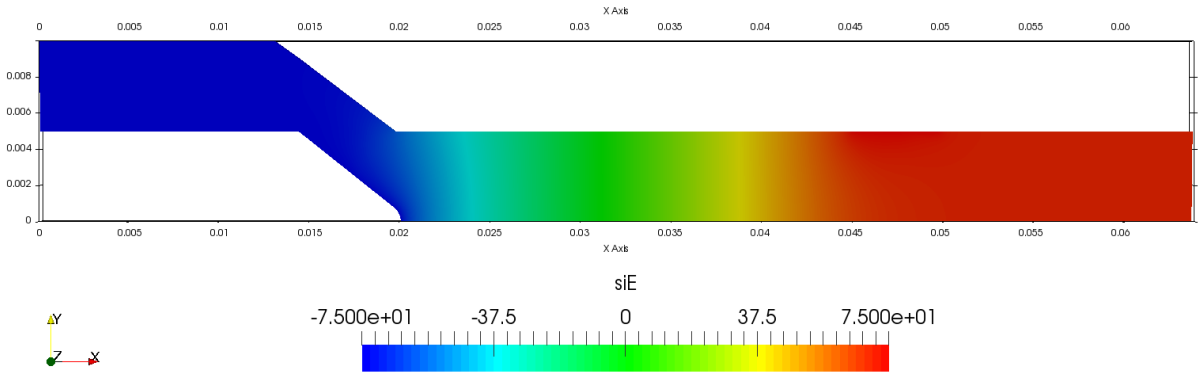


Figure 4.22: Electric Potential, 2.5 sec

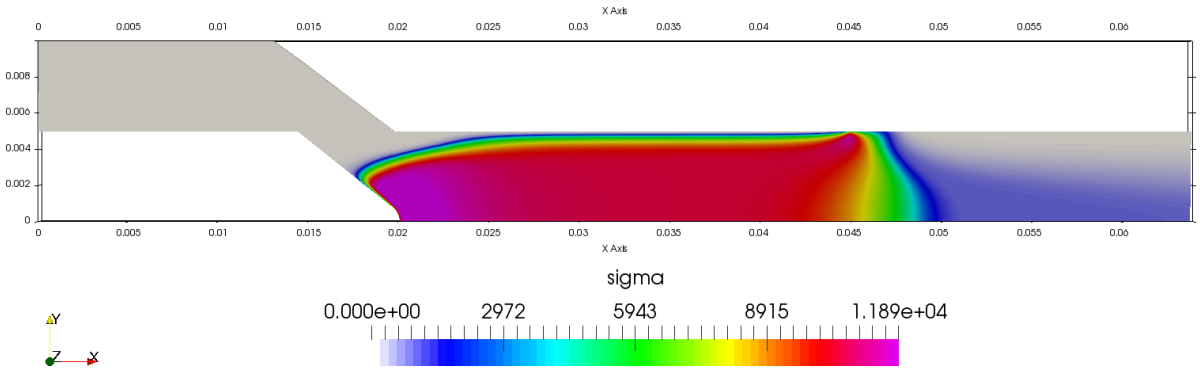


Figure 4.23: Electrical Conductivity, 2.5 sec

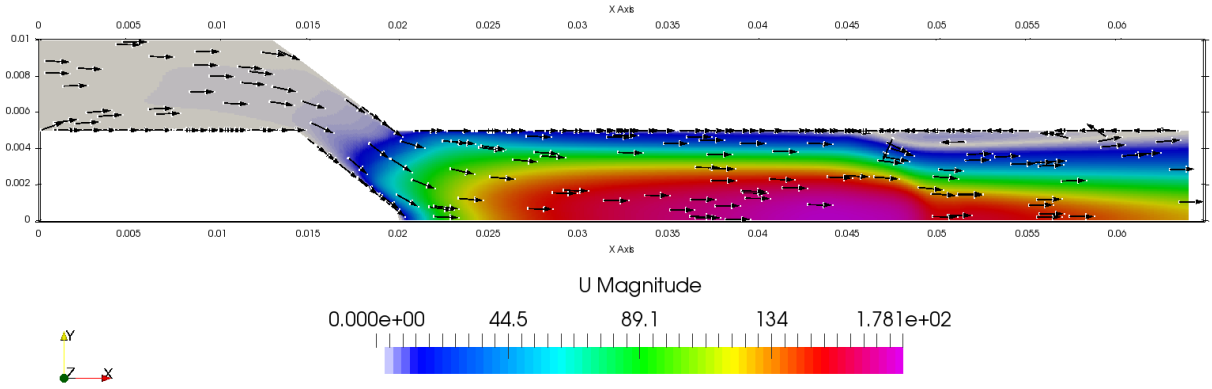


Figure 4.24: Velocity Contour, 2.5 sec

The Maximum value of current density in the case of air is 3.4×10^8 while in the case of argon it was 7.2×10^7. So it is clear we will get a higher value of current for Air. Also the temperature is limited to 20,000K in Air while the maximum temperature obtained was 17,720K for Argon. As a result the radiation losses are going to be much more in the case of Air. These trends are also confirmed by conductivity contours whose maximum value is 9,330 S/m for Argon and 11,890 S/m for Air. This will automatically reduce the efficiency of the tunnel. Thus, we can conclude that tunnel operating with 150V is highly inefficient. Next, As done earlier for Argon, A comparison of current density for a range of inlet velocity from 0.5-1.75 m/sec is made. At 2m/sec Arc could not sustain itself and blows away. Below are the contours obtained for Current Density:

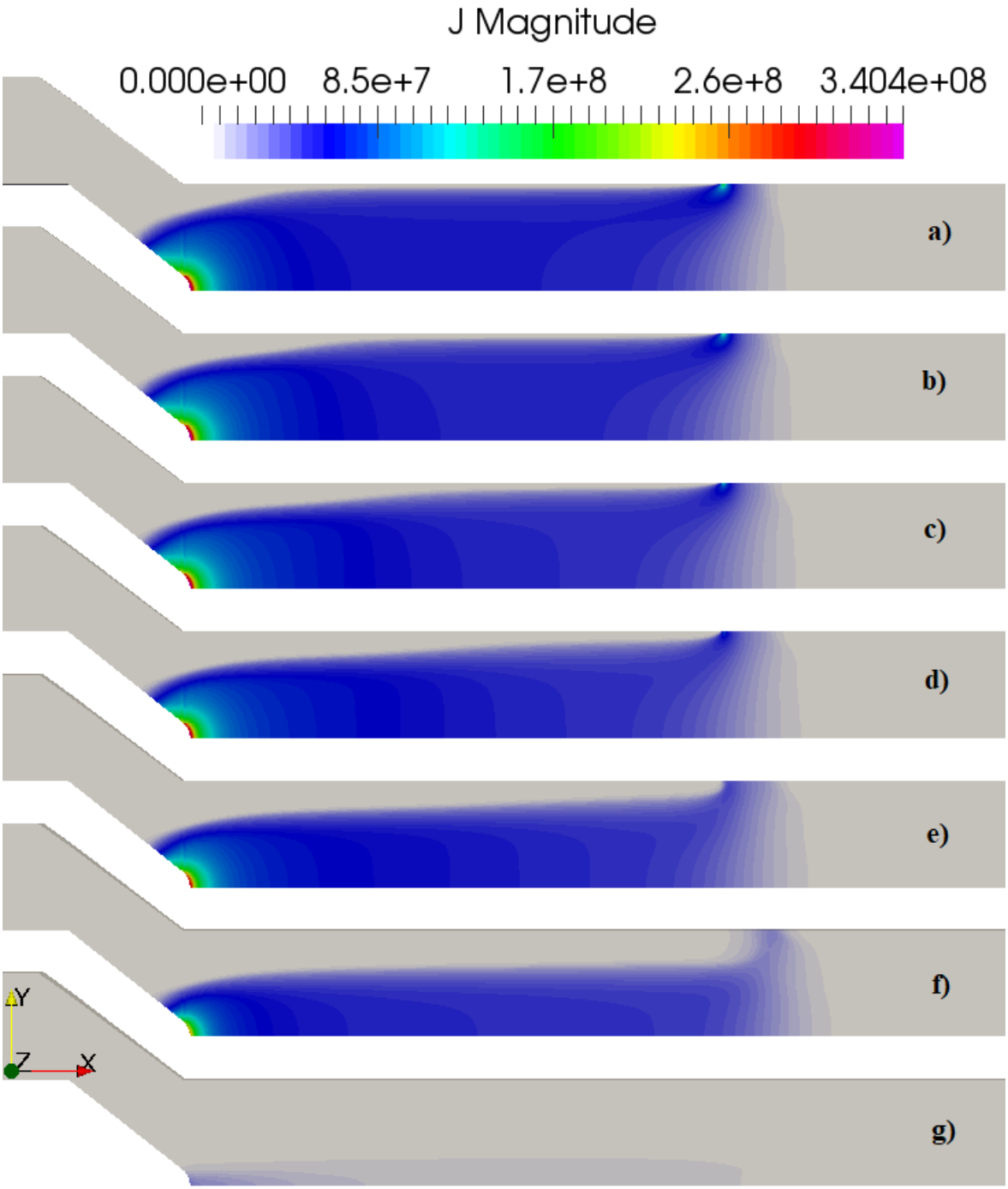


Figure 4.25: Current Density contours for Air at 2.5 sec; a) v=0.5m/sec, b) v=0.75m/sec, c) v=1m/sec, d) v=1.25m/sec, e) v=1.5m/sec, f) v=1.75m/sec, g) v=2m/sec

4.4 Post processing & Verification

For Verification Purposes, mass and electric current conservation were checked. Both of these were calculated by two different processes. Firstly, as we had done till now i.e using sample file extracting the cell centre values along a line, then writing a python script to evaluate desired properties for axisymmetric model. Secondly, using the inbuilt integrate variables filter and calculator of para-View utilising the cell centre values. Second method was rather fast and less cumbersome. Also the integration errors were less compared to the first method where we were using trapezoidal method to evaluate line integrals. So, Mass flow rate was evaluated at inlet and outlet using following formulation created within para-View:

$$\iint_S \rho(y,z) \times V_x(y,z)\, dy\, dz \qquad Mass flowrate\ Calculation \tag{4.1}$$

120 is multiplied since our wedge is of 3 degree and we are considering the axisymmetric model.

	Mass Flow Rate (gm/sec)		
	Inlet	Outlet	Percentage difference
Argon	0.188444	0.188445	Nil
Air	0.137217	0.137235	0.013

Table 4.4: Mass Flow rate for Argon and Air with 0.5 m/sec inlet velocity

In a similar manner Current at two different sections (Near cathode and anode) on YZ plane is evaluated.

$$\iint_S J(y,z)\, dy\, dz, \qquad Current\ Calculation \tag{4.2}$$

	Current (Ampere)		
	X =0.021	X =0.044	Percentage difference
Argon	539.68	542.227	0.47
Air	3367.33	3365.088	0.067

Table 4.5: Current near Cathode and Anode for Argon & Air with 0.5 m/s inlet velocity

The trend is same as expected from the Contours of J for air and Argon. Current is more than six times for Air as compared to Argon. All because of high input voltage

applied and more temperature on main anode. A comparison for Argon with different inlet velocities is done in table 4.6. Efficiency of the tunnel is simply calculated by taking the ratio of Enthalpy gain to Power input.

Velocity(m/s)	0.5	0.75	1	1.25	1.5
Mass flow rate (gm/s)	0.188	0.283	0.379	0.475	0.573
Current (Amp)	540	497	464	426	378
Power Input(KW)	27	24.85	23.2	21.3	18.9
Avg. T at outlet(K)	6966.9	7963.9	8448.3	8755.4	8851.7
Avg. U at outlet(m/s)	39.13	66.42	93.54	120.7	146.36
Avg. ρ at outlet(kg/m^3)	0.089	0.0757	0.0696	0.066	0.0648
R_{e_Avg} at outlet	162.36	214.44	267.04	319.28	377.44
Enthalpy gain(W)	807.5	1505.73	2209.15	2921.47	3564.63
Radiation Loss (W)	20,158	18,311	16,887	15,254	13,179
Conduction Loss(W)	6034.5	5033.3	4103.8	3124.5	2156.9
Efficiency(%)	2.99	6.06	9.52	13.71	18.86

Table 4.6: Effect of increasing inlet Velocity of Argon gas

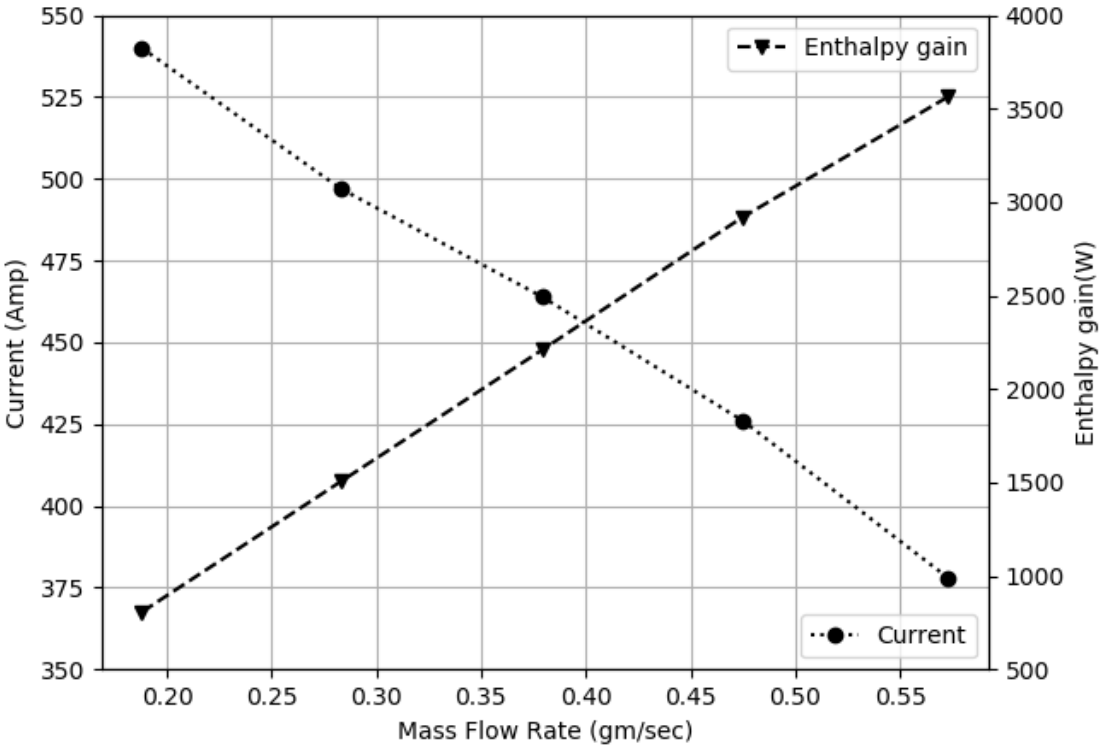


Figure 4.26: Variation of Current and Enthalpy gain with mass flow rate for 100% Argon

From post processed results, It can be seen that with an increment in mass flow rate Current, Power input, Radiation losses & Conduction losses reduce, as a consequence of which more enthalpy at outlet is obtained. It is also intuitional that with an increase in velocity the flow will carry away more heat with itself. Since losses are reducing and enthalpy gain is increasing, efficiency of the tunnel is increasing.

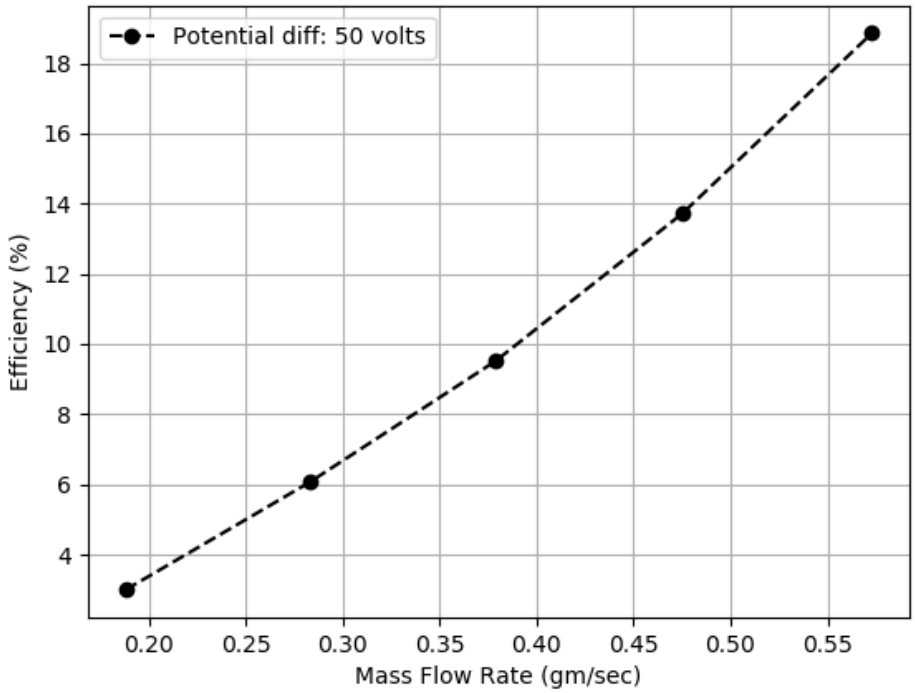


Figure 4.27: Efficiency Vs Mass Flow rate for Argon case

A similar comparison has also been done in Table 4.7 for Air and the effect of increasing inlet velocity on current, enthalpy gain, losses & average values at the outlet has been summarised.

The trends were same as obtained for Argon. Current, Power input, Radiation losses & Conduction losses reduce while Average Temperature, Velocity, and Enthalpy gain increase with increasing mass flow rate. The Efficiency is coming too low, far away from any practical application. This can be attributed to the high value of potential difference which we had picked, leading to a high input current and a large power input to the tunnel. Most of this power input is lost in radiation and conduction through walls. The radiation losses accounts for approx 85% of the power input on an average, thus making the arc-heater highly inefficient.

Velocity(m/s)	0.5	0.75	1	1.25	1.5	1.75
Mass flow rate (gm/s)	0.137	0.206	0.276	0.346	0.416	0.486
Current (Amp)	3366	3130	2842	2500	2070	1402
Power Input(KW)	504.9	469.5	426.3	375	310.5	210.3
Avg. T at outlet(K)	5976	6170.4	6305.6	6413.3	6518.4	6678.8
Avg. U at outlet(m/s)	35.63	55.56	75.25	94.56	113.27	129.09
Avg. ρ at outlet(kg/m^3)	0.0651	0.0625	0.0607	0.0595	0.0583	0.0553
R_{e_Avg} at outlet	134.46	196.12	253.44	307.85	356.62	378.68
Enthalpy gain(W)	3843	6345	8788	11127	13303	15127
Radiation Loss (KW)	427.554	398.985	366.63	326.946	272.97	181.948
Conduction Loss(KW)	73.503	64.17	50.882	36.927	24.227	13.225
Efficiency(%)	0.76	1.35	2.06	2.97	4.28	7.2

Table 4.7: Effect of increasing inlet Velocity of air

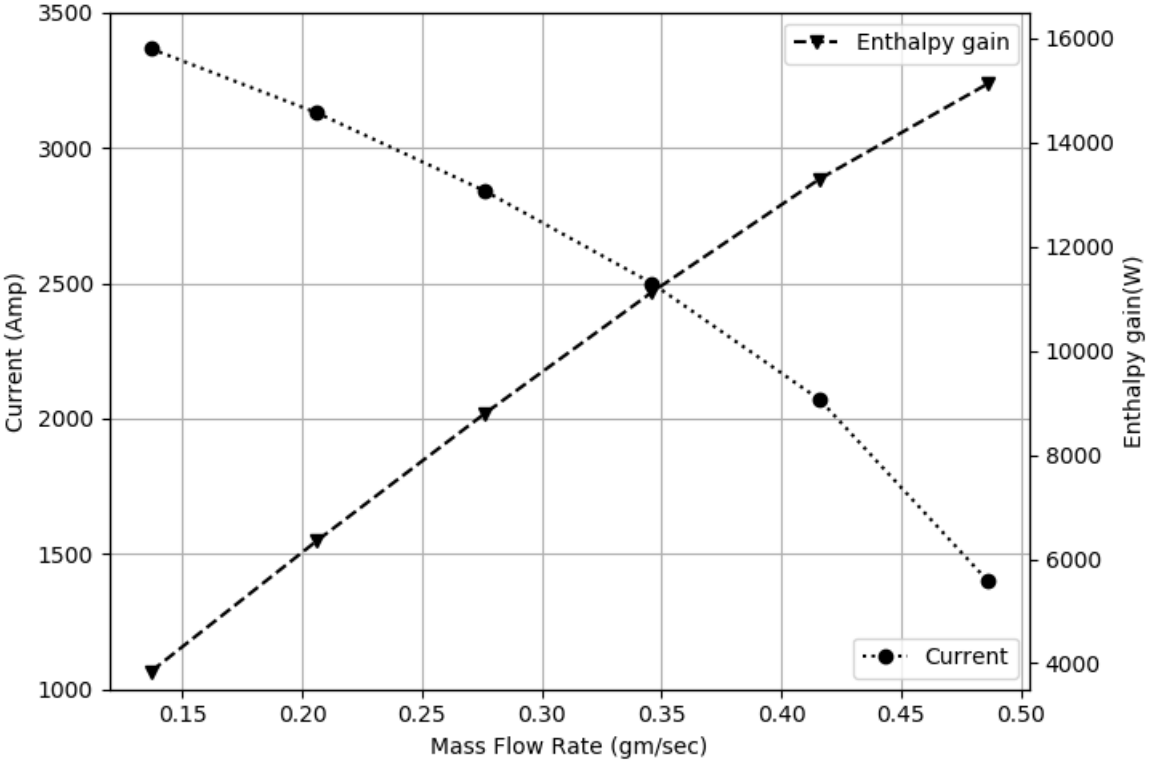


Figure 4.28: Variation of Current and Enthalpy gain with mass flow rate for Air

Chapter 5

3D simulation of Flow through Arc-Heater

After gaining an insight into 2D flow through Arc-heater, we jumped to the 3D flow simulation of the arc-Heater. Unlike the axisymmetric case, here we made the arc to strike at only one section of the main anode by bifurcating the main anode in two sections, one where the arc strikes (a 10 deg patch) and main section (a 350 deg patch). A comparison between the 2D and 3D results was made and VI characteristics of tunnel were studied.

5.1 Geometry and Mesh details

Both geometry and mesh were created using Gmsh, an opensource meshing tool compatible with OpenFOAM. 3D geometry was created by rotating below geometry about line AB, assuming that the flow is entering radially inwards.

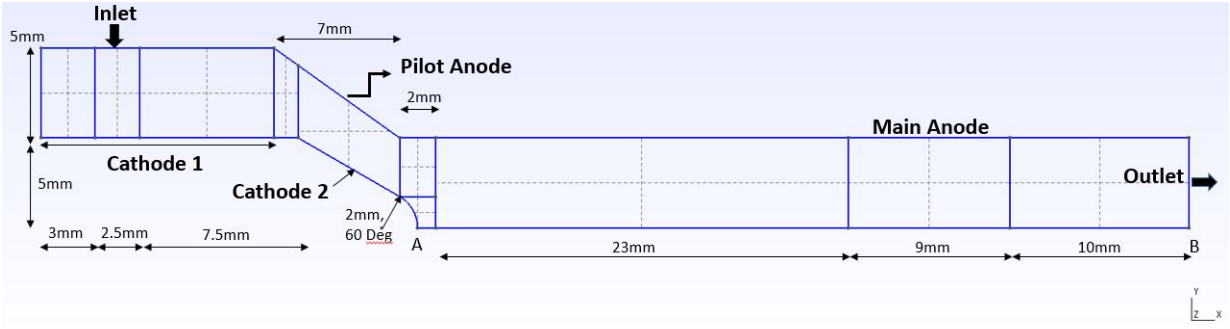


Figure 5.1: 2D view of the arc heater

Number of cells:

A structured mesh was created within Gmsh using the transfinite option, initially a coarser mesh was used with 5,97,240 hexahedra and 7,000 prisms which was later refined to give

18,26,994 hexahedra and 18,644 prisms. Initially number of divisions are 315 in axial direction, 64 in radial direction and 35 in tangential direction. The divisions are increased to 442 in axial direction, 80 in radial direction and 59 in tangential direction.

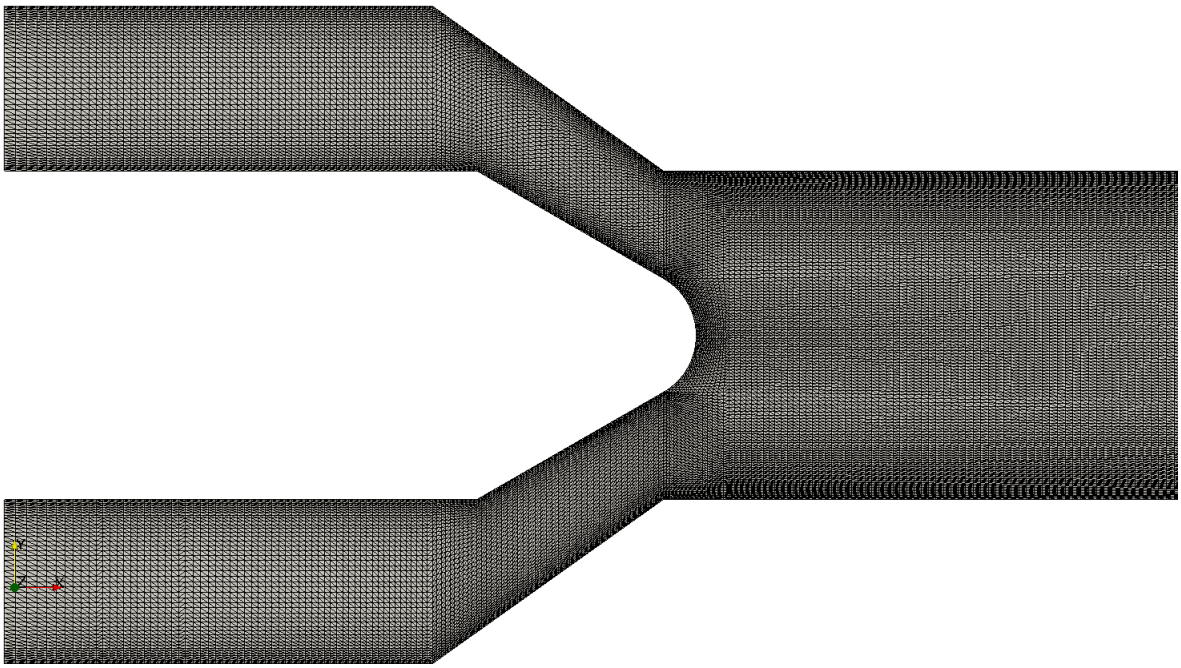
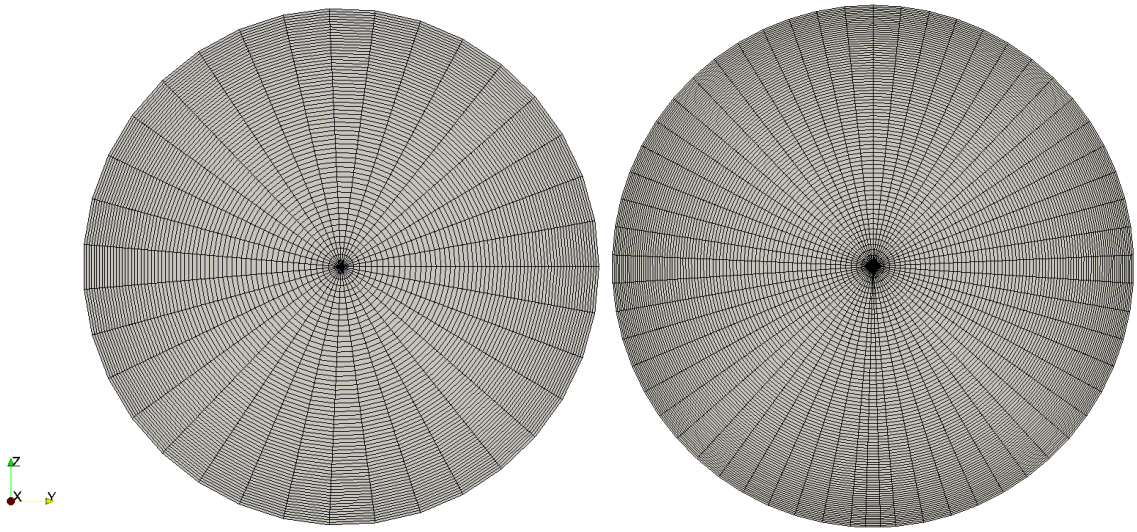


Figure 5.2: Sliced view of mesh along XY plane



(a) Coarser Mesh (sliced in YZ plane) (b) Finer Mesh (sliced in YZ plane)

Mesh Quality

Max aspect Ratio: 38.39

Max skewness :0.89

Max nonorthogonality: 56.79 (Avg. 8.78)

5.2 Case Setup

As previously discussed, a modified version of ArcFoam solver is used to simulate turbulent flow through the arc-heater. Argon gas is taken as working fluid with Perfect gas model for density calculation and Tabular Formulations for Viscosity, Enthalpy, Specific Heat and Thermal conductivity. A comparison between the 2D and 3D models is made. Initially the calculations were done neglecting the lorentz force. Later on, it was proved that it have very negligible effects on 2D simulations.

K-Epsilon in taken as the turbulent model. However the effect of turbulence is not significant and can be neglected, since inlet Reynolds number is too low i.e 815 with the given mass flow rate of 0.435gm/sec.

Work Flow

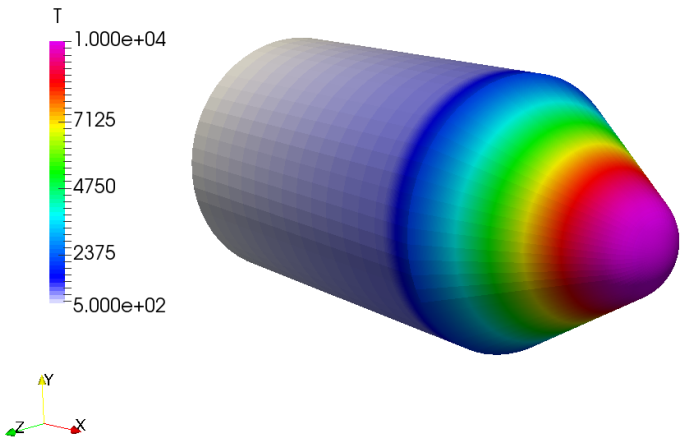
1. Initially the flow is heated through the tunnel using HT_sonicFoam solver which is a basic extension of sonicFoam to extend the temperature ranges upto 24,000K using tables of Enthalpy(J/kg), Specific Heat at constant Pressure(J/kg.K), Viscosity(kg/m.s) and thermal conductivity(W/m.K). At first a lesser mass flow rate of 0.25 gm/sec is taken to facilitate more heat diffusion and heating inside tunnel.

   Compatible wall functions like epsilonWallFunction, kLowReWallFunction, nutkWallFunction, compressible::alphatWallFunction are used along the wall boundaries with a calculated inlet value of $0.0081432 m^2/sec^{-3}$ for K and $0.1725 m^2/sec^{-2}$ for Epsilon.

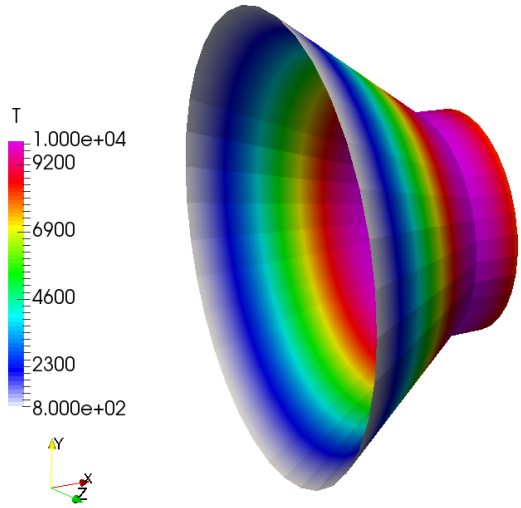
   GroovyBC Boundary conditions are used to give expressions of temperature at Top_wall, Cathode1, Cathode2, Pilot Anode and main anode. Calculations are done at atmospheric pressure condition. Gaussian distribution is used for temperature at both the Cathode2 and Anode . The solution got steady at 1.2 sec. Below are the boundary conditions used for Pressure, Temperature and Velocity:

	Pressure(Pa)	Temperature(K)	Velocity(m/sec)
Inlet	Z.G	300	mass flow Rate:0.00025
Outlet	10^5	Z.G	Z.G
Cathode1	Z.G	Linear Variation from [500-1000]	noSlip
Cathode2	Z.G	Gaussian Distribution Min:1000, Max:10,000, Centre: 0.021	noSlip
Topwall	Z.G	Linear Variation from [300-1000]	noSlip
pilot_anode	Z.G	Gaussian Distribution Min:800, Max:10,000, Centre: 0.02	noSlip
main_anode	Z.G	Gaussian in x, linear in y Min:2500, Max:10,000 Centre:(0.0454, -0.005)	noSlip
left_end_wall	Z.G	Z.G	noSlip
Internal field	101325	300	(0,0,0)

Table 5.1: Boundary Conditions for Pressure ,Temperature & Velocity



(a) Temperature distribution on Cathode



(b) Temperature distribution on Pilot Anode

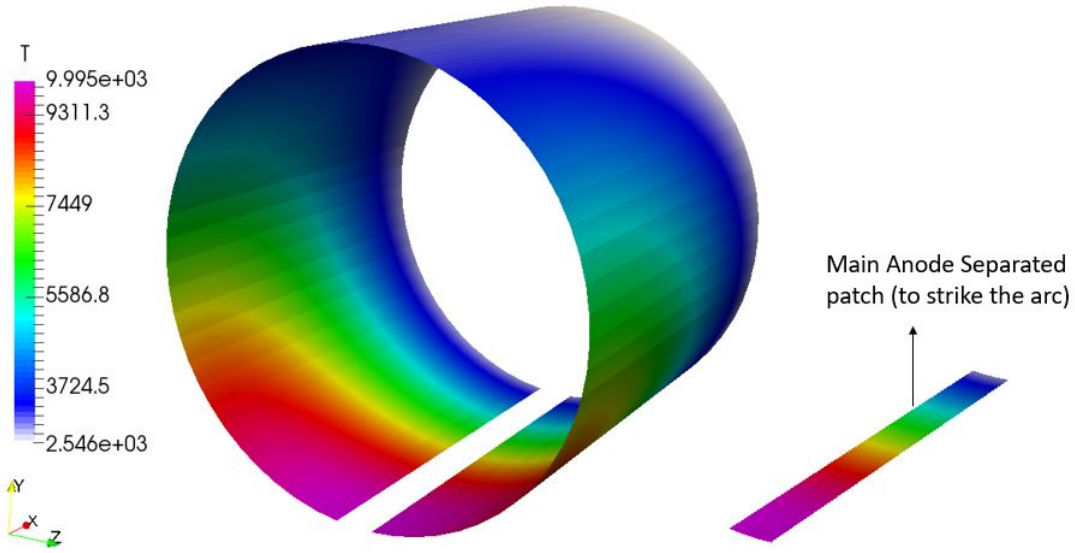


Figure 5.5: Temperature distribution on Main Anode

2. The steady state solution (P, T, U, K, epsilon, nut, alphat) of HT_sonicFoam is passed on to argon_arcFoam using mapfields command where a potential is provided and current density is calculated. This solver is for argon and is taking joule heating and radiation cooling into consideration. Lorentz force is neglected in the momentum equation. A short arc is formed first between Cathode and Pilot Anode. Below are the boundary conditions of Current Density(J), Potential(Sie), Sigma(σ) and Net Emission Coefficient (ϵ)

	Current Density (Amp/m^2)	Potential (Volt)	Sigma (S/m)	Net Emission Coefficient (W/m^3)
Inlet	(0,0,0)	Z.G	Z.G	Z.G
Outlet	Z.G	Z.G	Z.G	Z.G
Cathode1	Calculated	-25	Z.G	Z.G
Cathode2	Calculated	-25	Z.G	Z.G
Topwall	(0,0,0)	Z.G	0	Z.G
Pilot_anode	Calculated	25	Z.G	Z.G
Main_anode	Calculated	Z.G	Z.G	Z.G
Main_anode_ separated_patch	Calculated	5	Z.G	Z.G
Left_end_wall	(0,0,0)	Z.G	0	Z.G
Internal field	(0,0,0)	0	1000	0

Table 5.2: Boundary Conditions for J, Sie, Sigma and net Emission Coefficient

3. At 1.5 sec the arc is shifted from pilot anode to main anode (striking at main anode separated patch). Mass flow rate is changed to original 0.435 gm/sec. Also the values of K and epsilon are changed to 0.02126 m^2/sec^{-3} and 0.72766 m^2/sec^{-2}. Steady Solution was obtained at 2.15 sec. Below are the modifications done in the boundary conditions:

	Temperature	Potential	Current Density
Pilot_anode	Consistent with Top wall	Z.G	(0,0,0)
Main_anode_ separated_patch	No Change	25	No change

Table 5.3: Changes done in Boundary Conditions for shifting Arc

4. At 2.15 sec there were discrepancies between the cell centre values and the nodal values because of Coarser Mesh we were using. So a finer mesh was created with the details as mentioned in section 5.1. The results of Coarser mesh were mapped into the finer Mesh and the simulation was run till 2.3 sec.

5. Lastly, For a constant mass flow rate of 0.435 gm/sec, we studied the VI characteristics of tunnel from range of 40V-100V.

   In all the above simulations we used parallel processing. The domain was decomposed using decomposeParDict file present in system folder. The issue faced was discontinuities in the current density field which can be sorted out either by running the last time step in serial processing or by changing the decomposed domain at any intermediate time step. For Example: one can decompose initially in x direction and after few time steps can decompose in some other direction

5.3 Results

At 1.2 sec the solution of HT_sonicFoam got steady and residuals were with in limits. The flow was completely heated up inside the arc_heater and after this the fields are ready to be mapped in argon_arcFoam. Below is the contour plot of the Temperature:

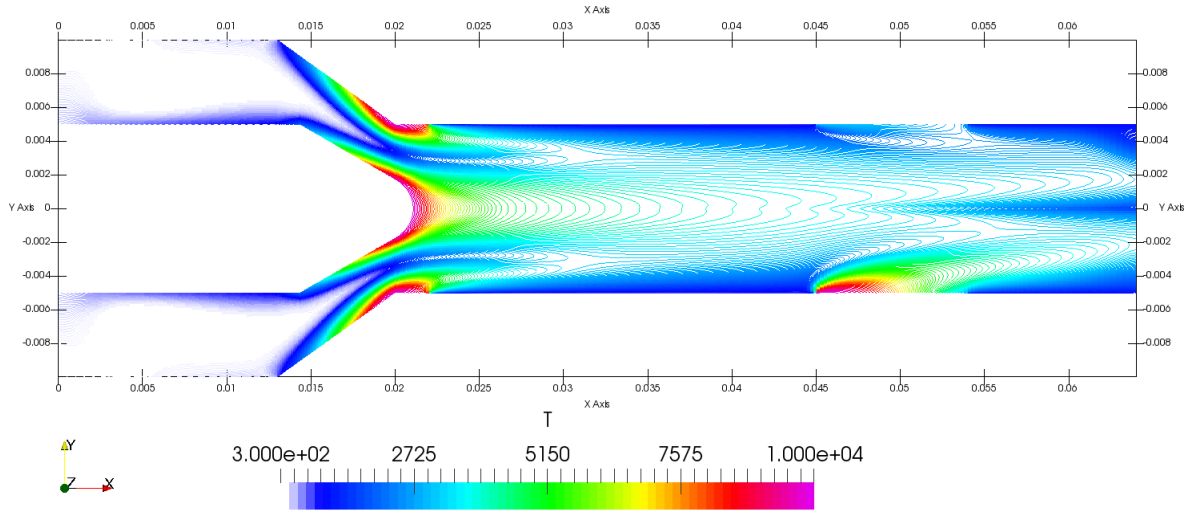


Figure 5.6: Temp. Contour,1.2 sec (sliced in xy plane)

At 1.2 sec boundary conditions of Current density, Potential, Electrical Conductivity and Net Emission coefficient were applied and a short arc between cathode and pilot anode is formed. In actual arc should strike at a point but since we are giving axisymmetric boundary conditions it is striking all over the pilot anode.

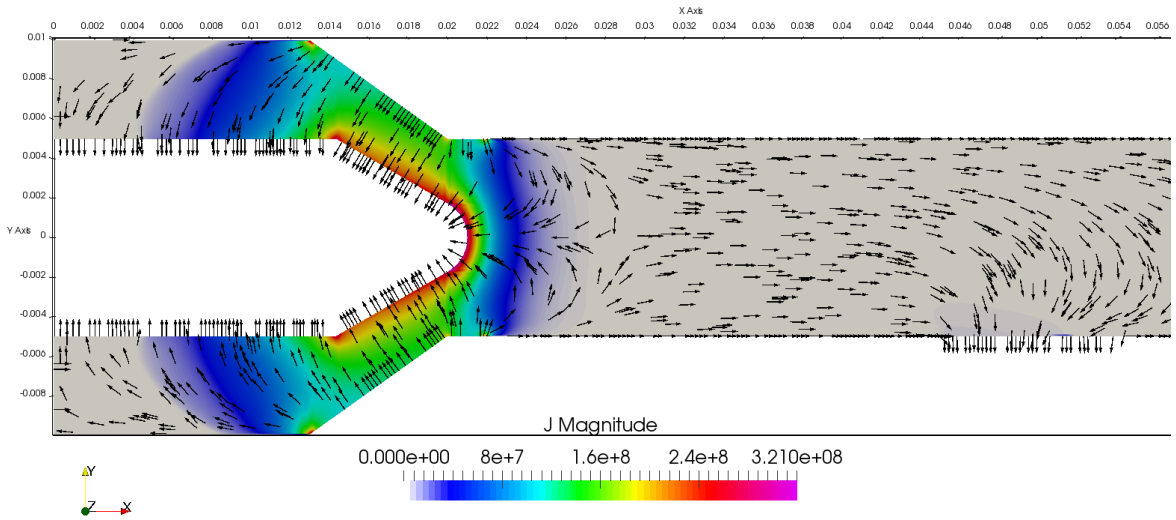


Figure 5.7: Current Density Contour,1.5 sec (sliced in xy plane)

The main purpose of this short arc is to heat the flow further inside the tunnel to facilitate the longer arc between the cathode and main anode. It is evident from the Contour plots of Temperature and Electrical Conductivity that the sufficient temperature is reached and now we can shift the arc. Below are the Contour Plot of Temperature and clipped view of Potential Field and Electrical Conductivity.

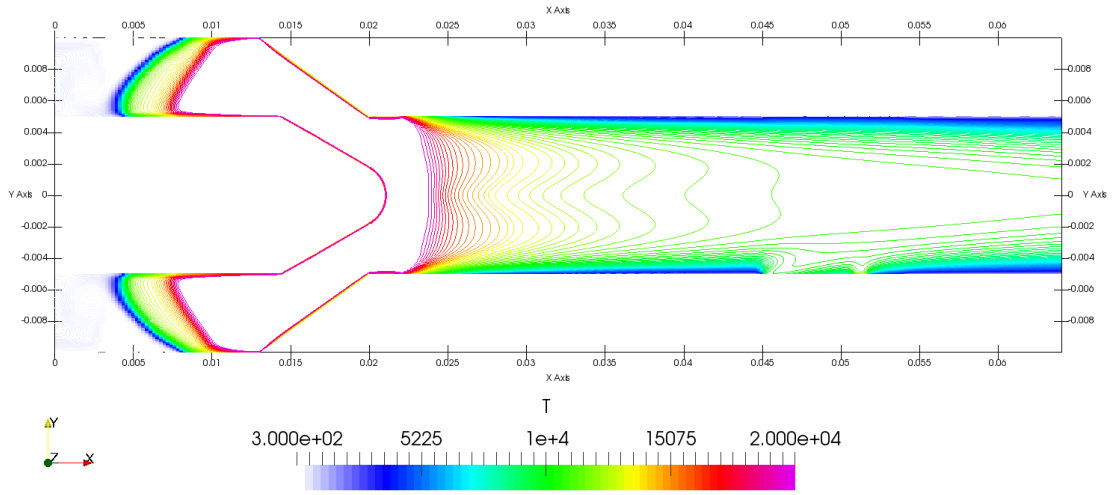


Figure 5.8: Temperature Contour,1.5 sec (sliced in xy plane)

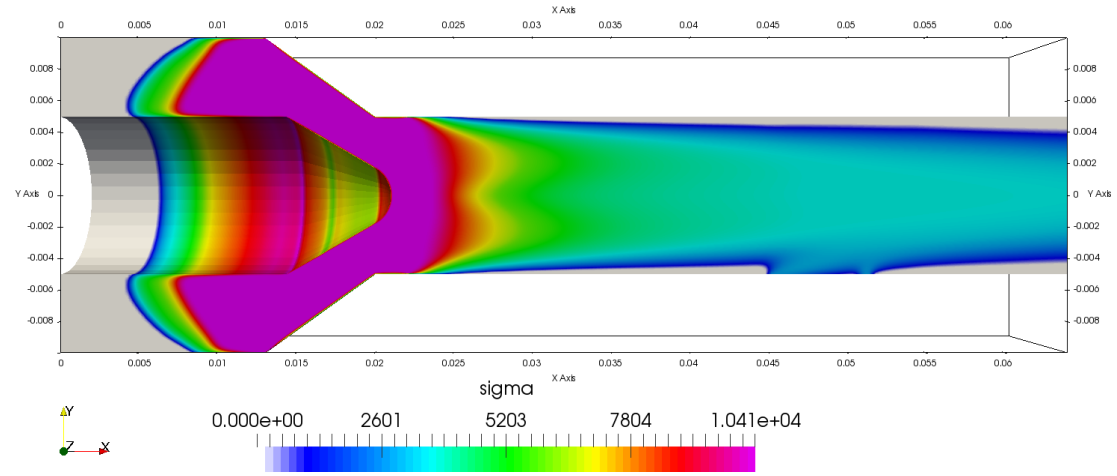


Figure 5.9: Electrical Conductivity ,1.5 sec (clipped in xy plane)

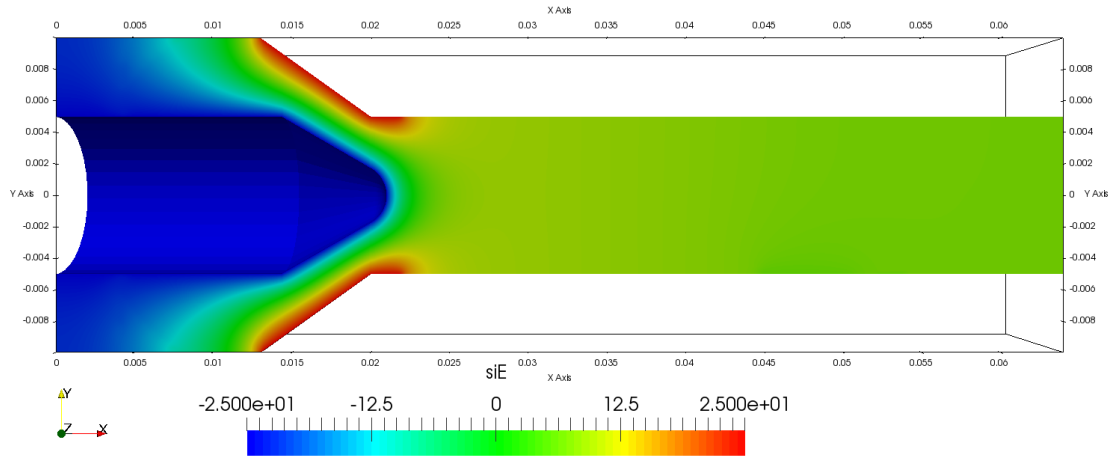


Figure 5.10: Electric Potential,1.5 sec (clipped in xy plane)

As previously mentioned, At 1.5 sec, the mass flow is increased to 0.435 gm/sec and the boundary conditions were changed to make the arc strike at a separated patch. The results obtained with the finer mesh are as below:

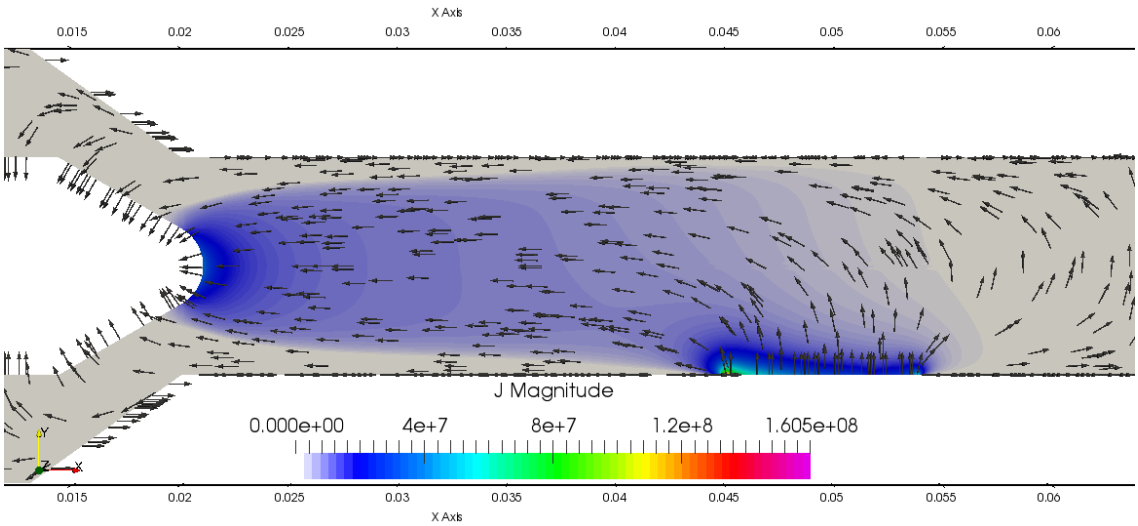


Figure 5.11: Current Density, 2.3 sec (sliced in xy plane)

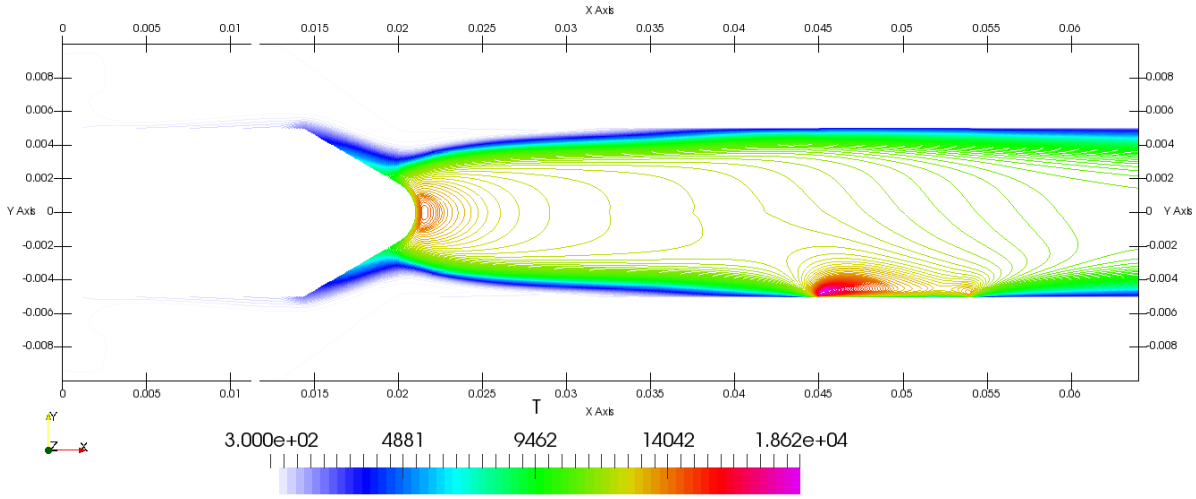


Figure 5.12: Temperature Contour, 2.3 sec (sliced in xy plane)

It can be seen that maximum current density is of order 10^8 and is occurring near electrodes. Also the temperature is as high as 18,620K near the main anode separated patch and is of order of 9,000 K behind the arc. Arc shape can be more easily visualized using Iso_Volumes. So a threshold of Electrical Conductivity from 4500-8000 S/m have been created in para-View.

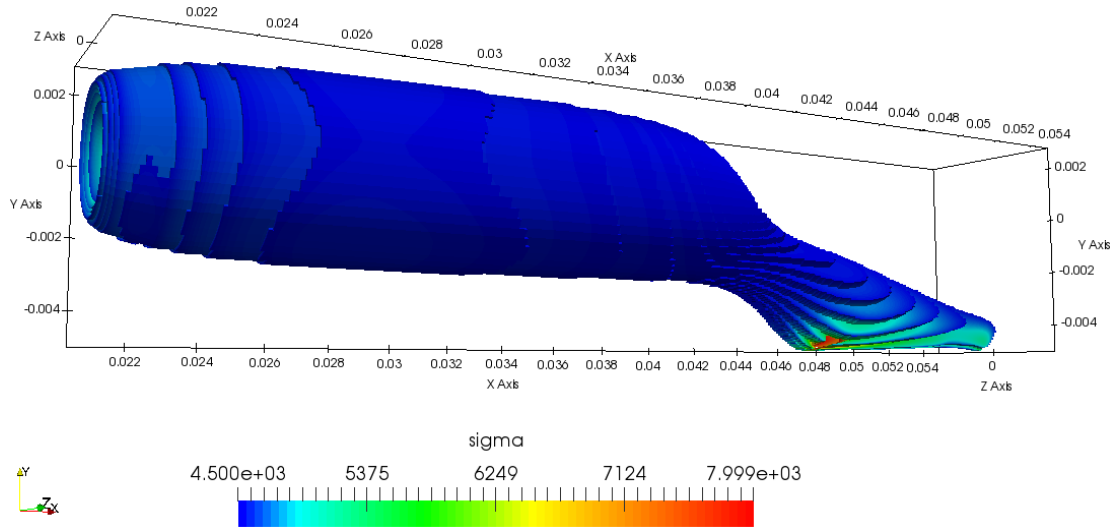
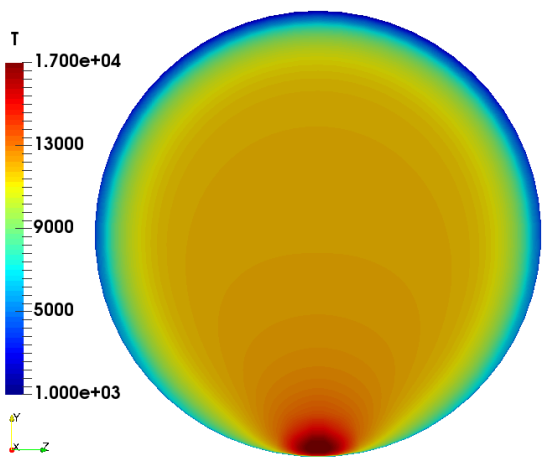
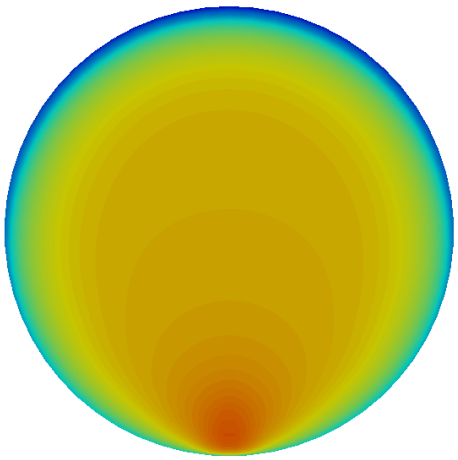


Figure 5.13: Electrical Conductivity Iso-Volume, 2.3 sec

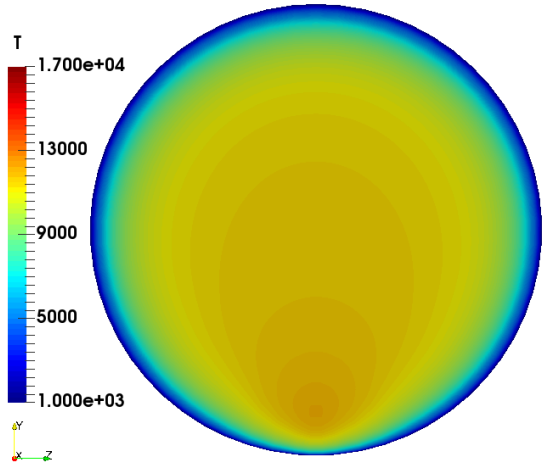
Arc is circular near the cathode which then shrink down to the anode patch forming a wedge type shape. More insight can be gained through the Temp. contours in YZ plane.



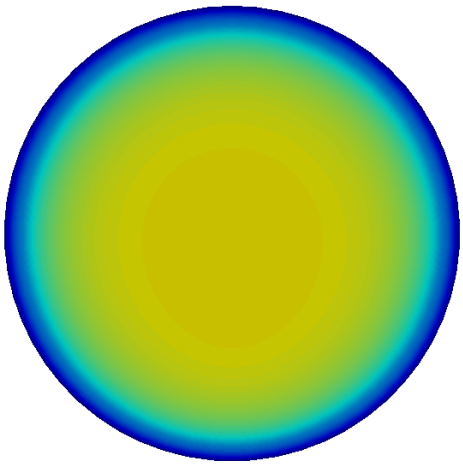
(a) YZ plane, x=0.045m,2.3 sec



(b) YZ plane, x=0.05m,2.3 sec



(c) YZ plane, x=0.055m,2.3 sec



(d) YZ plane, x=0.064m,2.3 sec

With Applied potential difference of 50 Volts, the contour plot of Electrical conductivity, Electric Potential and Velocity are as below:

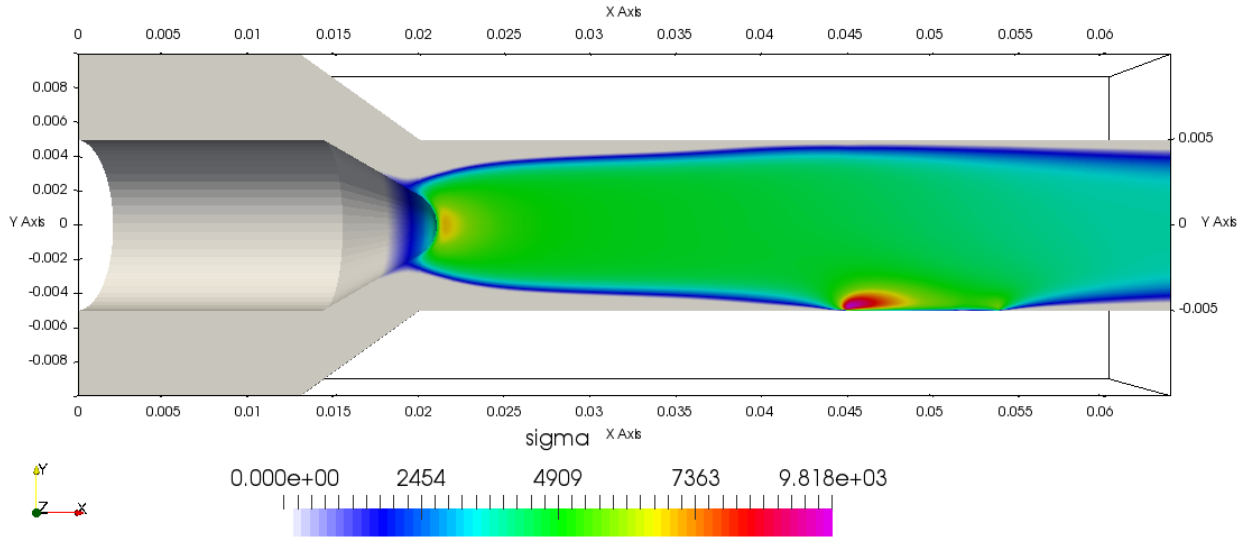


Figure 5.15: Electrical Conductivity, clipped view, 2.3 sec

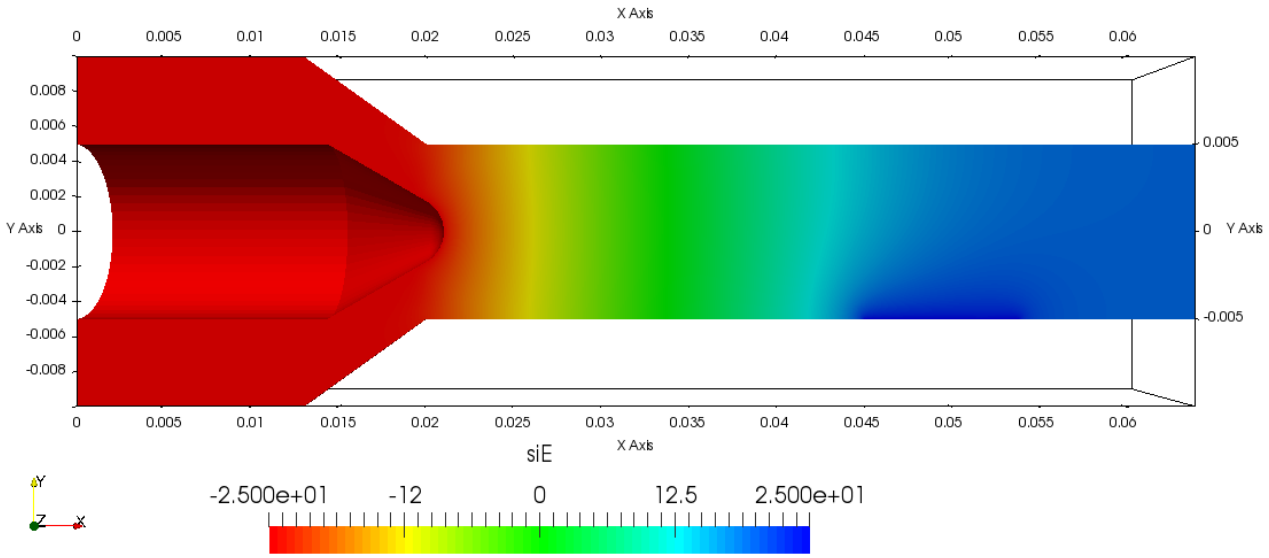


Figure 5.16: Electric Potential, clipped view, 2.3 sec

Sigma Contours show a very high region of conductivity close to 9,818 S/m near the main anode patch. Potential contours are rather continuous with a potential of 25 at anode and -25 at cathode. Inlet velocity is 1.7 m/sec with a pressure of 101000 pascal. Maximum velocity is reached at outlet and is having a value of 215 m/sec.

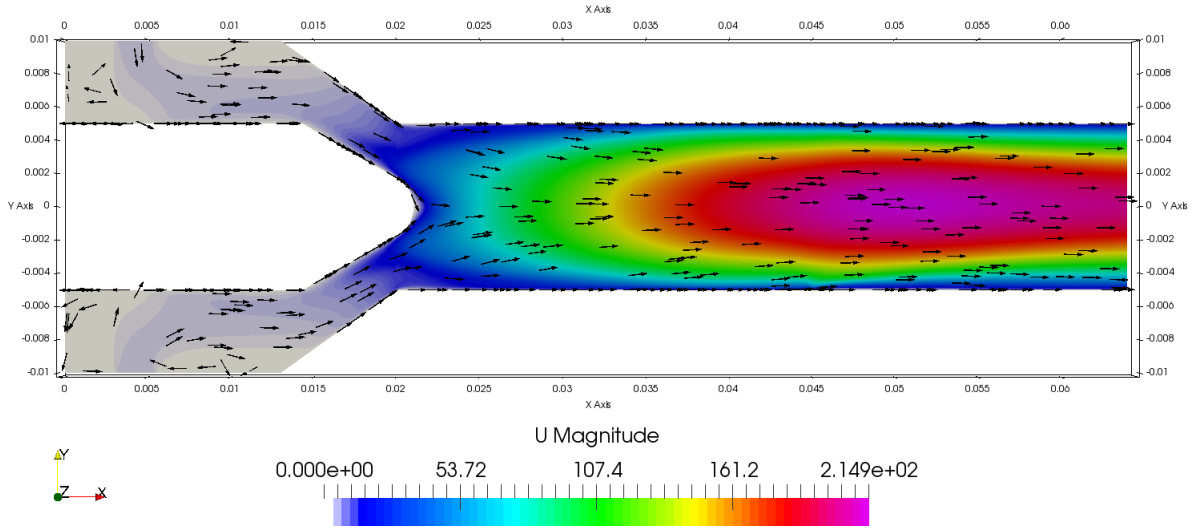


Figure 5.17: Velocity, 2.3 sec (sliced in xy plane)

5.4 Post Processing & Verification

For Verification Purposes, mass and electric current conservation were checked. Mass flow rate was evaluated at inlet & outlet using Cell data and Integrate variable filter of paraView. Inbuilt calculator was used & following formula was created within paraView:

$$\iint_S \rho(y,z) \times V_x(y,z)\, dy\, dz \qquad Mass flowrate\ Calculation\ at\ outlet \tag{5.1}$$

Mass Flow Rate (gm/sec)		
Inlet	Outlet	Percentage Error
0.435	0.43503	0.007

Table 5.4: Mass Flow rate at inlet and outlet

In a similar manner Current at two different sections (Near cathode and anode) on YZ plane is evaluated.

$$\iint_S J(y,z)\, dy\, dz, \qquad Current\ Calculation \tag{5.2}$$

Current (Ampere)		
x=0.021	x=0.044	Percentage difference
326.635	327.644	0.38

Table 5.5: Current near cathode and anode

A comparison between the 3D case results and 2D case (axisymmetric) results was done in table 5.6. Since in axisymmetric case it is assumed that arc is striking all over the anode, Current and radiation losses are coming significantly higher than the 3D case. However the Enthalpy gain is nearly same in both the cases. This leads to a lower efficiency in 2D axisymmetric case.

	2D Case (with Lorentz Force)	2D Case (without Lorentz Force)	3D case (without Lorentz Force)
Current (Amp)	481.1	481.8	327
Power Input(Watts)	24055	24090	16350
Joule Heating(Watts)	24085	24103	16409
Avg. T at outlet(K)	8658	8646	8738
Avg. U at outlet(m/s)	110.83	110.10	109.627
Avg. ρ at outlet(kg/m^3)	0.0672	0.0673	0.0663
R_{e_Avg} at outlet	300.67	298.96	291.7
Enthalpy gain	6183.01 kJ/kg = 2689.61 W	6113.37 kJ/kg = 2659.32 W	6125.36 kJ/kg = 2664.51 W
Radiation Losses (W)	17897	17896.37	10782.97
Conduction Losses (W)	3468.3	3534.3	2902.52
Efficiency(%)	11.18	11.04	16.29

Table 5.6: A comparison between axisymmetric and 3D results

Axisymmetric case with and without Lorentz force was also compared in table 5.6 and it is clearly visible that there is very minute difference between both.

	40 V	50 V	60 V	70 V	80 V	90 V	100 V
Current (Amp)	194	327	479	657	859	1099	1401
Power Input(Watts)	7,760	16,350	28,740	45,990	68,720	98,910	1,40,100
Avg. T at outlet(K)	8,542	8,738	8,816	8,978	9,066	9,145	9,234
Avg. U at outlet(m/s)	105.5	109.6	111.3	113.7	115.3	117.2	119.6
Avg. ρ at outlet(kg/m^3)	0.0683	0.0663	0.0657	0.0638	0.0627	0.0619	0.0611
R_{e_Avg} at outlet	293.34	291.7	291.3	286.2	283.3	283.1	283.4
Enthalpy gain(W)	2488.1	2664.5	2741.2	2855.7	2955.9	3081.3	3268.9
Radiation Losses(W)	4,260	10,783	20,450	34,476	52,429	75,423	1,04690
Efficiency(%)	32.06	16.29	9.54	6.21	4.3	3.12	2.33

Table 5.7: Effect of changing Input Voltage with constant mass flow rate

At last, we studied the VI characteristics of tunnel from range of 40-100V. Current, radiation losses and enthalpy gain are increasing with voltage. However, radiation losses are increasing at a significantly higher rate than enthalpy gain at outlet, resulting in the loss of efficiency of tunnel being maximum at 40 Volt.

At outlet, average temperature and velocity are increasing with voltage while density is decreasing. Viscosity of argon increases with temperature in the above region. Overall effect of which is a decreasing trend in Reynolds's number at outlet.

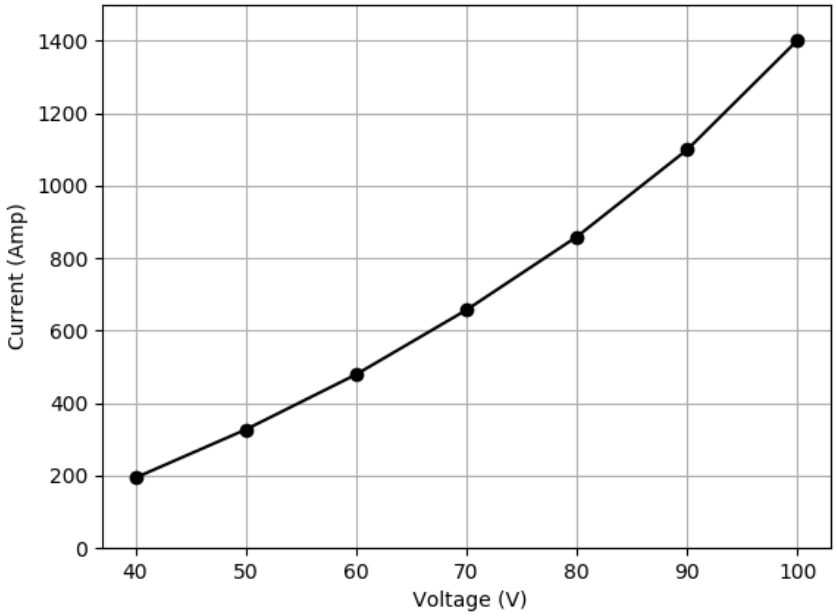


Figure 5.18: VI Characteristics

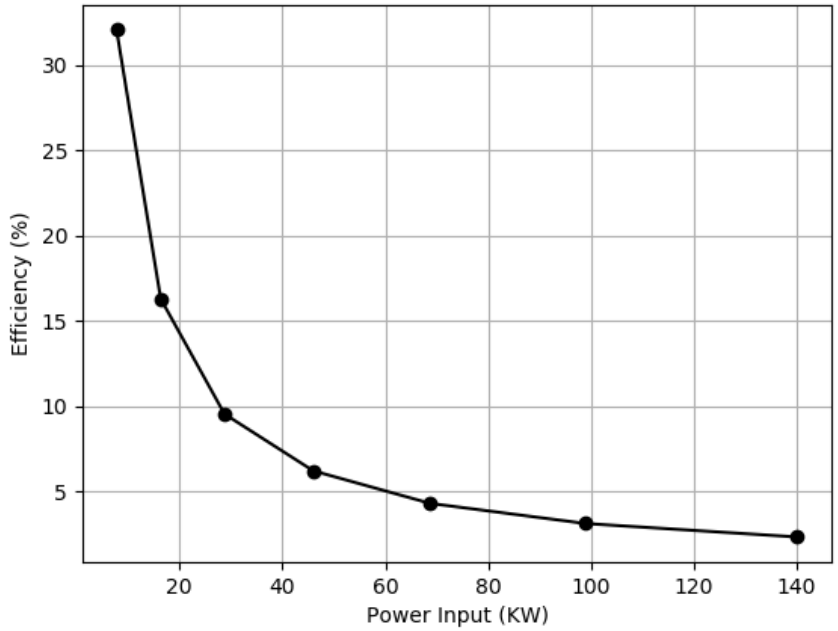


Figure 5.19: Efficiency Vs Power Input

Chapter 6

Conclusion and Future Work

6.1 Conclusion

In this dissertation we have successfully simulated flow through conical CD nozzle for inviscid & viscous cases, accounting for turbulence in the flow field. Results of nozzle flow with different wall conditions were verified either with quasi 1D theory or experimentally. In all the cases SonicFoam worked perfectly well and captured shock quite accurately. Then we moved to arc heated flow between two parallel plates, where we tried to simulate the flow with and without Lorentz force. Shockingly, as soon as Lorentz force was applied the flow get reversed and a supersonic jet directed towards inlet arise from electrodes. On investigation it was found that this was happening because of asymmetric Lorentz Force arising due to nonuniform magnetic Field.

Arc Physics depends largely on the electrode boundary conditions. So, we modified the thermal boundary conditions to account for convective cooling by external fluid and radiative heating loss. The solver has been modified to calculate current density based on fixed applied potential. Linear curve fits of electrical conductivity and net radiation coefficient were changed to polynomial curve fits. Also, solver is changed to take air as working fluid. Effects of increasing mass flow rate in 6.4 cm, VSSC arc heater with argon and air were investigated. For both air and argon, while Enthalpy gain increases with an increment in flow velocity, Current decreases. Lastly, a comparison between the axis-symmetric and 3D model for the VSSC arc heater was done. A higher value of current, thus more power input was coming in the axis-symmetric case. Although the enthalpy gain was comparable with each other. This was mostly because of high radiation and conduction losses in 2D model. VI characteristics for 3D case was studied and it was found that current increases almost linearly with Potential difference. Also to gain

maximum efficiency the tunnel should operate at minimum voltage and maximum mass flow rate.

6.2 Future Work

So far we were considering the ideal gas model for density calculation. It can be changed to Capitelli EOS. Following the process mentioned in chapter 4 & 5 arc can be shifted between the pilot cathode and main anode for 60cm arc heated tunnel. The effects of swirl on the arc remain untouched and can be studied next.

References

[1] Back, L.H., Massier, P.F. and Gier, H.L., 1963. Convective heat transfer in a convergent-divergent nozzle. International Journal of Heat and Mass Transfer, 7(5), pp.549-568.

[2] Back, L.H., Gier, H.L. and Massier, P.F., 1964. Comparisons of experimental with predicted wall static-pressure distributions in conical supersonic nozzles.

[3] Sass-Tisovskaya, M., 2009. Plasma arc welding simulation with OpenFOAM (Doctoral dissertation, Chalmers University of Technology).

[4] Ved Mukherjee,2017. Development of ArcFoam for Simulation of Arc Heated Wind Tunnels, AeSI symposium Bangalore

[5] Lim, S., Cheon, J.H., Son, M.J. and Shin, E.S.,2018. Quantitative analysis of surface recession on carbon-based ablators using a high-resolutionnon-contact profilometer,Materials & Design, 149, pp.73-80.

[6] Litchford, R., Cole, J., Lineberry, J., Chapman, J., Schmidt, H. and Lineberry, C., 2002. Magnetohydrodynamic augmented propulsion experiment: I. performance analysis and design. In 33rd Plasmadynamics and Lasers Conference (p. 2184).

[7] Marcantoni, L.G., Tamagno, J.P. and Elaskar, S.A., 2012. High speed flow simulation using openfoam. MecÃąnica Computacional, 31, pp.2939-2959.

[8] Lebouvier, A., Delalondre, C., Fresnet, F., Boch, V., Rohani, V., Cauneau, F. and Fulcheri, L., 2011. Three-dimensional unsteady MHD modeling of a low-current high-voltage nontransferred DC plasma torch operating with air. IEEE Transactions on plasma science, 39(9), pp.1889-1899.

[9] Xu, G., Hu, J. and Tsai, H.L., 2008. Three-dimensional modeling of the plasma arc in arc welding. Journal of Applied Physics, 104(10), p.103301.

[10] Capitelli, M., Colonna, G., Gorse, C. and d'Angola, A., 2000. Transport properties of high temperature air in local thermodynamic equilibrium. The European Physical Journal D-Atomic, Molecular, Optical and Plasma Physics, 11(2), pp.279-289.

[11] Kim, K.S., Park, J.M., Choi, S., Kim, J. and Hong, S.H., 2008. Comparative study of two-and three-dimensional modeling on arc discharge phenomena inside a thermal plasma torch with hollow electrodes. Physics of Plasmas, 15(2), p.023501.

[12] Trelles, J.P., Chazelas, C., Vardelle, A. and Heberlein, J.V.R., 2009. Arc plasma torch modeling. Journal of thermal spray technology, 18(5-6), p.728.

[13] Geuzaine, C. and Remacle, J.F., 2009. Gmsh: A 3D finite element mesh generator with builtin pre and post processing facilities. International journal for numerical methods in engineering, 79(11), pp.1309-1331.

[14] OpenFOAM User Guide The Open Source CFD Toolbox, https://www.openfoam.com/, version 4.0,2016

[15] Anderson Jr, J.D., 2010. Fundamentals of aerodynamics. Tata McGraw-Hill Education.